

From partial permutation information to Fourier bounds

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Abstract

We show how information about partial permutations controls full permutation laws. Let n tend to infinity through multiples of eight. If the images of a uniformly chosen $7n/8$ labels approach the uniform injection law in average total variation, and the sign mean tends to zero, then the product of two independent permutations with the given law converges to uniform on S_n .

1 What partial observations can determine

A permutation can look uniform on many labels while retaining information about the whole deck. Parity gives the simplest example: the uniform law on the alternating group has uniform images on any set of at most $n - 2$ labels. Thus partial observations require an additional argument before they can establish convergence on S_n . This paper develops that argument. Its input is quantitative information about partial permutations; its output is a bound on every nontrivial Fourier matrix, with the sign representation treated separately.

Let X be a set of n labels and $G = \text{Sym}(X)$. We regard $g \in G$ as the map from initial labels to final positions. For $M \subseteq X$, let $H_M = \text{Sym}(M)$ fix $X \setminus M$ pointwise. Two permutations have the same images on $X \setminus M$ precisely when they lie in the same left coset gH_M . Consequently an upper bound on the distribution of those images is exactly an upper bound on the masses of these cosets. The elements of gH_M are obtained by multiplication on the right; this convention matters when image observations are compared with inverse-image observations.

For a nonnegative measure f on G and an irreducible unitary representation $\rho : G \rightarrow U(V_\rho)$, write

$$\hat{f}(\rho) = \sum_{g \in G} f(g) \rho(g), \quad D_\rho = \dim V_\rho.$$

Our Hilbert–Schmidt norm is unnormalized: $\|A\|_{\text{HS}}^2 = \text{tr}(A^* A)$. Let D_λ denote the degree of the irreducible of S_m indexed by a partition $\lambda \vdash m$, and set

$$C_u = \sup_{m \geq 1} \sum_{\lambda \vdash m} D_\lambda^{-u} \quad (u > 0).$$

These constants are finite, and the sums with the row and column partitions removed tend to zero. A short proof is given in Lemma 2.1; Section 6 retains the distinct counting proofs that give more detailed information.

Theorem 1.1 (Complementary cosets control Fourier matrices). *Partition X into b nonempty sets $M_1, \dots, M_b$. Put $H_i = H_{M_i}$. Suppose f is a nonnegative measure on G of mass at most one and, for every $g \in G$ and i ,*

$$f(gH_i) \leq \frac{B}{[G : H_i]}. \quad (1.1)$$

For every $u > 0$ and every irreducible ρ of degree D ,

$$\|\widehat{f}(\rho)\|_{\text{HS}}^2 \leq bBC_u D^{-1+(u+2)/b}. \quad (1.2)$$

There is also an independent orbit-span proof of the bound

$$\|\widehat{f}(\rho)\|_{\text{op}}^2 \leq bBC_u D^{-1+(u+2)/b}. \quad (1.3)$$

No restriction is imposed on block sizes or on the multiplicities in $\rho|_{H_1 \times \dots \times H_b}$.

The Hilbert–Schmidt conclusion implies the operator-norm conclusion, but the two proofs use different information. The orbit proof follows one vector and controls the dimension of its orbit under a small subgroup type. The isotypic proof instead bounds the entire Fourier matrix by cosetwise Plancherel. We give the orbit proof first because it isolates the multiplicity issue in a concrete linear map, and then prove the stronger norm conclusion independently.

The principal consequence is a transfer from observations of $7n/8$ labels to the product of just two independent permutations. Suppose 8 divides n , the average total-variation error of these observations tends to zero, and the sign mean is $o(1)$. One can choose eight complementary blocks whose errors sum to $o(1)$, then retain a common subprobability of mass $1 - o(1)$ with coset cap one. With $b = 8$ and $u = 1$, Theorem 1.1 gives squared Hilbert–Schmidt norm at most $8C_1 D^{-5/8}$. Two convolutions have nonsign Plancherel contribution at most $(8C_1)^2 \sum D^{-1/4}$, which tends to zero. The discarded mass and sign coefficient also vanish. Corollary 3.6 gives the precise conclusion. The orbit proof alone gives four convolutions, as recorded in Corollary 3.3. The distinction is the dimension factor saved by controlling the whole matrix, not a change in the observed marginals.

1.1 The first proof and its extensions

Restriction to $H_1 \times \dots \times H_b$ decomposes an irreducible space into tensor-product types, each tensored with its multiplicity space. The product of the factor degrees is at most D . Hence some factor has degree at most $D^{1/b}$. Assign the whole type to one such factor. This produces an orthogonal decomposition of any test vector into b pieces. The orbit of one piece under its assigned subgroup has dimension at most the sum of the squares of the small factor degrees. Multiplicity causes no extra factor: on all copies of one type the group acts by the same matrix tensored with identity. The bound C_u now controls that sum. Averaging the projections onto translates of the orbit span, Schur’s lemma turns its dimension into a factor $1/D$. This proves (1.3).

The next proof retains more of the Fourier matrix. The central idempotent of a subgroup type projects onto all its copies. Cosetwise convolution bounds the squared Hilbert–Schmidt norm of the projected transform, summed over every ambient irreducible. The small subgroup types cover the original representation; taking a trace against a positive operator completes (1.2). This also explains why one must count types once rather than count their multiplicities.

The complete fixed-coset route is proved in Sections 2–3. Sections 4 and 5 compare alternative coefficient arguments and retained measures for the same marginal information. Section 6 develops the shape-sensitive degree estimates needed later, and Section 7 constructs one-dimensional characters in subgroups of controlled index. Supplementary tableau and hook proofs are grouped in Appendix A. Appendix C retains the projection-averaging and longest-line peeling methods, together with explicit parameter substitutions. These independent routes remain available without interrupting the main transfers.

The remaining results distinguish three kinds of additional information. First, image and inverse-image caps control opposite sides of a Fourier matrix. Section 8 places those sides together

in a product of two laws. Section 9 instead works with transition kernels: the sign contribution is then an operator on a multiplicity space, not merely the scalar sign mean of a group law.

Second, the quantifiers on an information bound determine which changes of law it permits. Section 10 constructs retained laws by selecting good omitted sets separately for each permutation, under the same averaged marginal hypothesis as the fixed-partition route. Section 11 assumes an entropy comparison for every law on one common event; it can therefore condition on a rare representation-coefficient event. Section 12 obtains analogous rare-event control from a pointwise bound on accumulated reveal errors. In contrast, the single-law entropy bounds of Section 13 give a growing coset cap by clipping. They control only sufficiently large dimensions; Section 14 supplies a precise hybrid conclusion when a separate law handles the remaining diagrams by long-row decay and exact vanishing on diagrams with more than $n/2$ rows.

Finally, Section 15 uses a conditional product structure: after fixing an environment, the two half-permutations are independent. A signed tabloid basis turns their image caps into matrix bounds, while a preceding law’s inverse-image caps average positive operators. Appendix B gives an independent arm–leg construction of the signed representation used there. These different hypotheses are kept explicit; no mixing conclusion is inferred by substituting one kind of partial information for another.

1.2 Relation to earlier methods

Fourier analysis of random walks on finite groups converts distributional convergence into estimates of matrix transforms. Diaconis and Shahshahani’s analysis of random transpositions is a fundamental example [3]; the Plancherel and total-variation steps used here belong to that framework. We keep matrix norms throughout because the laws considered here need not be central and their transforms need not be normal.

The representation-theoretic ingredients are classical. Specht modules, Young branching, induced representations from Young subgroups, and the Littlewood–Richardson rule supply the restriction and signed-tabloid constructions; see James [6] and Sagan [13]. The hook formula of Frame, Robinson, and Thrall [5] and elementary standard-tableau constructions supply degree lower bounds. The reciprocal-degree limits are part of the symmetric-group Witten-zeta theory; see Liebeck and Shalev [7, Proposition 2.5 and Theorem 2.6]. We give elementary proofs suited to the different transfer arguments and retain their more detailed tableau bounds. The orbit-span and coset estimates are proved explicitly. Relative entropy enters through its chain rule and its variational behavior under conditioning; the rare-event arguments specify the changed law explicitly.

One motivation is the Thorp shuffle, introduced in [14]. Entropy methods yielded an $O((\log n)^4)$ mixing bound for even decks and an $O((\log n)^3)$ bound for power-of-two decks [8, 9]. These bounds count elementary cut-and-interleave shuffles; on $n = 2^d$ cards, a complete coordinate sweep consists of d such steps. The entropy arguments reveal and compare card trajectories, providing a methodological antecedent to the partial-information hypotheses considered here.

More directly, Czumaj and Vöcking studied the joint randomization of a fixed fraction of labeled cards under the Thorp shuffle [2]. Czumaj subsequently obtained full-permutation sampling by recursively applying partial-permutation randomizers on successively smaller sets [1]. The present transfer has a different form: complementary quotient estimates for a single law control its noncentral Fourier matrices and, with separate sign control, imply mixing after a fixed number of independent compositions of that same law. No shuffle construction or physical-time bound is assumed in the abstract proofs.

Three companion papers construct inputs for the later transfers. The coordinate-frame paper proves the all-tilt entropy estimate in Lemma 12.13 and establishes the quotient and retained sign-multiplicity estimates within the proof of Theorem 14.2 [12]. The conditional-information paper supplies the fixed-list estimate in Proposition 3.3; the palette estimate and common-event construction in Proposition 13.2 and equations (13.2)–(13.3); the random-domain estimate in Theorem 12.2; the weak-entropy and sparse-Fourier inputs in Propositions 15.1 and 15.4; and the entropy assertion (16.1) of Theorem 16.1 [10]. The conditional-permutation paper constructs the two-half environment and its independent internal permutations in the proof of Theorem 7.1 and Lemma 7.2 [11]. These estimates are established before the cited papers apply the transfer results here. Their full-deck mixing conclusions are applications, not hypotheses of the present proofs; their discarded-mass, parity, and physical-time checks remain part of those applications.

2 Degree sums, common trimming, and normalization

We first supply the three ingredients needed to complete a transfer: a uniform count of small representation degrees, one measure satisfying all coset bounds at once, and the precise Fourier conversion to total variation. For a finite set, U denotes uniform probability. For a subgroup $H \leq G$, the pushforward of μ to G/H is denoted by μ_H , so $\mu_H(gH) = \mu(gH)$. Our convention is $\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_x |\mu(x) - \nu(x)|$ for probabilities.

Lemma 2.1 (A short reciprocal-degree proof). *For every $u > 0$, $C_u < \infty$ and*

$$Z_u(n) := \sum_{\lambda \vdash n, \lambda \notin \{(n), (1^n)\}} D_\lambda^{-u} \longrightarrow 0.$$

The excluded partitions form a set, so the unique partition of one is removed only once.

Proof. We use the classical tableau dimension and hook formulas: D_λ is the number of fillings of the diagram of λ by $1, \dots, n$ increasing along each row and column, and

$$D_\lambda = \frac{n!}{\prod_{x \in \lambda} h(x)},$$

where $h(x)$ counts the box x and the boxes strictly right of it or strictly below it [5, 6]. Write $p(k)$ for the number of partitions of k , with $p(0) = 1$. Euler's product at $x = e^{-1/\sqrt{k}}$ gives, for $k \geq 1$,

$$\log p(k) \leq \sqrt{k} + \sum_{j \geq 1} \frac{1}{j(e^{j/\sqrt{k}} - 1)} \leq \sqrt{k} \left(1 + \sum_{j \geq 1} j^{-2} \right) \leq 3\sqrt{k}.$$

If a Young diagram has a top row of length a and $j \leq a$ lower boxes, its number of standard tableaux is at least $\binom{a}{j}$. Put the first j labels in the first j top-row positions, choose the j lower labels from the remaining a labels, and fill the lower diagram in one fixed standard relative order. Its width is at most j , so every top-row predecessor is smaller. Fill the rest of the row increasingly. A tableau of any Young subdiagram extends to the full diagram by adding larger labels in an order respecting row and column predecessors.

Orient a longest row or column as the first row and let its length be $n - j$. When $1 \leq j \leq n/3$, at most $2p(j)$ diagrams have this value and

$$D_\lambda \geq \binom{n-j}{j} \geq 2^j.$$

For each fixed $j > 0$ the binomial coefficient diverges, and $2p(j)2^{-uj}$ is summable. Dominated convergence handles this range.

In the remaining range put $a = \max(\lambda_1, \lambda'_1) < 2n/3$. If $a \geq n/10$, retain the first row and a lower Young order ideal of $\lfloor a/2 \rfloor$ boxes; there are enough lower boxes. Extension and the preceding construction give $D_\lambda \geq 2^a/(a+1)$, exponential in n . If $a < n/10$, every hook has length at most $2a$, and the hook formula gives $D_\lambda \geq n!/(2a)^n \geq (5/e)^n$. Both bounds beat $p(n) \leq e^{3\sqrt{n}}$. Thus $Z_u(n) \rightarrow 0$. Adding the row and column shows that the full sum tends to two; finitely many smaller values are finite. $\square$

Lemma 2.2 (Selecting one partition). *If n is divisible by b and a law μ on S_n satisfies*

$$\mathbb{E}_{|M|=n/b} \|\mu_{H_M} - U_{G/H_M}\|_{\text{TV}} \leq \delta,$$

there is a partition into b equal blocks for which the sum of the complementary coset errors is at most $b\delta$.

Proof. Every block of a uniformly chosen ordered equal partition is itself a uniform n/b -subset. The expected sum of the errors is at most $b\delta$, so one partition has sum no larger than that expectation. $\square$

Lemma 2.3 (Two common trimming operations). *Let $H_1, \dots, H_b$ be subgroups of a finite group G , let μ be a probability, and put $\delta_i = \|\mu_{H_i} - U_{G/H_i}\|_{\text{TV}}$ and $\delta = \sum_i \delta_i$. There is a subprobability $f \leq \mu$ with*

$$f(G) \geq 1 - \delta, \quad f(gH_i) \leq [G : H_i]^{-1} \quad (g \in G, i \leq b).$$

For any $c > 1$, deleting the union of all cosets with original density greater than c loses mass at most $c\delta/(c-1)$. If that loss is $\eta < 1$, the conditional law ν has

$$\|\nu - \mu\|_{\text{TV}} = \eta, \quad \nu(gH_i) \leq \frac{c}{1-\eta} [G : H_i]^{-1}.$$

In particular $c = 2$ loses at most 2δ .

Proof. Process the subgroup coset systems successively. Within each coset whose current mass exceeds its uniform mass, multiply all current point masses by the ratio of the cap to the current mass. The loss at stage i is at most the original positive excess on the i th quotient, namely δ_i , because all current masses are below μ . Later decreases preserve earlier caps.

There is also a simultaneous construction with the same guarantee. For each i , trim μ separately to a measure f_i , and set $f(g) = \min_i f_i(g)$. Then $f \leq f_i$ ensures every cap, while $\mu - f \leq \sum_i (\mu - f_i)$ bounds the loss by δ . Thus these two constructions need not produce the same measure, but prove exactly the same assertion.

If a coset Q has $\mu(Q) > cU(Q)$, then $\mu(Q) \leq c(\mu(Q) - U(Q))/(c-1)$. Summing over its heavy cosets and then over subgroup systems bounds the removed mass. A surviving coset has original mass at most $cU(Q)$; division by $1-\eta$ gives the displayed cap. Conditioning deletes mass η and adds that same mass on the retained set, proving the variation equality. $\square$

Lemma 2.4 (Plancherel with a mass deficit). *For every complex mass function a on a finite group G ,*

$$|G| \sum_g |a(g)|^2 = \sum_{\rho \in \text{Irr}(G)} D_\rho \|\widehat{a}(\rho)\|_{\text{HS}}^2.$$

If $f \geq 0$ has mass $z \leq 1$, then

$$|G| \sum_g |f(g) - zU(g)|^2 = \sum_{\rho \neq \mathbf{1}} D_\rho \|\widehat{f}(\rho)\|_{\text{HS}}^2, \quad (2.1)$$

$$\left(\sum_g |f(g) - U(g)| \right)^2 \leq (1 - z)^2 + \sum_{\rho \neq \mathbf{1}} D_\rho \|\widehat{f}(\rho)\|_{\text{HS}}^2. \quad (2.2)$$

If $f \leq \mu$ for a probability μ , its normalization, when $z > 0$, is at variation distance at most $1 - z$ from μ .

Proof. Expand the squared Hilbert–Schmidt norms. The regular-character identity

$$\sum_{\rho} D_\rho \chi_\rho(h^{-1}g) = |G| \mathbf{1}_{g=h}$$

proves Plancherel. Apply it to $f - zU$, whose trivial coefficient is zero, and to $f - U$, whose trivial coefficient is $z - 1$. Cauchy–Schwarz gives (2.2). Finally, $\sum |f/z - f| = 1 - z = \sum |\mu - f|$, so the triangle inequality proves the normalization claim. $\square$

With convolution $(f * k)(g) = \sum_x f(x)k(x^{-1}g)$, transforms multiply in the displayed order. If $f \leq \mu$, positivity implies $f^{*r} \leq \mu^{*r}$ and the missing mass is $1 - z^r \leq r(1 - z)$. If μ has sign mean s , then $|\widehat{f}(\text{sgn})| \leq |s| + 1 - z$. For nearby probabilities, the corresponding sign perturbation is at most twice their total-variation distance.

Proposition 2.5 (Fourier amplification). *Let μ_n be probabilities on S_n with $\widehat{\mu}_n(\text{sgn}) = o(1)$. Suppose either f_n is a nonnegative subprobability with $f_n \leq \mu_n$ and mass tending to one, or f_n is a probability with $\|f_n - \mu_n\|_{\text{TV}} = o(1)$. If, uniformly for every irreducible of degree $D > 1$,*

$$\|\widehat{f}_n\|_{\text{op}} \leq AD^{-\beta},$$

where $A, \beta > 0$ are fixed, then $\|\mu_n^{*r} - U\|_{\text{TV}} \rightarrow 0$ for every fixed integer r with $2 - 2r\beta < 0$. If instead $\|\widehat{f}_n\|_{\text{HS}}^2 \leq AD^{-\gamma}$, the sufficient condition is $1 - r\gamma < 0$.

Proof. The sign term tends to zero by the preceding mass bounds. In the operator case,

$$D \|\widehat{f}_n^r\|_{\text{HS}}^2 \leq D^2 \|\widehat{f}_n\|_{\text{op}}^{2r} \leq A^{2r} D^{2-2r\beta}.$$

In the Hilbert–Schmidt case submultiplicativity gives $D \|\widehat{f}_n^r\|_{\text{HS}}^2 \leq A^r D^{1-r\gamma}$. The sums tend to zero by Lemma 2.1. The trivial coefficient tends to one. Lemma 2.4 gives convergence for f_n^{*r} ; the missing mass or a telescoping replacement of the r probability factors gives convergence for μ_n^{*r} . No normality or diagonalization is used. $\square$

3 The complete transfer from disjoint coset systems

The coset bounds are now available on one common measure. We first turn them into an operator estimate by following the orbit of one vector. For eight complementary coset systems, this gives convergence after four convolutions. Central projections then retain the whole Fourier matrix and reduce that count to two. Both arguments keep every restriction multiplicity; they differ in what they average.

Lemma 3.1 (The orbit of a single vector). *Let H act unitarily on V , and suppose a vector v has components only in H -types belonging to a set $\Lambda \subseteq \text{Irr}(H)$. Then*

$$\dim \text{span}\{\rho(h)v : h \in H\} \leq \sum_{\tau \in \Lambda} (\dim \tau)^2.$$

All copies of a type are included in this estimate.

Proof. Regroup the entire τ -isotypic space as $V_\tau \otimes \mathcal{A}_\tau$, where H acts as $\tau \otimes I$. If v_τ is the component of v there, its orbit lies in the image of

$$\text{End}(V_\tau) \longrightarrow V_\tau \otimes \mathcal{A}_\tau, \quad Q \longmapsto (Q \otimes I)v_\tau.$$

The image has dimension at most $(\dim \tau)^2$, independently of $\dim \mathcal{A}_\tau$. Equivalently, writing $v_\tau = \sum_{j=1}^{\dim \tau} e_j \otimes a_j$, the orbit lies in $V_\tau \otimes \text{span}(a_j)$. Sum over inequivalent types. $\square$

Write $a = \dim \tau$, $W_\tau = \text{span}\{\rho(h)v_\tau : h \in H\}$, and $T(Q) = (Q \otimes I)v_\tau$. The key point is the map in Figure 1: multiplicity enlarges the ambient space, but not the matrix space that moves a fixed vector.

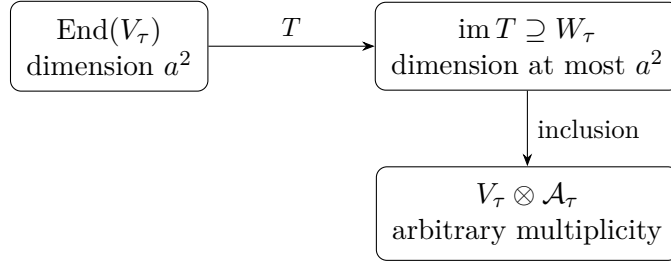


Figure 1: The orbit of one vector in all copies of a subgroup type of degree a . The restriction of the group action factors through a matrix space of dimension a^2 .

Proposition 3.2 (Orbit-span transfer). *Under the assumptions of Theorem 1.1, for every $u > 0$,*

$$\|\hat{f}(\rho)\|_{\text{op}} \leq \sqrt{bBC_u} D^{-1/2+(u+2)/(2b)}.$$

Proof. The disjoint subgroups form a direct product $H = H_1 \times \cdots \times H_b$. Its isotypic decomposition in V_ρ is an orthogonal sum of spaces

$$(V_{\tau_1} \otimes \cdots \otimes V_{\tau_b}) \otimes \mathcal{A}_\tau.$$

Every occurring tensor product has dimension $\prod_i \dim \tau_i \leq D$. Assign its whole isotypic space to the least index i for which $\dim \tau_i \leq D^{1/b}$. Group the assigned spaces to obtain $V_\rho = E_1 \oplus \cdots \oplus E_b$. Equivalently, let P_i select the small H_i -types. These projections commute and $\prod_i (I - P_i) = 0$. The projections $Q_i = P_i \prod_{j < i} (I - P_j)$ are orthogonal and sum to identity: on each product type Q_i is identity exactly when i is its first small factor. Thus the explicit splitting $v_i = Q_i v$ is precisely the least-index assignment above.

Fix $v_i \in E_i$ and put $W_i = \text{span} \rho(H_i)v_i$. Lemma 3.1 and the definition of C_u give

$$\dim W_i \leq \sum_{\tau \in \text{Irr}(H_i): \dim \tau \leq D^{1/b}} (\dim \tau)^2 \leq C_u D^{(u+2)/b}. \quad (3.1)$$

The second inequality follows term by term from $(\dim \tau)^2 \leq D^{(u+2)/b}(\dim \tau)^{-u}$. It is uniform in the block size.

The remaining task is to use this dimension bound on a coefficient. Let Π_i be the orthogonal projection onto W_i . Since W_i is H_i -invariant, the projection onto $\rho(g)W_i$ depends only on gH_i . The coset cap therefore gives, for every w ,

$$\begin{aligned} \sum_g f(g) \|\Pi_{\rho(g)W_i} w\|^2 &\leq B \mathbb{E}_{g \sim U_G} \|\Pi_{\rho(g)W_i} w\|^2 \\ &= B \frac{\dim W_i}{D} \|w\|^2. \end{aligned}$$

Indeed $\mathbb{E}_U \rho(g) \Pi_i \rho(g)^*$ commutes with $\rho(G)$ and has trace $\dim W_i$, so Schur's lemma identifies it as $(\dim W_i/D)I$.

Since $\rho(g)v_i \in \rho(g)W_i$ and f has mass at most one, Cauchy–Schwarz yields

$$|\langle w, \hat{f}(\rho)v_i \rangle| \leq \|v_i\| \left(\sum_g f(g) \|\Pi_{\rho(g)W_i} w\|^2 \right)^{1/2} \leq \sqrt{BC_u} D^{-1/2+(u+2)/(2b)} \|w\| \|v_i\|.$$

For $v = \sum_i v_i$, orthogonality gives $\sum_i \|v_i\| \leq \sqrt{b} \|v\|$. Summing proves the claim. In particular no dimension of a multiplicity space has been discarded. $\square$

Corollary 3.3 (Four convolutions by the orbit argument). *Let $n \rightarrow \infty$ through multiples of eight. Suppose probabilities μ_n on S_n have sign mean $o(1)$ and*

$$\mathbb{E}_{|M|=n/8} \|\mu_{n,H_M} - U_{G/H_M}\|_{\text{TV}} = o(1).$$

*Then $\|\mu_n^{*4} - U_G\|_{\text{TV}} \rightarrow 0$.*

Proof. Lemma 2.2 selects an eight-block partition with summed marginal error $o(1)$. Lemma 2.3 supplies a common subprobability $f_n \leq \mu_n$ of mass $1 - o(1)$ and coset cap one. Proposition 3.2 at $u = 1$ gives $\|\hat{f}_n(\rho)\|_{\text{op}} \leq \sqrt{8C_1} D^{-5/16}$. The nonsign Plancherel contribution after four factors is at most $(8C_1)^4 Z_{1/2}(n)$, because $2 - 8(5/16) = -1/2$. Sign and missing mass tend to zero by Proposition 2.5. This completes a route from averaged observations to the full law. $\square$

3.1 Central projections retain the whole Fourier matrix

The orbit argument controls one vector at a time. To estimate the Hilbert–Schmidt norm without paying an extra ambient dimension, we instead convolve with a central idempotent of the subgroup. Its cosetwise squared norm controls all Fourier matrix entries at once.

For an irreducible H -type τ of degree a , define the mass function

$$e_{H,\tau}(h) = \frac{a}{|H|} \overline{\chi_\tau(h)} \quad (h \in H), \quad e_{H,\tau} = 0 \text{ off } H.$$

Character orthogonality shows that $\hat{e}_{H,\tau}(\rho)$ is the orthogonal projection $P_{\rho,H,\tau}$ onto the full τ -isotypic subspace in $\rho|_H$. Thus every copy is included.

Lemma 3.4 (Cosetwise isotypic estimate). *Let $H \leq G$ and let $f \geq 0$ have mass $z \leq 1$ and satisfy $f(gH) \leq B/[G : H]$. For an irreducible H -type τ of degree a ,*

$$\sum_{\rho \in \text{Irr}(G)} D_\rho \|\hat{f}(\rho) P_{\rho,H,\tau}\|_{\text{HS}}^2 \leq Bza^2. \quad (3.2)$$

Proof. Convolution with $e = e_{H,\tau}$ acts separately in each left coset $Q = xH$. Since $\sum_{h \in H} |e(h)|^2 = a^2/|H|$, the triangle inequality for a positive weighted sum of translates gives

$$\sum_{h \in H} |(f * e)(xh)|^2 \leq f(Q)^2 \frac{a^2}{|H|}.$$

One may equivalently normalize the weights by $f(Q)$ and use Jensen; the zero-mass coset contributes zero. Also $\sum_Q f(Q)^2 \leq Bz/[G : H]$. Plancherel on G proves (3.2).

There is a second exact expression for the same calculation. Let $f_Q(h) = f(xh)$ and $\hat{f}_Q(\tau) = \sum_h f_Q(h)\tau(h)$. Subgroup Plancherel yields

$$\sum_{\rho} D_{\rho} \|\hat{f}(\rho) P_{\rho,H,\tau}\|_{\text{HS}}^2 = [G : H] \sum_Q a \|\hat{f}_Q(\tau)\|_{\text{HS}}^2.$$

Unitarity gives $\|\hat{f}_Q(\tau)\|_{\text{HS}} \leq \sqrt{a} f(Q)$ and recovers the same bound. This identity makes the normalization of the Hilbert–Schmidt norm explicit.

There is also a positive-projection proof of the same bound. Right convolution by e is an orthogonal projection, since $e * e = e$ and $e^*(h) = \overline{e(h^{-1})} = e(h)$. With the counting inner product,

$$\begin{aligned} |G| \|f * e\|_2^2 &= |G| \langle f, f * e \rangle \\ &\leq a^2 |G| \langle f, f * U_H \rangle \leq Bza^2. \end{aligned} \tag{3.3}$$

Indeed $|e(h)| \leq a^2 U_H(h)$, so positivity of f bounds the absolute convolution pointwise; and $(f * U_H)(g) = f(gH)/|H| \leq B/|G|$. This proves the estimate from idempotence and positivity, independently of the cosetwise norm calculation. $\square$

Proposition 3.5 (Isotypic transfer). *Under the assumptions of Theorem 1.1,*

$$\|\hat{f}(\rho)\|_{\text{HS}}^2 \leq bBC_u D^{-1+(u+2)/b}.$$

In fact the right side can be multiplied by $f(G)$.

Proof. For each i , take all H_i -isotypic projections with degree at most $D^{1/b}$. On every product type in the restriction to $\prod_i H_i$, at least one such projection is identity. The projections commute and include all multiplicities, so

$$I \preceq \sum_{i=1}^b \sum_{\tau: \dim \tau \leq D^{1/b}} P_{\rho, H_i, \tau}. \tag{3.4}$$

Trace this inequality against the positive operator $\hat{f}(\rho)^* \hat{f}(\rho)$, apply Lemma 3.4 after discarding other ambient irreducibles, and use (3.1). The resulting inequality is $D \|\hat{f}(\rho)\|_{\text{HS}}^2 \leq Bf(G)bC_u D^{(u+2)/b}$. $\square$

This proves Theorem 1.1. The improvement from operator to Hilbert–Schmidt norm changes the convolution count because the regular-representation weight is then D , rather than the D^2 obtained by bounding $\|A\|_{\text{HS}}^2$ by $D\|A\|_{\text{op}}^2$.

Corollary 3.6 (Two convolutions from eight complements). *Under the hypotheses of Corollary 3.3,* $\|\mu_n^{*2} - U_G\|_{\text{TV}} \rightarrow 0$.

Proof. Use the same partition and subprobability as in that corollary. Proposition 3.5 with $u = 1$ gives HS squared at most $8C_1 D^{-5/8}$. Two powers have nonsign contribution at most $(8C_1)^2 Z_{1/4}(n)$, since $1 - 2(5/8) = -1/4$. Proposition 2.5 handles sign and mass. $\square$

The last corollary is an abstract consequence of the stated marginal hypotheses. A shuffle application must supply those hypotheses at its own time and preserve its own time convention; a numerical mixing statement is not part of this abstract deduction.

4 Coefficient spaces and finer fixed-partition estimates

The central-isotypic estimate already controls the whole Fourier matrix. This section compares three features of alternative fixed-coset arguments: evaluation in a coefficient space, pointwise character control, and the number of restriction types allowed by the ambient diagram. The first two give weaker bounds but require different intermediate estimates; the branching refinement retains information that the uniform constant C_u suppresses. Throughout this section $X = M_1 \sqcup \cdots \sqcup M_b$, $H_i = \text{Sym}(M_i)$ fixes the complement, and $f \geq 0$ has mass at most one and satisfies $f(gH_i) \leq B/[G : H_i]$. The constants C_u have the meaning fixed in the introduction.

4.1 Comparison of the information and the conclusions

The first three rows of Table 1 compare methods under the same fixed-coset hypothesis; the remaining rows locate transfers with different orientations, quantifiers, or conditional structure. A fixed family of coset bounds controls one common measure. An estimate averaged over omitted sets first needs a choice of partition or a permutation-dependent modification. An entropy inequality valid for every change of law can be applied to a rare coefficient event; an entropy estimate for the reference law alone cannot justify that step. For kernels, small quotient error must be supplemented by control of the full sign multiplicity operators.

4.2 Evaluation of coefficient functions

Lemma 4.1 (Evaluation in a coefficient space). *Let H be a finite group and let $\Lambda \subseteq \text{Irr}(H)$. If F is a linear combination of matrix coefficients of the types in Λ , then*

$$\max_{h \in H} |F(h)|^2 \leq \left(\sum_{\tau \in \Lambda} (\dim \tau)^2 \right) \mathbb{E}_{h \sim U_H} |F(h)|^2.$$

Proof. Schur orthogonality makes the functions $\sqrt{\dim \tau} \tau(h)_{jk}$ an orthonormal basis of this coefficient space. At every h , the sum of their squared absolute values is $\sum_{\tau \in \Lambda} (\dim \tau)^2$, because each $\tau(h)$ is unitary. Cauchy–Schwarz for expansion in this basis proves the assertion. Multiple copies of a representation give the same coefficient functions and do not enlarge the space. $\square$

Proposition 4.2 (Coefficient-evaluation transfer). *For every $u > 0$ and irreducible ρ of degree D ,*

$$\|\widehat{f}(\rho)\|_{\text{op}} \leq \sqrt{bBC_u} D^{-1/2+(u+2)/(2b)}.$$

Proof. Let P_i project onto all H_i -types of degree at most $D^{1/b}$. The P_i commute, and $\prod_i (I - P_i) = 0$ by the product-degree bound. Thus $Q_i = P_i \prod_{j < i} (I - P_j)$ are pairwise orthogonal projections summing to identity. Write $v_i = Q_i v$.

Input or method	Orientation and quantifier	Mass treatment	Fourier conclusion
Central isotypic projection	gH_i , every coset of one law	Subprobability allowed	HS squared $bBC_u D^{-1+(u+2)/b}$
Coefficient-space projection	Same fixed cosets	Same retained law	HS squared $bB^2 C_u D^{-1+(u+2)/b}$
Pointwise character bound	Same fixed cosets	Same retained law	HS squared $bB^2 C_u D^{-1+(u+4)/b}$
Two-sided observations	Image and inverse-image cosets	Two retained factors	Joint-type bound, Section 8
Uniform random omitted set	Average marginal TV error	Nearby conditional law	Alternative retained laws; Sections 5 and 10
Entropy on a common event	Every law supported on that event	Condition on the event	Tail $CD^{-1/256}$; Theorem 11.1
Equivariant quotient kernels	Mean rows, or both rows and columns	Subkernels with explicit loss	Product-type HS bounds; Theorems 9.1 and 9.2
Conditional half-products	Inverse-image cap followed by conditional image caps	Product subprobabilities	Operator $\sqrt{22}D^{-1/40}$; Theorem 15.1

Table 1: Inputs that lead to different Fourier transfers. The dimension is that of the ambient irreducible except in the kernel product-type estimates. All Hilbert–Schmidt norms are unnormalized.

For fixed g, w , the function $h \mapsto \langle w, \rho(gh)v_i \rangle$ on H_i uses only the selected types. Lemma 4.1 gives

$$|\langle w, \rho(g)v_i \rangle|^2 \leq C_u D^{(u+2)/b} \mathbb{E}_{h \sim U_{H_i}} |\langle w, \rho(gh)v_i \rangle|^2.$$

Average over g with mass $f(g)$. Right averaging replaces f by $f * U_{H_i}$, whose density relative to U_G is at most B . Schur orthogonality on G gives

$$\sum_g f(g) |\langle w, \rho(g)v_i \rangle|^2 \leq BC_u D^{-1+(u+2)/b} \|w\|^2 \|v_i\|^2.$$

Jensen for a measure of mass at most one and $\sum_i \|v_i\| \leq \sqrt{b}\|v\|$ finish the proof. This argument proves the same numerical inequality as Proposition 3.2, using evaluation of functions instead of dimensions of orbit subspaces. $\square$

4.3 Two more isotypic estimates

Lemma 4.3 (Coefficient and pointwise character estimates). *Let $H \leq G$, let $f \geq 0$ have mass at most one and coset cap $B/[G : H]$, and let $\tau \in \text{Irr}(H)$ have degree a . For every $\rho \in \text{Irr}(G)$,*

$$D_\rho \|\widehat{f}(\rho) P_{\rho, H, \tau}\|_{\text{HS}}^2 \leq B^2 a^2, \quad (4.1)$$

$$D_\rho \|\widehat{f}(\rho) P_{\rho, H, \tau}\|_{\text{HS}}^2 \leq B^2 a^4. \quad (4.2)$$

The first bound comes from coefficient projection on each coset; the second comes from a pointwise bound on the character idempotent.

Proof. Put $F = |G|f$. On xH , project $h \mapsto F(xh)$ onto the conjugates of the matrix coefficients of τ . Its uniform squared norm is

$$a \|\mathbb{E}_{h \in H} F(xh) \tau(h)\|_{\text{HS}}^2 \leq B^2 a^2,$$

because the matrix in the norm has operator norm at most the nonnegative function's mean, which is at most B . The entries of $\rho(xh)P_{\rho, H, \tau}$ belong to the τ coefficient space. Replacing F by this projection therefore preserves the tested Fourier matrix. Averaging the bound over cosets and applying full-group Plancherel proves (4.1).

For the second proof, the character bound $|\chi_\tau(h)| \leq a$ gives

$$|(f * e_{H, \tau})(g)| \leq \frac{a^2}{|H|} f(gH) \leq \frac{Ba^2}{|G|}.$$

The density of this convolution has squared $L^2(U_G)$ norm at most $B^2 a^4$. Plancherel gives (4.2). No bound on the conditional law inside a coset was required in either proof. $\square$

Proposition 4.4 (Coefficient and character Hilbert–Schmidt transfers). *For the disjoint-block data of this section and every $u > 0$,*

$$\|\hat{f}(\rho)\|_{\text{HS}}^2 \leq B^2 b C_u D^{-1+(u+2)/b}, \quad (4.3)$$

$$\|\hat{f}(\rho)\|_{\text{HS}}^2 \leq B^2 b C_u D^{-1+(u+4)/b}. \quad (4.4)$$

Proof. Trace the covering inequality (3.4) against $\hat{f}^* \hat{f}$. Sum the corresponding estimate of Lemma 4.3 over all selected subgroup types. For $j = 2, 4$,

$$\sum_{\dim \tau \leq D^{1/b}} (\dim \tau)^j \leq C_u D^{(u+j)/b}.$$

This proves both bounds, including every multiplicity. $\square$

Although (3.2) gives a stronger bound, (4.4) records the method that uses only pointwise character control. For $u = 2$, large block size makes the complete subgroup reciprocal sum at most three, so cap $B = 4$ gives the concrete bound $48bD^{-1+6/b}$.

Proposition 4.5 (Low-type central-projector estimates). *Let η be a probability satisfying the coset caps of this section. Let P_i select its H_i -types of degree at most $D^{1/b}$ and put $R_i = \sum_{\dim \tau \leq D^{1/b}} (\dim \tau)^2$. Then*

$$\|\hat{\eta}(\rho)P_i\|_{\text{op}} \leq BD^{-1/2}R_i, \quad D\|\hat{\eta}(\rho)P_i\|_{\text{HS}}^2 \leq B^2 R_i.$$

Consequently

$$\begin{aligned} \|\hat{\eta}(\rho)\|_{\text{op}} &\leq BC_u \sqrt{b} D^{-1/2+(u+2)/b}, \\ \|\hat{\eta}(\rho)\|_{\text{op}} &\leq B\sqrt{bC_u} D^{-1/2+(u+2)/(2b)}. \end{aligned}$$

Proof. On H_i take the central projector density $k_i(h) = \sum_{\dim \tau \leq D^{1/b}} (\dim \tau) \overline{\chi_\tau(h)}$, relative to uniform subgroup measure. Its transform is P_i . Pointwise $|k_i| \leq R_i$, so convolution of the density $|G|\eta$ with k_i is bounded in absolute value by BR_i . Plancherel yields the first displayed bound. Alternatively project onto all selected coefficient spaces on each coset. They are orthogonal; their squared mean sums to at most $B^2 R_i$ by the first proof of Lemma 4.3. This gives the second bound.

Use the orthogonal $Q_i = P_i \prod_{j < i} (I - P_j)$, apply either bound to $Q_i v$, and sum their norms. The inequalities $\sum_i \|Q_i v\| \leq \sqrt{b} \|v\|$ and $R_i \leq C_u D^{(u+2)/b}$ give the conclusions. $\square$

4.4 Counting only types allowed by branching

Let $p(j)$ again denote the partition number, and define $M(k) = \sum_{j=0}^k p(j)$. For $\lambda \vdash n$ put $k_\lambda = n - \max(\lambda_1, \lambda'_1)$.

Proposition 4.6 (Branching refinement). *For an irreducible ρ_λ of degree D , the disjoint-block caps give*

$$\|\widehat{f}(\rho_\lambda)\|_{\text{HS}}^2 \leq BbM(k_\lambda)D^{-1+2/b}.$$

The blocks may have unequal sizes.

Proof. Young branching says that each diagram τ occurring under restriction to a block symmetric group is contained in λ ; conjugating a block to the standard subgroup does not change the set of occurring types [6, Theorem 9.2]. Orient a longest row or column of λ horizontally, transposing the restricted diagrams too if necessary. Each τ then has at most k_λ boxes below its first row. Given those lower boxes and the block size, its first row is determined. Thus at most $M(k_\lambda)$ types occur, regardless of multiplicity. Apply (3.2) to each selected small type and bound its squared degree by $D^{2/b}$ in (3.4). $\square$

Appendix C develops two averages of low-type projections, a four-factor coefficient decomposition, and explicit parameter substitutions. Those alternative estimates retain their proofs, while the tools above suffice for the transfers that follow.

5 Keeping the partition random

Selecting one partition and imposing common coset caps already handles the averaged marginal hypothesis. Here we construct a different retained law: each retained permutation is good for many partitions, without choosing one partition on which the law must satisfy every cap. The partition therefore varies inside the coefficient estimate. Its event of simultaneous goodness must be inserted before decomposing a test vector.

Proposition 5.1 (Random-partition lift). *Let n tend to infinity through multiples of a fixed integer $q > 3$, and let μ be probability laws on S_n . For a uniform n/q -subset C , put $H_C = \text{Sym}(C)$, fixing its complement pointwise. Suppose*

$$\mathbb{E}_C \|\mu_{H_C} - (U_G)_{H_C}\|_{\text{TV}} = o(1), \quad \widehat{\mu}(\text{sgn}) = 0.$$

There is a conditional probability law ν with $\|\nu - \mu\|_{\text{TV}} = o(1)$ such that, for every irreducible ρ of degree D ,

$$\|\widehat{\nu}(\rho)\|_{\text{op}} \leq \frac{2}{M} \sqrt{2qC_1} D^{-(1-3/q)/2}, \quad (5.1)$$

*where $M = 1 - o(1)$ is the retained mass and C_1 is from Lemma 2.1. In particular for $q = 8$, $\|\mu^{*8} - U_G\|_{\text{TV}} = o(1)$.*

Proof. Put $h_0 = |H_C|/|G|$. Say g is good for C if $\mu(gH_C) \leq 2h_0$. The total μ -mass of bad cosets is at most twice their marginal total-variation distance. Thus the set

$$R = \{g : \mathbb{P}_C(g \text{ is bad for } C) \leq 1/(2q)\}$$

has mass $M = 1 - o(1)$ by Markov's inequality. For every $g \in R$ a uniform ordered equal partition $(C'_1, \dots, C'_q)$ has probability at least $1/2$ that g is good for all blocks. Define $\nu = \mu(\cdot \mid R)$.

We use Lemma 4.1: if a function F on a finite group is in the span of matrix coefficients of a collection Λ of irreducibles, then

$$\max |F|^2 \leq \left(\sum_{\tau \in \Lambda} (\dim \tau)^2 \right) \mathbb{E}_U |F|^2. \quad (5.2)$$

The sum counts distinct types, since equivalent copies have the same space of coefficient functions.

Fix a partition and restrict ρ to its direct product of block subgroups. Decompose orthogonally into tensor-product constituents, including all multiplicity spaces. Every constituent has product of factor degrees at most D . Assign it to one factor with degree at most $D^{1/q}$. This splits a vector $v = \sum_i v_i$ orthogonally, with v_i using only $H_{C'_i}$ -types of degrees at most $D^{1/q}$. Fix a test vector w and set $F_i(g) = \langle w, \rho(g)v_i \rangle$. On a coset $gH_{C'_i}$ it belongs to the coefficient space in (5.2), whose dimension is at most

$$\sum_{b \leq D^{1/q}} b^2 \leq C_1 D^{3/q}.$$

The sum counts each irreducible type once; multiplicities do not increase the function space of matrix coefficients. Summing the coset maximum bound only over good cosets, and using Schur orthogonality on G , yields

$$\sum_{g \text{ good for } C'_i} \mu(g) |F_i(g)|^2 \leq 2C_1 D^{3/q-1} \|w\|^2 \|v_i\|^2. \quad (5.3)$$

Only a bound on total coset mass was used here.

For $g \in R$ insert the all-good partition event, whose probability is at least $1/2$, before splitting the coefficient. It follows that

$$|\langle w, \hat{\nu}(\rho)v \rangle| \leq \frac{2}{M} \mathbb{E}_{(C'_i)} \sum_i \sum_{g \text{ good for } C'_i} \mu(g) |F_i(g)|.$$

Cauchy–Schwarz and (5.3), followed by $\sum_i \|v_i\| \leq \sqrt{q} \|v\|$, prove (5.1). The dependence of v_i on the partition is harmless because the estimate holds for each fixed partition.

The sign coefficient obeys $|\hat{\nu}(\text{sgn})| \leq (1-M)/M = o(1)$, since the original sign coefficient is zero. The same argument works when that coefficient is $o(1)$. Fourier Plancherel in the normalization of Lemma 2.4 gives

$$4\|\nu^{*r} - U_G\|_{\text{TV}}^2 \leq \sum_{\rho \neq 1} D_\rho \|\hat{\nu}(\rho)^r\|_{\text{HS}}^2.$$

For $q = r = 8$, the nonsign sum is at most an absolute constant times

$$\sum_{\rho \neq 1, \text{sgn}} D_\rho^{2-r(1-3/q)} = \sum_{\rho \neq 1, \text{sgn}} D_\rho^{-3} = o(1)$$

by Lemma 2.1. The sign term also vanishes, and $\|\mu^{*8} - \nu^{*8}\|_{\text{TV}} \leq 8(1-M) = o(1)$. $\square$

6 Symmetric-group degrees: the quantitative toolkit

The lifting estimates involve sums of negative powers of dimensions. The short proof in Lemma 2.1 already supplies every positive inverse power. Here we develop the quantitative estimates used by the later transfers: tableau and hook constructions, exponential growth in the deficit from a longest line, and sharper bounds near a row or column.

Theorem 6.3 combines summability with the uniform deficit bound. Lemmas 6.4 and 6.5 then control near-row dimensions and the number of restriction types. Appendix A retains the independent fixed-power proofs, including routes that use only coarse partition counts or avoid the hook formula.

For a partition $\lambda \vdash n$, write $D_\lambda = f^\lambda$ for the dimension of its complex irreducible representation and λ' for its transpose. We use the classical standard-tableau dimension theorem and hook-length formula [13, 4]:

$$D_\lambda = \#\{\text{standard tableaux of shape } \lambda\} = \frac{n!}{\prod_{x \in \lambda} h(x)}.$$

We set $D_\emptyset = 1$ and $\binom{a}{-1} = 0$. A standard tableau increases along rows and columns; $h(x)$ is one plus the number of boxes strictly to the right of or below x . Transposition preserves dimension. Put

$$L(\lambda) = \max(\lambda_1, \lambda'_1), \quad k(\lambda) = n - L(\lambda), \quad Z_u(n) = \sum_{\lambda \vdash n, \lambda \notin \{(n), (1^n)\}} D_\lambda^{-u}.$$

The excluded diagrams are a set, so coincide when $n = 1$. They are exactly the one-dimensional types: transpositions are conjugate and generate S_n , so a one-dimensional character takes the same value 1 or -1 on all transpositions.

6.1 Counting constructions

Let $p(j)$ be the number of partitions of j , with $p(0) = 1$. We will use each of the elementary bounds

$$p(j) \leq 2^{j-1} \leq 2^j, \quad p(j) \leq (j+1)^{2\lceil \sqrt{j} \rceil}, \quad p(j) \leq e^{3\sqrt{j}} \quad (j \geq 1). \quad (6.1)$$

The first follows by counting compositions. For the second, record the multiplicities of parts at most $\lceil \sqrt{j} \rceil$ and a zero-padded list of the at most $\sqrt{j}$ larger parts. Each entry has at most $j+1$ possibilities. For the third, evaluate the partition generating product at $x = e^{-1/\sqrt{j}}$. Its logarithm is

$$\sum_{r \geq 1} \frac{1}{r(e^{r/\sqrt{j}} - 1)} \leq \sqrt{j} \sum_{r \geq 1} r^{-2} \leq 2\sqrt{j},$$

and extracting the coefficient of x^j costs $e^{\sqrt{j}}$. At a fixed positive value of $k(\lambda)$ there are at most $2p(k)$ shapes, since deleting a longest row or column leaves a partition of k . Also $L(\lambda) \geq \sqrt{n}$, because the diagram is contained in a square of side $L(\lambda)$.

Lemma 6.1 (Tableau constructions). *Suppose a diagram has first row of length a , lower diagram γ of size k , and second row of length c (put $c = 0$ if $k = 0$).*

1. *Every tableau on a Young subdiagram extends to the whole diagram. For $0 \leq b \leq \min(a, k)$,*

$$D_\lambda \geq \binom{a}{b}, \quad D_\lambda \geq \binom{a+k-c}{k}. \quad (6.2)$$

2. *If $k \leq a$, ballot interleavings give*

$$D_\lambda \geq \binom{a+k}{k} - \binom{a+k}{k-1} = \binom{a+k}{k} \frac{a-k+1}{a+1} \geq \frac{1}{k+1} \binom{2k}{k}. \quad (6.3)$$

In particular a two-row rectangle of width b has $\binom{2b}{b}/(b+1)$ tableaux.

3. If the nonempty diagonals of constant row-plus-column index have sizes $t_1, \dots, t_q$, then

$$D_\lambda \geq \prod_{j=1}^q t_j! \geq 2^{n-q}, \quad D_\lambda \geq \left(\frac{n}{qe}\right)^n. \quad (6.4)$$

Thus width and height at most r give $q \leq 2r - 1$ and $D_\lambda \geq 2^{n-2r+1}$.

Proof. A Young subdiagram is closed under moving up and left. Add its remaining boxes in row-major order with successive larger labels; every predecessor is already present. To prove the first bound, retain the whole first row and a Young subdiagram of b lower boxes. Such a subdiagram is obtained by removing lower corners. Fill the first b top boxes with $1, \dots, b$; choose the b lower labels from the remaining a labels and fill them in the relative order of one fixed standard tableau. All lower boxes lie in the first b columns, so filling the rest of the top row increasingly is valid. For the second bound, reserve only the first c top labels and choose all k lower labels from the remaining $a + k - c$ labels.

For the ballot bound, fix a standard order of the lower boxes and interleave their additions with those of the top row. Require each prefix to have at least as many top-row additions as lower additions. The j th lower box lies in a column at most j , so its top predecessor has been inserted. Reflection at the first violating prefix counts the valid words by the displayed binomial difference. A valid balanced word with k additions of each kind, followed by $a - k$ top additions, gives the last bound. For an actual two-row rectangle the word condition is also necessary, proving the exact count.

Finally, put successive batches of labels on increasing diagonals, in arbitrary order within each diagonal. Every row or column predecessor lies on an earlier diagonal. This gives the product of factorials. The first estimate uses $t! \geq 2^{t-1}$; the second uses $t! \geq (t/e)^t$ and convexity of $t \log t$ to obtain $\sum_j t_j \log t_j \geq n \log(n/q)$. $\square$

Lemma 6.2 (Separating the first-row hooks). *With the notation of Lemma 6.1, let b_j be the number of lower boxes in column j . Then*

$$D_\lambda = \binom{a+k}{k} D_\gamma \prod_{j=1}^a \left(1 + \frac{b_j}{a-j+1}\right)^{-1} \quad (6.5)$$

$$\geq \binom{a+k}{k} D_\gamma \exp\left(-\frac{k(1+\log a)}{a}\right). \quad (6.6)$$

If $k \leq a$, a second lower bound is

$$D_\lambda \geq \binom{a+k}{k} D_\gamma \exp\left(-\frac{k}{a-k+1}\right). \quad (6.7)$$

Proof. The lower hooks are unchanged, and the top hooks are $a - j + 1 + b_j$, which proves the identity. The sequence b_j decreases while $(a - j + 1)^{-1}$ increases. The opposite-order rearrangement inequality therefore gives

$$\sum_{j=1}^a \frac{b_j}{a-j+1} \leq \frac{k}{a} \sum_{j=1}^a \frac{1}{j} \leq \frac{k(1+\log a)}{a}.$$

For completeness, the rearrangement inequality follows by expanding $\sum_{i,j} (b_i - b_j)(c_i - c_j) \leq 0$ for oppositely ordered sequences b_i, c_i . Apply $\log(1+x) \leq x$ to the product. For the last estimate, $b_j = 0$ for $j > k$, so every relevant denominator is at least $a - k + 1$, and $\sum_j b_j = k$. $\square$

6.2 All positive powers and the first-row deficit

Theorem 6.3 (Every positive inverse power). *For every fixed real $u > 0$,*

$$C_u := \sup_{n \geq 1} \sum_{\lambda \vdash n} D_\lambda^{-u} < \infty, \quad Z_u(n) \longrightarrow 0. \quad (6.8)$$

Moreover there are absolute constants $c_1, c_2 > 0$ such that, for all sufficiently large n and every $\lambda \vdash n$,

$$D_\lambda \geq c_1 e^{c_2 k(\lambda)}. \quad (6.9)$$

For each fixed positive k , these dimensions tend uniformly to infinity among shapes with $k(\lambda) = k$.

Proof. Put $L = L(\lambda)$ and $k = n - L$. If $L \leq n/16$, every hook is at most $2L$, so $D_\lambda \geq (n/(2eL))^n \geq 2^n$. Otherwise orient the diagram with first row L , and for $k \geq 1$ retain $b = \min(k, \lfloor L/3 \rfloor)$ lower boxes. For sufficiently large n , $b \geq 1$, and Lemma 6.1 gives

$$D_\lambda \geq \binom{L}{b} \geq (L/b)^b \geq 3^b, \quad b \geq k/48 - 1.$$

Indeed $L > n/16$ implies $\lfloor L/3 \rfloor > n/48 - 1 \geq k/48 - 1$. Thus one may take $c_1 = 1/3$ and $c_2 = (\log 3)/48$ after enlarging the size threshold. For a fixed positive k , eventually $b = k$, and $D_\lambda \geq \binom{n-k}{k} \rightarrow \infty$ uniformly.

There are at most $2p(k)$ shapes at deficit k . By (6.1) and (6.9), their total inverse u th powers are bounded by a constant depending on u times $e^{3\sqrt{k} - uc_2 k}$. This is summable in k , and each fixed positive- k contribution tends to zero. Dominated convergence proves $Z_u(n) \rightarrow 0$. The full sum tends to two; its finitely many initial values are finite, including the value one at $n = 1$. $\square$

There are two other ways to obtain the full positive-power range. The next proof uses ballot words; the short proof from Lemma 2.1 uses a central-binomial subdiagram.

A ballot proof of the summability assertions. For $k \geq 1$ and $L = n - k \geq n/2$, (6.3) gives

$$D_\lambda \geq \binom{n}{k} \frac{n - 2k + 1}{n - k + 1} \geq \frac{2^k}{k + 1}.$$

For the second inequality, the ballot fraction is at least $1/(k + 1)$ and $\binom{n}{k} \geq (n/k)^k \geq 2^k$. The majorant $2p(k)(k + 1)^u 2^{-uk}$ is summable, and the first binomial expression diverges for each fixed $k \geq 1$. If $L < n/2$ but $L \geq n/10$, retain a row of length L and L lower boxes. The ballot count $\binom{2L}{L}/(L + 1)$ is exponential in n . If $L < n/10$, the hook formula gives $(5/e)^n$. The subexponential bound for $p(n)$ handles both remaining ranges. Adding the one-dimensional terms proves the uniform bound. $\square$

The third all-positive-power route is the one already proved in Lemma 2.1. Its three ranges are $1 \leq k \leq n/3$, $k > n/3$ with $L \geq n/10$, and $L < n/10$. The respective bounds are $\binom{n-k}{k} \geq 2^k$, $2^L/(L + 1)$, and $(5/e)^n$. Thus it uses the same initial-row construction with a central-binomial subdiagram, rather than the deficit-exponential or ballot estimates.

Lemma 6.4 (Near-row dimensions and inverse-first summability). *For $1 \leq k = k(\lambda) \leq n/4$, with γ the lower diagram after orienting a longest line as the first row,*

$$D_\lambda \geq e^{-1/2} \binom{n}{k} D_\gamma \geq \frac{1}{2} \binom{n}{k}. \quad (6.10)$$

For $k > n/4$ and all sufficiently large n , $D_\lambda \geq e^{n/100}$. These estimates independently give $Z_1(n) \rightarrow 0$ and $C_1 < \infty$.

Proof. Equation (6.7) has exponent $k/(n - 2k + 1) \leq 1/2$, proving (6.10). If $k > n/4$ and $L \geq n/8$, keep the first row and $b = \lfloor n/32 \rfloor$ lower boxes. For large n , $1 \leq b \leq L/4$. The same hook bound gives dimension at least $\frac{1}{2} \binom{L+b}{b} \geq 4^b/2$. If $L < n/8$, every hook is shorter than $n/4$, giving $(4/e)^n$. Both exceed $e^{n/100}$ for sufficiently large n . The near-row inverse sum is bounded by $4 \sum_{1 \leq k \leq n/4} p(k)/\binom{n}{k}$. Each term vanishes at fixed k and is bounded by $4p(k)4^{-k}$; the remaining contribution is at most $p(n)e^{-n/100} = o(1)$. This proves both assertions about inverse-first sums. $\square$

Lemma 6.5 (Absorbing the number of tail types). *Let $M(k) = \sum_{j=0}^k p(j)$. For every fixed $B \geq 1$ and all sufficiently large n , uniformly over $\lambda \vdash n$ with $k(\lambda) \geq 1$,*

$$4BM(k(\lambda)) \leq D_\lambda^{1/4}. \quad (6.11)$$

Proof. The partition bounds give $\log M(k) = O(\sqrt{k} \log(k+1)) = o(k)$. For $k > n/4$, Lemma 6.4 gives $\log D_\lambda \geq n/100$, whereas $\log M(k) = o(n)$ uniformly for $k \leq n$. For $k \leq n/4$, use $D_\lambda \geq \frac{1}{2} \binom{n}{k} \geq \frac{1}{2} 4^k$ once k exceeds a fixed sufficiently large cutoff. The finitely many positive k below that cutoff are handled by the uniform divergence of $\binom{n}{k}$. One size threshold works for all shapes. $\square$

7 One-dimensional characters in subgroups of controlled index

A coset cap also controls a Fourier projection obtained from a one-dimensional subgroup character. This gives a different route to the full isotypic estimates: first find one nonzero character space, then average it over conjugates. The construction here assigns boxes to rows or columns. Appendix C gives an independent construction by successively removing longest lines, together with its reciprocal-degree proof and transfer.

Lemma 7.1 (A character from a row–column assignment). *For every irreducible complex representation of S_n , of degree D , there are a subgroup $J \leq S_n$ and a character $\chi : J \rightarrow \{1, -1\}$ occurring in its restriction such that*

$$[S_n : J] \leq D^{1024}.$$

Proof. We use the polytabloid realization of the Specht module of shape λ ; see [13, 4]. Fix a labeling of its boxes. A row tabloid remembers which labels lie in each row while forgetting their order within that row; its polytabloid is the signed sum of its translates under the column stabilizer. Assign every box either to its row or to its column. Let R permute the assigned set in each row and let C permute the assigned set in each column. These sets are disjoint, so $J = R \times C$. Give R the trivial character and each factor of C the sign character.

Let e be the polytabloid of the labeled diagram. It transforms by sign under the full column stabilizer, hence under C . The average $v = |R|^{-1} \sum_{r \in R} r e$ is fixed by R and still transforms by sign under C . It is nonzero. Indeed the original row tabloid has coefficient one in e , since the full row and column stabilizers intersect trivially. Every element of R fixes that tabloid, so its coefficient is also one in v . This proves character occurrence for every assignment.

It remains to choose the assignment with controlled index. The arm and leg of a box count boxes strictly to its right and strictly below it. Assign a box to its row when its arm is at least its leg, and to its column otherwise. Its hook length is at most twice the length l of the line receiving

it. If that line receives a boxes, then $a! \geq (a/e)^a$; we give a line with $a = 0$ zero contribution below. Consequently

$$\log \frac{\prod_{x \in \lambda} h(x)}{\prod_{\text{assigned lines}} a!} \leq n \log 2 + \sum_{\text{lines}} \{a + a \log(l/a)\} \leq (\log 2 + 1 + 2/e)n \leq 3n.$$

Here $a \log(l/a) \leq l/e$ and the total length of all rows and columns is $2n$. The hook-length formula therefore gives an assignment with index

$$I = [S_n : J] \leq De^{3n}. \quad (7.1)$$

We convert the exponential factor into a power of D . Put $L = \max(\lambda_1, \lambda'_1)$. Suppose first that $n \geq 64$ and $L \leq 3n/4$. If $L \leq n/8$, all hooks are at most $2L$, and $D \geq (n/(2eL))^n \geq (4/e)^n \geq e^{n/64}$. Otherwise transpose so that the first row has length L , and retain a Young order ideal of $s = \lfloor L/4 \rfloor$ boxes below it. Such an ideal exists because $n - L \geq L/3$; moreover $s \geq n/64$. For the retained diagram, let c_j count its lower boxes in column j . Dividing its hook formula by the lower diagram's formula gives

$$D \geq \binom{L+s}{s} \prod_{j=1}^L \left(1 + \frac{c_j}{L-j+1}\right)^{-1} \geq \binom{L+s}{s} \exp\left(-\frac{s}{L-s+1}\right).$$

The first inequality also uses extension of tableaux from a subdiagram. For the second, $c_j = 0$ for $j > s$ and $\sum_j c_j = s$. Since $s \leq L/4$, the logarithm of the last expression is at least $s \log 5 - s/(L-s+1) \geq s$. Thus $\log D \geq n/64$ throughout this case. Equation (7.1) now gives $I \leq D^{193}$.

Suppose instead that $L > 3n/4$. Orient a longest line as the top row, of length L , and write ζ for the lower diagram of size $s = n - L$. Assign the entire top row to its row and use the preceding arm-leg assignment on ζ . Hook comparison and (7.1) applied to the tail give

$$I \leq \binom{n}{s} D_\zeta e^{3s}, \quad D \geq \binom{n}{s} D_\zeta \exp\left(-\frac{s}{L-s+1}\right).$$

If $s \geq 1$, then $\binom{n}{s} \geq (n/s)^s \geq 4^s$ and $L - s + 1 \geq 2$. Hence

$$\log D \geq s(\log 4 - 1/2), \quad \log(I/D) \leq \frac{7}{2}s.$$

These bounds imply $I \leq D^{1024}$. Transposing the assignment back preserves the subgroup index and interchanges trivial and sign factors. If $s = 0$, take the full group and its trivial or sign character. Finally, for $n < 64$ and $D > 1$, the trivial subgroup has index $n! \leq 64^{64} < 2^{1024} \leq D^{1024}$. Degree-one representations again use the full group. This covers every case. $\square$

7.1 From a character space to an isotypic space

The index bound is useful because a subgroup character gives an orthogonal projection through a signed average. Positivity of the original mass function controls that signed average. The next lemma also explains why repeated copies of a subgroup representation cause no loss.

Lemma 7.2 (Averaging character projections). *Let G be a finite group, $H \leq G$, and let $f \geq 0$ have mass $z \leq 1$ and satisfy $f(gH) \leq B/[G : H]$ for every $g \in G$. Let $\tau \in \text{Irr}(H)$ have degree d_τ .*

Suppose $J \leq H$ has a one-dimensional unitary character χ occurring in $\tau|_J$, and put $I = [H : J]$. For $\rho \in \text{Irr}(G)$ of degree D , let Q_τ be the full τ -isotypic projection in $\rho|_H$. Then

$$D\|\hat{f}(\rho)Q_\tau\|_{\text{HS}}^2 \leq BzId_\tau. \quad (7.2)$$

All copies of τ are included in Q_τ .

Proof. Define the complex mass function $e_{J,\chi}$ to be $\overline{\chi(j)}/|J|$ on J and zero elsewhere. Its Fourier transform is the orthogonal projection P onto vectors satisfying $\rho(j)v = \chi(j)v$ for all $j \in J$. With U_H, U_J denoting uniform probability masses on the subgroups, positivity gives

$$|f * e_{J,\chi}| \leq f * U_J \leq I(f * U_H) \leq \frac{IB}{|G|}, \quad \sum_g |(f * e_{J,\chi})(g)| \leq z.$$

Plancherel therefore implies $D\|\hat{f}(\rho)P\|_{\text{HS}}^2 \leq IBz$. Every conjugate of (J, χ) inside H satisfies the same bound.

Average the corresponding projections, writing $\overline{P} = |H|^{-1} \sum_{h \in H} \rho(h)P\rho(h)^*$. On the full H -isotypic space $V_\eta \otimes \mathcal{A}_\eta$, a group-algebra element acts on V_η tensored with the identity on the multiplicity space $\mathcal{A}_\eta$. Schur's lemma thus gives

$$\overline{P}|_{V_\eta \otimes \mathcal{A}_\eta} = \frac{\text{rank}(P|_{V_\eta})}{d_\eta} I.$$

These scalars are nonnegative and the scalar for $\eta = \tau$ is at least $1/d_\tau$. Hence $\overline{P} \geq d_\tau^{-1}Q_\tau$. Test this positive-operator inequality against $\hat{f}(\rho)^*\hat{f}(\rho)$, and average the preceding Plancherel bounds. The result is (7.2). $\square$

Theorem 7.3 (Character-projection transfer). *Let $H_i = \text{Sym}(M_i) \leq S_n$, $1 \leq i \leq b$, for disjoint nonempty sets M_i . Let $f \geq 0$ have mass $z \leq 1$ and suppose*

$$f(gH_i) \leq \frac{B}{[S_n : H_i]} \quad (g \in S_n, 1 \leq i \leq b).$$

For every irreducible ρ of degree D ,

$$\|\hat{f}(\rho)\|_{\text{HS}}^2 \leq BzbC_1D^{-1+1026/b}, \quad (7.3)$$

where $C_1 = \sup_{m \geq 1} \sum_{\tau \in \text{Irr}(S_m)} d_\tau^{-1}$. More generally, if $a \geq 0$ and every type of every H_i has a one-dimensional subcharacter of index at most d_τ^a , the exponent $1026/b$ in (7.3) may be replaced by $(a+2)/b$.

Proof. For each type τ of H_i , Lemma 7.2 and the assumed subcharacter give

$$D\|\hat{f}(\rho)Q_{i,\tau}\|_{\text{HS}}^2 \leq Bzd_\tau^{a+1}.$$

Restrict ρ to the product of the H_i . In any occurring joint type, the product of the factor dimensions is at most D , including when that type has multiplicity greater than one. Some factor therefore has dimension at most $R = D^{1/b}$. The sum of the commuting projections $Q_{i,\tau}$ with $d_\tau \leq R$ dominates the identity. Taking its trace against $\hat{f}(\rho)^*\hat{f}(\rho)$ gives

$$D\|\hat{f}(\rho)\|_{\text{HS}}^2 \leq Bz \sum_{i=1}^b \sum_{d_\tau \leq R} d_\tau^{a+1} \leq BzbC_1R^{a+2}.$$

The last inequality uses $d_\tau^{a+1} \leq R^{a+2}d_\tau^{-1}$. Lemma 7.1 supplies $a = 1024$. $\square$

For example, take $b = 8192$ and $B = 2$. If $f_n \leq \mu_n$ has mass tending to one and $\widehat{\mu}_n(\text{sgn}) = o(1)$, then (7.3) and Proposition 2.5 give $\|\mu_n^{*4} - U_{S_n}\|_{\text{TV}} \rightarrow 0$. Indeed $\varepsilon = 1026/8192 < 1/4$, and the nonlinear Plancherel sum for four factors is bounded by $(2bC_1)^4 \sum_{D_\lambda > 1} D_\lambda^{1-4(1-\varepsilon)}$, which tends to zero even using only the inverse-first-degree sum. This statement requires only the displayed caps, mass and sign input.

8 Joint restriction types and multiplication on two sides

For a fixed collection of disjoint block subgroups, we can retain the number of distinct joint restriction types rather than summing all small factor degrees. This gives an operator estimate by orbit spans. It also gives a product estimate when one measure is capped on left cosets and another is capped on right cosets. These two conclusions use the same type count, but different Fourier estimates.

Let $M_1, \dots, M_b$ be disjoint nonempty subsets of $\{1, \dots, n\}$, and put $H_i = \text{Sym}(M_i)$, acting trivially on the complement. On an irreducible V_λ of S_n , restriction to $H = H_1 \times \dots \times H_b$ has the form

$$V_\lambda|_H = \bigoplus_{\tau} (V_{\tau_1} \otimes \dots \otimes V_{\tau_b}) \otimes \mathcal{A}_\tau. \quad (8.1)$$

The sum ranges over distinct occurring joint types; the nonzero space $\mathcal{A}_\tau$ records their arbitrary multiplicities. Write K_λ for the number of terms in this sum and Q_τ for their orthogonal projections.

Lemma 8.1 (Counting joint types). *Put $k = n - \max(\lambda_1, \lambda'_1)$ and $M(k) = \sum_{j=0}^k p(j)$, where $p(0) = 1$ and $p(j)$ is the partition number. Then*

$$K_\lambda \leq M(k)^b. \quad (8.2)$$

For every fixed b and every fixed $\eta > 0$,

$$\sum_{\lambda \vdash n: D_\lambda > 1} K_\lambda^2 D_\lambda^{-\eta} \rightarrow 0. \quad (8.3)$$

The statement is uniform over the disjoint sets M_i and does not require them to have equal sizes.

Proof. By branching, a type of H_i has shape $\nu \subseteq \lambda$ and size $|M_i|$. Orient λ so that a longest line is a row. Every such ν has at most k boxes below its first row. For a fixed lower size j , the lower partition determines the first row because $|\nu| = |M_i|$ is fixed. There are therefore at most $M(k)$ possibilities for each factor. If the longest line was a column, transpose all diagrams. This proves (8.2); absent tuples only reduce the count.

There are at most $2p(k)$ ambient diagrams with this deficit. The partition bound $p(j) \leq e^{3\sqrt{j}}$ gives, for a constant C_b ,

$$2p(k)M(k)^{2b} \leq e^{C_b\sqrt{k}} \quad (k \geq 1).$$

By Theorem 6.3, $D_\lambda \geq c_1 e^{c_2 k}$ and, at each fixed positive k , the dimensions tend uniformly to infinity. Thus the contribution at deficit k is bounded by $c_1^{-\eta} e^{C_b\sqrt{k} - \eta c_2 k}$, a summable sequence, and tends to zero for fixed k . Dominated convergence proves (8.3). Multiplicities do not affect either counting step. $\square$

8.1 An operator estimate retaining the joint-type count

Theorem 8.2 (Joint-type orbit transfer). *Let $f \geq 0$ be a mass function on S_n , of total mass $z \leq 1$, such that $f(gH_i) \leq B/[S_n : H_i]$ for every g, i . With K_λ as in (8.1),*

$$\|\hat{f}(\lambda)\|_{\text{op}} \leq \sqrt{BzK_\lambda} D_\lambda^{-1/2+1/b}. \quad (8.4)$$

Proof. First let v lie in a single H_i -isotypic space of type τ_i . The orbit span $W = \text{span}\{\rho_\lambda(h)v : h \in H_i\}$ has dimension at most $d_{\tau_i}^2$. To see this in the presence of multiplicity, write the full H_i -isotypic space as $V_{\tau_i} \otimes \mathcal{A}$; the orbit lies in the image of the linear map $A \mapsto (A \otimes I)v$ from $\text{End}(V_{\tau_i})$.

For a test vector w ,

$$|\langle w, \rho_\lambda(g)v \rangle| \leq \|v\| \|P_{\rho_\lambda(g)W} w\|.$$

The squared projected norm is constant on each gH_i , and its uniform mean is $(\dim W/D_\lambda)\|w\|^2$: the average of the conjugate projections is scalar by Schur's lemma, and its trace is $\dim W$. Cauchy–Schwarz for the mass function f , followed by its coset cap, gives

$$|\langle w, \hat{f}(\lambda)v \rangle|^2 \leq zB \frac{\dim W}{D_\lambda} \|v\|^2 \|w\|^2 \leq \frac{zBd_{\tau_i}^2}{D_\lambda} \|v\|^2 \|w\|^2.$$

Thus this single-type bound holds before any joint decomposition. In a joint type of (8.1), the factor degrees satisfy $\prod_i d_{\tau_i} \leq D_\lambda$, since the entire tensor representation occurs at least once. Choose i with $d_{\tau_i} \leq D_\lambda^{1/b}$ and apply the preceding bound. For an arbitrary v , decompose it into its K_λ orthogonal joint components and sum their bounds. Their norms have sum at most $\sqrt{K_\lambda}\|v\|$, proving (8.4). $\square$

The following two specializations keep two useful ways of comparing the joint-type count with the ambient dimension. The four-block argument uses a uniform linear lower bound in the broad-diagram range. The eight-block argument uses a different cutoff for the longest line and a direct hook estimate below that cutoff.

Corollary 8.3 (Four-block transfer). *Partition n points into four sets of size $n/4$, and let H_i permute those sets. For all sufficiently large such n , every subprobability f with $f(gH_i) \leq 2/[S_n : H_i]$ satisfies*

$$\|\hat{f}(\lambda)\|_{\text{op}} \leq D_\lambda^{-1/8} \quad (D_\lambda > 1).$$

Proof. Theorem 8.2 gives $\|\hat{f}(\lambda)\|_{\text{op}} \leq \sqrt{2K_\lambda} D_\lambda^{-1/4}$, and Lemma 8.1 gives $\log(2K_\lambda) = O(\sqrt{k})$. We verify directly that $\log D_\lambda/\sqrt{k} \rightarrow \infty$ uniformly over $k \geq 1$.

Orient a longest line as the first row, of length $a = n - k$. For $1 \leq k \leq n/3$, the tableau construction of Lemma 6.1 gives

$$D_\lambda \geq \binom{n-k}{k} \geq ((n-k)/k)^k.$$

For k above a fixed cutoff, divide the logarithm by $\sqrt{k}$ and use $(n-k)/k \geq 2$; for the finitely many smaller positive k , the same expression tends to infinity with n . For $k > n/3$ and $a \leq n/10$, the hook formula gives $D_\lambda \geq (n/(2ea))^n \geq (5/e)^n$. If instead $a > n/10$, retain the first row and $r = \lfloor a/2 \rfloor$ lower boxes; there are enough lower boxes because $k > n/3 > a/2$. Their tableaux extend to the full diagram, and the same construction gives $D_\lambda \geq \binom{a}{r} \geq 2^r \geq e^{cn}$ for an absolute $c > 0$ and all large n . This proves the asserted uniform divergence. Consequently $2K_\lambda \leq D_\lambda^{1/4}$ eventually, which yields the displayed operator bound. $\square$

Corollary 8.4 (Eight-block transfer). *Partition n points into eight sets of size $n/8$, and let H_i permute those sets. For all sufficiently large such n , every subprobability f with $f(gH_i) \leq 2/[S_n : H_i]$ satisfies*

$$\|\widehat{f}(\lambda)\|_{\text{op}} \leq D_\lambda^{-1/4} \quad (D_\lambda > 1).$$

Proof. Here the joint-type estimate is $\sqrt{2K_\lambda}D_\lambda^{-3/8}$. For $k \leq n/3$, use $K_\lambda \leq M(k)^8$ and $D_\lambda \geq \binom{n-k}{k}$ exactly as above to get $\log(2K_\lambda)/\log D_\lambda \rightarrow 0$ uniformly. For $k > n/3$, the equal block sizes give the coarser count $K_\lambda \leq p(n/8)^8$, whose logarithm is $O(\sqrt{n})$. Let a again be the longest line. If $a \geq n^{3/4}$, retain that line and $r = \lfloor a/3 \rfloor$ lower boxes, possible because $k > n/3 > a/3$. Tableau extension gives $D_\lambda \geq \binom{a}{r} \geq 3^r$, so $\log D_\lambda \geq cn^{3/4}$. If $a < n^{3/4}$, all hooks are at most $2n^{3/4}$, and

$$D_\lambda \geq \frac{n!}{(2n^{3/4})^n} \geq \left(\frac{n^{1/4}}{2e}\right)^n.$$

Thus $2K_\lambda \leq D_\lambda^{1/4}$ uniformly for all large n , and the result follows. $\square$

These are exact specializations of the joint-type orbit argument, including its multiplicity treatment. For reference, the four-block proof also gives $\sum_{D_\lambda > 1} D_\lambda^{-4} = o(1)$ directly: at $k \leq n/3$ use the summable majorant $2e^{3\sqrt{k}}2^{-4k}$ and fixed- k divergence; at $k > n/3$ use the linear-in- n logarithmic dimension bound against $p(n)$. The eight-block proof gives the inverse-third-power statement by the same summation, with majorant $2e^{3\sqrt{k}}2^{-3k}$ and its $n^{3/4}$ lower bound in the remaining range. With mass and sign input, these reproduce respectively the 24-factor and 10-factor Fourier completions through Proposition 2.5: the nonlinear Plancherel exponents are respectively -4 and -3 . Alternatively, a final independent probability law with zero sign mean removes the sign coefficient exactly. Its Fourier matrices have operator norm at most one, so this extra convolution preserves both nonlinear estimates. This alternative does not require small sign mean in the law supplying the coset caps.

8.2 A product with caps in opposite orientations

For a mass function α , a cap on $H_i g$ controls left multiplication of its Fourier matrix by an H_i -isotypic projector. A cap on gH_i controls right multiplication. The product below places these two controlled sides together. There is no assertion that either cap implies the other.

Theorem 8.5 (Two-sided Fourier multiplication). *Let $H_1, \dots, H_b$ be the disjoint symmetric block subgroups above. Let $\alpha, \beta \geq 0$ have masses $a, c \leq 1$ and satisfy*

$$\alpha(H_i g) \leq \frac{B_L}{[S_n : H_i]}, \quad \beta(g H_i) \leq \frac{B_R}{[S_n : H_i]} \quad (g \in S_n, 1 \leq i \leq b). \quad (8.5)$$

Then every irreducible $\lambda \vdash n$ satisfies

$$D_\lambda \|\widehat{\beta}(\lambda) \widehat{\alpha}(\lambda)\|_{\text{HS}}^2 \leq B_L B_R a c K_\lambda^2 D_\lambda^{-1+4/b}. \quad (8.6)$$

In particular, for eight blocks and $B_L = B_R = 2$, the bound is $4K_\lambda^2 D_\lambda^{-1/2}$.

Proof. Use densities $A = |S_n|\alpha$ and $B = |S_n|\beta$ with respect to uniform probability on $G = S_n$. Write $\int_G$ and $\int_{H_i}$ for normalized counting averages. The two caps read

$$\int_{H_i} A(h^{-1}g) dh \leq B_L, \quad \int_{H_i} B(gh^{-1}) dh \leq B_R. \quad (8.7)$$

Indeed these averages are respectively $[G : H_i]\alpha(H_i g)$ and $[G : H_i]\beta(gH_i)$. The density transform $\int_G A(g)\rho_\lambda(g) dg$ equals the mass transform $\widehat{\alpha}(\lambda)$, and similarly for B . These conventions give $\widehat{\beta * \alpha} = \widehat{\beta}\widehat{\alpha}$.

For a type τ of H_i of degree d_τ and character χ_τ , its full isotypic projector is

$$P_{i,\tau} = d_\tau \int_{H_i} \overline{\chi_\tau(h)} \rho_\lambda(h) dh.$$

Centrality makes this scalar on each irreducible carrier, and character orthogonality identifies the scalar as one on type τ and zero otherwise. It acts as the identity on all multiplicity copies of τ . Consider the density

$$C(g) = d_\tau \int_{H_i} \overline{\chi_\tau(h)} A(h^{-1}g) dh.$$

Weighted Cauchy–Schwarz and nonnegativity give

$$|C(g)|^2 \leq d_\tau^2 \left(\int_{H_i} A(h^{-1}g) dh \right) \left(\int_{H_i} |\chi_\tau(h)|^2 A(h^{-1}g) dh \right).$$

The first factor is at most B_L . The average over G of the second factor is $a \int_{H_i} |\chi_\tau|^2 = a$. The Fourier matrix of C is $P_{i,\tau}\widehat{\alpha}(\lambda)$. Plancherel, retaining the λ term, therefore yields

$$D_\lambda \|P_{i,\tau}\widehat{\alpha}(\lambda)\|_{\text{HS}}^2 \leq B_L a d_\tau^2, \quad D_\lambda \|\widehat{\beta}(\lambda)P_{i,\tau}\|_{\text{HS}}^2 \leq B_R c d_\tau^2. \quad (8.8)$$

For the second inequality use $d_\tau \int_{H_i} \overline{\chi_\tau(h)} B(gh^{-1}) dh$, whose transform is $\widehat{\beta}(\lambda)P_{i,\tau}$, and the second cap in (8.7).

The projectors for the different disjoint factors commute, and $Q_\tau = \prod_i P_{i,\tau_i}$ are the orthogonal joint projections in (8.1). On a nonzero joint space, $\prod_i d_{\tau_i} \leq D_\lambda$, so some $d_{\tau_i} \leq D_\lambda^{1/b}$. Since $Q_\tau \leq P_{i,\tau_i}$, the two bounds in (8.8) imply

$$\begin{aligned} \|\widehat{\beta}Q_\tau\widehat{\alpha}\|_{\text{HS}} &\leq \|\widehat{\beta}Q_\tau\|_{\text{HS}} \|Q_\tau\widehat{\alpha}\|_{\text{HS}} \\ &\leq \sqrt{B_L B_R a c} D_\lambda^{-1+2/b}. \end{aligned}$$

Sum over the K_λ joint types before squaring, and then multiply by D_λ . This gives (8.6). The factor K_λ^2 is retained; no multiplicity-free restriction has been assumed. $\square$

Corollary 8.6 (Two-factor completion). *Fix $b > 4$ and finite constants B_L, B_R . Suppose probabilities μ_n, ν_n on S_n dominate subprobabilities α_n, β_n , respectively, whose masses tend to one and which satisfy (8.5) for b disjoint nonempty blocks. If $\widehat{\nu}_n(\text{sgn})\widehat{\mu}_n(\text{sgn}) \rightarrow 0$, then*

$$\|\nu_n * \mu_n - U_{S_n}\|_{\text{TV}} \rightarrow 0.$$

In particular this applies to a single law with small sign mean when separate subprobabilities supply its two cap orientations.

Proof. Use Lemma 8.1 with $\eta = 1 - 4/b > 0$ in Theorem 8.5. It makes the sum of the nonlinear Plancherel terms for $\beta_n * \alpha_n$ tend to zero. Its trivial coefficient is the product of the two masses, tending to one. Domination bounds the error in each sign coefficient by the corresponding lost mass, so its sign coefficient also tends to zero. Lemma 2.4 now gives convergence to uniform in ℓ^1 . Finally, $\beta_n * \alpha_n \leq \nu_n * \mu_n$ and the difference has mass one minus the product of the retained masses. Adding that mass proves the conclusion. $\square$

9 Equivariant kernels and their multiplicity spaces

Partial-permutation information can be imposed on a transition kernel without first selecting a starting permutation. This preserves a group action on the entire state space. The resulting Fourier decomposition has two parts: an irreducible carrier, on which the kernel acts as the identity, and a multiplicity space, on which it may act nontrivially. The latter space can be large even when the carrier is one-dimensional. We prove two amplification results that retain these multiplicities.

Throughout this section kernels act on functions by

$$(Af)(x) = \sum_y A(x, y)f(y).$$

All Hilbert–Schmidt norms use counting measure and the ordinary trace. For a finite state space Ω , let U_Ω be the matrix with every entry $1/|\Omega|$. Thus U_Ω is the orthogonal projection onto the normalized constant vector. A nonnegative matrix with row and column sums at most one is a contraction on $\ell^2(\Omega)$: indeed,

$$\sum_x \left| \sum_y A(x, y)f(y) \right|^2 \leq \sum_{x, y} A(x, y)|f(y)|^2 \leq \sum_y |f(y)|^2.$$

We will use this fact for kernels obtained by deleting entries of a doubly stochastic kernel.

9.1 Sixteen separately truncated factors

Let A and V be two sets of size n , and let $\Omega = \text{Bij}(A, V)$ be the set of bijections from labels to positions. Partition A into sixteen blocks $A_1, \dots, A_{16}$, of sizes m_i , and write

$$H_i = S_{A_i}, \quad H = \prod_{i=1}^{16} H_i.$$

These groups act on Ω on the right by composition: $x \cdot h = x \circ h$. The quotient $X_i = \Omega/H_i$ records the positions of every label outside A_i . Its size is $L_i = n!/m_i!$, and each of its fibers has $m_i!$ elements.

If a kernel K obeys $K(xh, yh) = K(x, y)$ for $h \in H$, its quotient kernel on X_i is well defined by

$$q_i(a, b) = \sum_{y: \pi_i(y)=b} K(x, y), \quad \pi_i(x) = a,$$

where $\pi_i : \Omega \rightarrow X_i$ is the quotient map. Equivariance under H_i makes the right side independent of the representative x . Fix a number $\varepsilon > 0$ and define a retained kernel by

$$T_i(x, y) = K(x, y) \mathbf{1} \left\{ q_i(\pi_i(x), \pi_i(y)) \leq \frac{1 + \varepsilon}{L_i} \right\}. \quad (9.1)$$

Since H_i is normal in H , the quotient has a right H -action. Equivariance of K gives $q_i(ah, bh) = q_i(a, b)$ for $h \in H$. The cutoff therefore preserves the action of all of H .

We specify one further hypothesis on T_i . In the orthogonal H_i -isotypic decomposition

$$\ell^2(\Omega) = \bigoplus_{\alpha \in \widehat{H}_i} V_\alpha \otimes \mathcal{M}_{i, \alpha}, \quad T_i = \bigoplus_{\alpha} I_{V_\alpha} \otimes T_i^\alpha,$$

$\mathcal{M}_{i, \alpha}$ is the full multiplicity space, including all quotient fibers. In particular T_i^{sgn} denotes the operator on the sign multiplicity space, not a single scalar Fourier coefficient.

Theorem 9.1 (Sixteen equivariant subkernels). *Consider a sequence of the preceding state spaces and partitions with $\min_i m_i \rightarrow \infty$. Let K be a doubly stochastic, H -equivariant kernel. Suppose that, for every i ,*

$$\frac{1}{L_i} \sum_{a \in X_i} \frac{1}{2} \sum_{b \in X_i} |q_i(a, b) - L_i^{-1}| \leq \delta, \quad (9.2)$$

where $\delta \rightarrow 0$. Choose $\varepsilon \rightarrow 0$ with $\delta/\varepsilon \rightarrow 0$, form the kernels (9.1), and assume also that

$$\beta_i := \|T_i^{\text{sgn}}\|_{\text{HS}}^2 \rightarrow 0 \quad (1 \leq i \leq 16). \quad (9.3)$$

Then

$$\frac{1}{|\Omega|} \sum_{x \in \Omega} \|K^{16}(x, \cdot) - U_\Omega(x, \cdot)\|_{\text{TV}} \rightarrow 0.$$

If K is equivariant under a transitive group of permutations of Ω , the same conclusion holds for the maximum over starting states.

More precisely, put $T = T_1 \cdots T_{16}$. For a product type $\alpha = (\alpha_1, \dots, \alpha_{16})$, set $f_i = \dim \alpha_i$ and $D = \prod_i f_i$. If $T_{i,\alpha}$ is the operator on its full H -multiplicity space, then

$$\|T_{i,\alpha}\|_{\text{HS}}^2 \leq (1 + \varepsilon) \frac{f_i^2}{D}, \quad D \|T_{1,\alpha} \cdots T_{16,\alpha}\|_{\text{HS}}^2 \leq (1 + \varepsilon)^{16} D^{-13}. \quad (9.4)$$

Proof. We first bound the deleted mass, then use each of the sixteen quotient caps on its corresponding factor. Write

$$\gamma_i = \frac{1}{|\Omega|} \sum_{x,y} (K(x, y) - T_i(x, y)).$$

On an atom with $q_i(a, b) > (1 + \varepsilon)/L_i$, its positive excess over uniform is at least $\varepsilon q_i(a, b)/(1 + \varepsilon)$. Summing over such atoms and then averaging over a gives

$$\gamma_i \leq \frac{1 + \varepsilon}{\varepsilon} \delta = o(1). \quad (9.5)$$

Every T_i is row- and column-substochastic. A uniform starting distribution remains dominated by uniform after any product of these kernels. The average mass lost in T is consequently at most $\Gamma = \sum_i \gamma_i$. Also $0 \leq T \leq K^{16}$ entrywise. The triangle inequality followed by Cauchy–Schwarz over all $|\Omega|^2$ entries gives

$$\frac{1}{|\Omega|} \sum_x \|K^{16}(x, \cdot) - U_\Omega(x, \cdot)\|_{\text{TV}} \leq \frac{\Gamma}{2} + \frac{1}{2} \|T - U_\Omega\|_{\text{HS}}. \quad (9.6)$$

Fix i and first decompose only under H_i . Each quotient fiber is a regular H_i -set. After choosing one representative per fiber, the block of T_i between fibers a, b is convolution by nonnegative weights of total mass

$$t_i(a, b) = q_i(a, b) \mathbf{1}\{q_i(a, b) \leq (1 + \varepsilon)/L_i\}.$$

On the multiplicity space of a regular-fiber type α , of degree f_α , this block is a weighted sum of unitary f_α by f_α matrices. Its Hilbert–Schmidt norm is at most $\sqrt{f_\alpha} t_i(a, b)$. Since each row of t_i has mass at most one, the cap gives

$$\|T_i^\alpha\|_{\text{HS}}^2 \leq f_\alpha \sum_{a,b} t_i(a, b)^2 \leq (1 + \varepsilon) f_\alpha. \quad (9.7)$$

Now decompose under the whole product H . Its α -isotype is

$$\left(\bigotimes_{j=1}^{16} V_{\alpha_j} \right) \otimes \mathcal{M}_\alpha.$$

Viewed only as an H_i -space, it contributes D/f_i copies of $\mathcal{M}_\alpha$ to the multiplicity of α_i . Thus (9.7) implies

$$\frac{D}{f_i} \|T_{i,\alpha}\|_{\text{HS}}^2 \leq (1 + \varepsilon) f_i.$$

Multiplying these inequalities and using Hilbert–Schmidt submultiplicativity proves (9.4).

For $u > 0$, write $\mathcal{Z}_u(m) = \sum_{\alpha \in \widehat{S}_m} (\dim \alpha)^{-u}$. The degree estimate of Theorem 6.3 gives $\mathcal{Z}_{13}(m) \rightarrow 2$. The total squared Hilbert–Schmidt contribution of all product types with $D > 1$ is therefore at most

$$(1 + \varepsilon)^{16} \left(\prod_{i=1}^{16} \mathcal{Z}_{13}(m_i) - 2^{16} \right) = o(1). \quad (9.8)$$

The subtracted types are precisely the products of trivial and sign representations, since eventually every $m_i \geq 2$.

It remains to handle these 2^{16} scalar carrier types; their multiplicity spaces have not been discarded. If a product type has a sign coordinate i , its T_i block has squared Hilbert–Schmidt norm at most β_i by (9.3). All the other factors are operator contractions. Hence its product block has squared Hilbert–Schmidt norm at most β_i . The sum over these finitely many types tends to zero.

For the all-trivial type, let $T_{i,0}$ be its multiplicity operator and let $u = |\Omega|^{-1/2} \mathbf{1}$ be the constant vector in that space. The restriction of U_Ω is $P_u = uu^*$. Equation (9.7) for the trivial H_i type gives $\|T_{i,0}\|_{\text{HS}}^2 \leq 1 + \varepsilon$, whereas

$$\langle u, T_{i,0} u \rangle = 1 - \gamma_i.$$

Consequently

$$\|T_{i,0} - P_u\|_{\text{HS}}^2 = \|T_{i,0}\|_{\text{HS}}^2 + 1 - 2\langle u, T_{i,0} u \rangle \leq \varepsilon + 2\gamma_i. \quad (9.9)$$

Telescoping the product against $P_u^{16} = P_u$, with operator contractions on either side of each difference, gives

$$\|T_{1,0} \cdots T_{16,0} - P_u\|_{\text{HS}} \leq \sum_{i=1}^{16} \sqrt{\varepsilon + 2\gamma_i} = o(1).$$

This is a bound on the entire all-trivial multiplicity space.

The three estimates prove $\|T - U_\Omega\|_{\text{HS}} \rightarrow 0$, so (9.6) proves average mixing. Finally, equivariance under a transitive group makes all row distances of K^{16} from uniform equal. That additional symmetry gives the worst-start conclusion. $\square$

The sign hypothesis concerns the retained factors themselves. A probabilistic construction can verify it by cancellation inside each retained fiber block. The quotient total-variation estimate alone supplies the mass loss and the general type bound, but does not control the sign multiplicity operator.

9.2 Three groups and two-sided quotient control

The next result uses only three label groups, at the cost of controlling every quotient row and column. All three cutoffs are imposed on one kernel. Fourier analysis then gives one matrix for each product type, and fifteen powers of that matrix supply the summable dimension saving.

Let $\mathcal{S}$ be a nonempty finite set of size M , let $H_i = S_{m_i}$ for $1 \leq i \leq 3$, and put $H = H_1 \times H_2 \times H_3$ and $\Omega = \mathcal{S} \times H$. The right H -action is $(s, g)h = (s, gh)$. Its equivariant kernels have the form

$$K((s, g), (s', g')) = p_{s, s'}(g'g^{-1}). \quad (9.10)$$

Indeed simultaneous right multiplication reduces the source group coordinate to the identity. We always take the functions $p_{s, s'} : H \rightarrow [0, \infty)$ to satisfy

$$\sum_{s', h} p_{s, s'}(h) = 1 \quad (s \in \mathcal{S}), \quad \sum_{s, h} p_{s, s'}(h) = 1 \quad (s' \in \mathcal{S}). \quad (9.11)$$

These are exactly the row and column normalizations of K . For $H_{-i} = \prod_{j \neq i} H_j$, define

$$q_i(s, s', h_{-i}) = \sum_{h_i \in H_i} p_{s, s'}(h), \quad M_i = M|H_{-i}|.$$

The associated quotient state consists of s and the two retained group coordinates. Its uniform transition probability is $1/M_i$. Transposition of K replaces the fibers by $\bar{p}_{s', s}(h) = p_{s, s'}(h^{-1})$. Thus a quotient row estimate for the transpose is exactly the quotient column estimate below, after inversion of h_{-i} .

Theorem 9.2 (Three-group two-sided truncation). *Consider a sequence of kernels (9.10) satisfying (9.11), with $\min_i m_i \rightarrow \infty$. Suppose $\delta \rightarrow 0$ and, for every i ,*

$$\sum_{s', h_{-i}} |q_i(s, s', h_{-i}) - M_i^{-1}| \leq 2\delta \quad (s \in \mathcal{S}), \quad (9.12)$$

$$\sum_{s, h_{-i}} |q_i(s, s', h_{-i}) - M_i^{-1}| \leq 2\delta \quad (s' \in \mathcal{S}). \quad (9.13)$$

Let $\chi(h) = \prod_{i=1}^3 \text{sgn}(h_i)$, and assume that the untrimmed all-sign matrix

$$A_\chi(s, s') = \sum_{h \in H} p_{s, s'}(h) \chi(h^{-1})$$

has absolute row and column sums at most a , where $a \rightarrow 0$. Then K^{15} mixes in average row total variation:

$$\frac{1}{|\Omega|} \sum_x \|K^{15}(x, \cdot) - U_\Omega(x, \cdot)\|_{\text{TV}} \rightarrow 0.$$

The maximum row distance also tends to zero if K is equivariant under a transitive group of permutations of Ω .

Specifically, retain the fibers

$$\tilde{p}_{s, s'}(h) = p_{s, s'}(h) \prod_{i=1}^3 \mathbf{1}\{q_i(s, s', h_{-i}) \leq 2/M_i\}, \quad (9.14)$$

and let $\widetilde{K}$ be the corresponding kernel. At most 6δ mass is removed from every row and column. For an irreducible product type $\rho = \rho_1 \otimes \rho_2 \otimes \rho_3$, put $D_i = \dim \rho_i$ and $D = \prod_i D_i$. Its multiplicity matrix is the M by M block matrix

$$B_\rho(s, s') = \sum_{h \in H} \widetilde{p}_{s, s'}(h) \rho(h^{-1}), \quad (9.15)$$

with blocks of size D by D , and it satisfies

$$\|B_\rho\|_{\text{HS}}^2 \leq 4 \frac{D_i}{\prod_{j \neq i} D_j} \quad (1 \leq i \leq 3), \quad D \|B_\rho^{15}\|_{\text{HS}}^2 \leq 2^{30} D^{-4}. \quad (9.16)$$

Proof. For fixed s , the entries of each q_i sum to one. The same is true for fixed s' by (9.11). Therefore the positive excess over uniform in either a quotient row or a quotient column is at most δ . On an atom exceeding twice uniform, its mass is at most twice its positive excess. Each cutoff loses at most 2δ in every row and column. Taking the union of the three deleted sets proves the loss bound 6δ . The retained kernel is row- and column-substochastic, so its blocks are operator contractions. Also its i th quotient marginal is everywhere bounded by $2/M_i$: an original atom above that value has been entirely deleted.

We check the Fourier orientation before estimating the blocks. The unitary regular decomposition of the right H -action has carrier V_ρ and multiplicity D on each fiber s . In the matrix-coefficient basis $\rho(g^{-1})_{ab}$, left multiplication $g \mapsto hg$ replaces

$$\rho(g^{-1}) \quad \text{by} \quad \rho(g^{-1}h^{-1}) = \rho(g^{-1})\rho(h^{-1}).$$

For each fixed carrier index a , the index b , together with s , is therefore acted on by (9.15). Each such matrix occurs D times. In particular composition of the deck kernels gives powers of B_ρ , with no adjoint or reversal of the factors.

Fix s, s' and an index i . First sum over its group coordinate:

$$F(h_{-i}) = \sum_{h_i} \widetilde{p}_{s, s'}(h) \rho_i(h_i^{-1}).$$

The quotient cap and unitarity give $\|F(h_{-i})\|_{\text{HS}} \leq 2\sqrt{D_i}/M_i$. Applying Plancherel entry by entry on H_{-i} yields

$$\begin{aligned} \sum_{\sigma \in \widehat{H}_{-i}} (\dim \sigma) \left\| \sum_{h_{-i}} \sigma(h_{-i}^{-1}) \otimes F(h_{-i}) \right\|_{\text{HS}}^2 \\ = |H_{-i}| \sum_{h_{-i}} \|F(h_{-i})\|_{\text{HS}}^2 \leq \frac{4D_i}{M^2}. \end{aligned}$$

Keep the term $\sigma = \bigotimes_{j \neq i} \rho_j$ and sum over the M^2 pairs (s, s') . A unitary permutation of tensor factors identifies the resulting matrix with B_ρ , proving the first bound in (9.16). Choose an index with $D_i \leq D^{1/3}$. Then

$$\|B_\rho\|_{\text{HS}}^2 \leq 4D^{-1/3}, \quad D \|B_\rho^{15}\|_{\text{HS}}^2 \leq D(4D^{-1/3})^{15} = 2^{30} D^{-4}.$$

Hilbert–Schmidt submultiplicativity suffices here; no normality is assumed. By Theorem 6.3, the sum over $D > 1$ is at most

$$2^{30} \left(\prod_{i=1}^3 \mathcal{Z}_4(m_i) - 8 \right) = o(1).$$

We have controlled every carrier of dimension greater than one. The eight scalar carriers remain: one is all-trivial, six contain both a trivial and a sign factor, and one is all-sign. In every case the corresponding M by M matrix has Hilbert–Schmidt norm at most two, by (9.16).

Consider first one of the six mixed types. Choose an index i where ρ_i is trivial, so that its matrix entries are the signed sums of the retained marginal $\tilde{q}_i$. Replacing that marginal by $1/M_i$ gives zero, since at least one of the other two factors is sign and its group sum vanishes. In each row and column,

$$\sum |\tilde{q}_i - M_i^{-1}| \leq \sum |q_i - M_i^{-1}| + \sum (q_i - \tilde{q}_i) \leq 8\delta.$$

Thus B_ρ has absolute row and column sums at most 8δ , and the same Cauchy–Schwarz argument used for nonnegative kernels, applied to absolute entries, gives $\|B_\rho\|_{\text{op}} \leq 8\delta$. Hence

$$\|B_\rho^{15}\|_{\text{HS}} \leq 2(8\delta)^{14} = o(1).$$

For the all-trivial type, uniform replacement instead gives the matrix J with every entry $1/M$. Write $B_0 = J + E$. The preceding row and column estimate gives $\|E\|_{\text{op}} \leq 8\delta$, while $\|E\|_{\text{HS}} \leq \|B_0\|_{\text{HS}} + \|J\|_{\text{HS}} \leq 3$. Expand $(J + E)^{15}$ into ordered words. The word containing only J equals J , because J is an orthogonal projection. Every word containing both letters has a factor J of Hilbert–Schmidt norm one and at least one factor E of operator norm at most 8δ , so its Hilbert–Schmidt norm tends to zero. The word E^{15} has norm at most $3(8\delta)^{14}$. There are only 2^{15} words. This proves

$$\|B_0^{15} - J\|_{\text{HS}} = o(1)$$

without bounding the size M of the trivial multiplicity space.

Finally consider the all-sign type. Its untrimmed matrix is A_χ . Deleting at most 6δ mass from each row and column changes each corresponding absolute sum by at most 6δ . Consequently

$$\|B_\chi\|_{\text{op}} \leq a + 6\delta, \quad \|B_\chi^{15}\|_{\text{HS}} \leq 2(a + 6\delta)^{14} = o(1).$$

This is precisely where the separate all-sign hypothesis is used.

The uniform kernel has only the all-trivial block J . Summing all blocks with their carrier multiplicities D now gives $\|\widetilde{K}^{15} - U_\Omega\|_{\text{HS}} = o(1)$. For all sufficiently large members of the sequence $6\delta < 1$. Every row of $\widetilde{K}^{15}$ has mass at least $(1 - 6\delta)^{15}$ and is dominated by the corresponding row of K^{15} . Thus, as in (9.6),

$$\frac{1}{|\Omega|} \sum_x \|K^{15}(x, \cdot) - U_\Omega(x, \cdot)\|_{\text{TV}} \leq 45\delta + \frac{1}{2} \|\widetilde{K}^{15} - U_\Omega\|_{\text{HS}} = o(1).$$

The final transitive-equivariance assertion follows because the untrimmed kernel then has the same row distance at every state. $\square$

The two-sided assumptions have distinct roles. Quotient rows bound the forward mass removed; quotient columns bound the reverse mass removed and the operator norms of the scalar error matrices. The all-sign estimate is additional data. In applications where it is proved by cancellation at the end of a time block, the distribution just before that canceling step must also be controlled in both orientations.

10 Selecting coefficient tests and omitted sets

The common coset truncation of Section 3 already applies to the marginal hypotheses considered here. This section gives two different constructions of a retained measure. With a fixed partition, we delete points detected by large coefficient tests and control the union of all those tests. With an average over omitted sets, we instead retain permutations for which most restrictions have bounded density. The good omitted sets then depend on the permutation; we do not select a fixed partition on which the retained law has all its coset caps. Recovering a vector from their subgroup projections is the new step, because projections for overlapping sets need not commute.

Write $[n] = \{1, \dots, n\}$ and let U_n be uniform on S_n . For $R \subseteq [n]$, put $H_R = \text{Sym}(R)$, acting trivially outside R , and write $g_{-R} = g|_{[n] \setminus R}$. Its fibers are the left cosets gH_R : two permutations have the same restriction precisely when they differ by right composition with an element of H_R . For a probability μ on S_n , let μ_{-R} and U_{-R} be the restriction laws under μ and U_n , respectively. Every average over R below is uniform over subsets of its stated size.

10.1 Pruning coefficient tests for a fixed partition

The first argument uses each marginal estimate once, on a union of tests chosen during a finite deletion procedure. Applying the marginal estimate separately at each deletion would multiply its error by the number of tests and lose the conclusion.

Theorem 10.1 (Spectral pruning). *Let $n \rightarrow \infty$ through multiples of 16. For each n , partition $[n]$ into $b = 16$ equal blocks R_i , and let μ_n be a probability on S_n satisfying*

$$\|(\mu_n)_{-R_i} - U_{-R_i}\|_{\text{TV}} \leq \delta_n \quad (1 \leq i \leq 16), \quad \delta_n \rightarrow 0.$$

There is a nonnegative measure $0 \leq \nu_n \leq \mu_n$ of mass $1 - \gamma_n$, with $\gamma_n \rightarrow 0$, such that

$$\|\widehat{\nu}_n(\rho)\|_{\text{op}} \leq D^{-1/8} \quad \text{for every irreducible } \rho \text{ of degree } D > 1.$$

The operator bound has constant one, and ν_n is not normalized.

Proof. Put $a = 1/8$ and $K_i = H_{R_i}$. Fix an irreducible ρ of degree $D > 1$. Every tensor type in its restriction to $\prod_i K_i$ has product of factor degrees at most D . Assign each entire isotypic space to one factor whose degree is at most $D^{1/b}$. This yields an orthogonal decomposition of any unit vector v as $v = \sum_i v_i$, where v_i contains only these small K_i -types. For $W_i = \text{span } \rho(K_i)v_i$, Lemma 3.1 counts each type once, including all its multiplicities. Together with Lemma 6.4, it gives

$$\dim W_i \leq C_1 D^{3/b}, \quad C_1 = \sup_{m \geq 1} \sum_{\tau \in \widehat{S_m}} (\dim \tau)^{-1}.$$

For fixed unit u, v , the score

$$H_{\rho, u, v, i}(g) = \|\text{Proj}_{\rho(g)W_i} u\|$$

depends only on g_{-R_i} . Its uniform second moment is $\dim W_i/D$ by Schur averaging [4, Proposition 3.14]. Consequently

$$U_n\{H_{\rho, u, v, i} > D^{-a}/(2b)\} \leq 4b^2 C_1 D^{-1+2a+3/b}. \quad (10.1)$$

If $|\langle u, \rho(g)v \rangle| > D^{-a}/2$, at least one score exceeds $D^{-a}/(2b)$, because $\|v_i\| \leq 1$ and $|\langle u, \rho(g)v_i \rangle| \leq H_{\rho, u, v, i}(g)$.

Start with the measure $\nu = \mu_n$. Whenever its claimed operator bound fails, choose ρ, u, v with $|\langle u, \widehat{\nu}(\rho)v \rangle| > D^{-a}$, record this test, and delete all remaining mass on

$$\{g : |\langle u, \rho(g)v \rangle| > D^{-a}/2\}.$$

The mass deleted is greater than $D^{-a}/2$: outside this event the coefficient has absolute value at most $D^{-a}/2$, everywhere it has absolute value at most one, and the current measure has mass at most one. Since all deleted pieces are disjoint, any one representation of degree D can be selected at most $2D^a$ times. There are finitely many irreducibles, so the procedure terminates with all the required bounds. The choices are made from deterministic current measures; thus the eventual finite list of tests is fixed, not selected anew for a sampled permutation.

For each i , let E_i be the union of its score events in (10.1) over the complete list of recorded tests. It is a single event determined by g_{-R_i} . Counting tests for each representation gives

$$U_n(E_i) \leq 8b^2C_1 \sum_{\rho: \dim \rho > 1} D^{-1+3a+3/b} = 8b^2C_1 \sum_{\rho: \dim \rho > 1} D^{-7/16} = o(1),$$

by Theorem 6.3. Apply the marginal hypothesis to this entire union to get $\mu_n(E_i) \leq U_n(E_i) + \delta_n$. Every deleted point belongs to some E_i , so the total mass loss is at most $\sum_i \mu_n(E_i) = o(1)$. This proves the theorem. $\square$

10.2 A common truncation and its low-dimensional types

We first give the truncation and the projection estimates with a variable number of omitted blocks. This identifies exactly which part of the two transfers below is shared.

Lemma 10.2 (Good domains and averaged projections). *Let $q \geq 2$ divide n , put $r = n/q$, and let μ be a probability on S_n . Define*

$$\delta = \mathbb{E}_{|R|=r} \|\mu_{-R} - U_{-R}\|_{\text{TV}}, \quad e_R(g) = \mathbf{1}\{\mu(gH_R) \leq 2/[S_n : H_R]\},$$

and

$$E = \{g : \mathbb{E}_R e_R(g) \geq 7/8\}.$$

Then $\mu(E) \geq 1 - 16\delta$. If $\mu(E) \geq 1/2$, the probability $\nu = \mu(\cdot | E)$ satisfies $\|\nu - \mu\|_{\text{TV}} = 1 - \mu(E) \leq 16\delta$.

Fix an irreducible unitary representation ρ of S_n on V , with $D = \dim V > 1$, and fix $\xi > 0$. Let P_R project onto the sum of the H_R -isotypic spaces whose irreducible type has degree at most D^ξ . Then

$$\mathbb{E}_R P_R = cI, \quad c \geq 1 - \frac{1}{q^\xi}. \quad (10.2)$$

For $z \in V$ with $\|z\| \leq 1$, put $W_R(z) = \text{span}\{\rho(h)P_R z : h \in H_R\}$. With the absolute constant

$$C_2 = \sup_{m \geq 1} \sum_{\tau \in \widehat{S_m}} (\dim \tau)^{-2}$$

from Lemma A.2, one has

$$\dim W_R(z) \leq C_2 D^{4\xi}. \quad (10.3)$$

If $\mu(E) \geq 1/2$, then for every unit $u \in V$,

$$\mathbb{E}_{g \sim \nu} e_R(g) |\langle u, \rho(g)P_R z \rangle| \leq 2\sqrt{C_2} D^{-1/2+2\xi}. \quad (10.4)$$

Proof. For any probability a and uniform probability b on a finite set,

$$\sum_{x:a(x)>2b(x)} a(x) \leq 2 \sum_{x:a(x)>b(x)} (a(x) - b(x)) = 2\|a - b\|_{\text{TV}}.$$

Apply this to each restriction law. Under $g \sim \mu$, the expected fraction of sets with $e_R(g) = 0$ is at most 2δ . Markov's inequality gives $\mu(E^c) \leq 16\delta$. Conditioning on E changes the law in total variation by exactly $\mu(E^c)$.

Conjugating H_R by $a \in S_n$ replaces it by H_{aR} and conjugates P_R by $\rho(a)$. Hence the average projection commutes with $\rho(S_n)$ and is scalar. To estimate its trace, choose one partition $(R_1, \dots, R_q)$ into equal blocks. The restriction of V to $\prod_i H_{R_i}$ is an orthogonal sum of tensor types, with arbitrary multiplicities. If a type has factor degrees $d_1, \dots, d_q$, then $\prod_i d_i \leq D$, since at least one copy of that type occurs in V . Consequently at most $1/\xi$ factors have degree greater than D^ξ . On this tensor type, $I - P_{R_i}$ is either the identity or zero according to that inequality. Summing traces, including multiplicities, gives

$$\sum_{i=1}^q \text{rank}(I - P_{R_i}) \leq D/\xi.$$

All these ranks agree by conjugacy. This proves (10.2). Commutation was used only for the disjoint subgroups in this rank calculation.

For the orbit bound, Lemma 3.1 gives, with all multiplicities included,

$$\dim W_R(z) \leq \sum_{\dim \tau \leq D^\xi} (\dim \tau)^2 \leq D^{4\xi} \sum_{\tau} (\dim \tau)^{-2},$$

which is (10.3).

It remains to turn the orbit estimate into a coefficient estimate. The space $W = W_R(z)$ is H_R -invariant, so $\rho(g)W$ depends only on gH_R . For unit u , Schur averaging and the trace give

$$\mathbb{E}_{g \sim U_n} \|\text{Proj}_{\rho(g)W} u\|^2 = \frac{\dim W}{D}.$$

On a coset with $e_R = 1$, the ν -mass is at most four times its uniform mass: the original cap is two and normalization costs at most two. Since $|\langle u, \rho(g)P_R z \rangle| \leq \|\text{Proj}_{\rho(g)W} u\|$, Cauchy–Schwarz gives

$$\mathbb{E}_\nu e_R(g) |\langle u, \rho(g)P_R z \rangle| \leq \left(4 \frac{\dim W}{D}\right)^{1/2}.$$

This proves (10.4). □

10.3 A normalized expansion for 1024 omitted blocks

Both transfers below use the event E and conditional law ν constructed in the preceding lemma, with their respective values of q . The first uses the small lost rank in (10.2) to normalize the average of the retained projections. Its complement is then the average of only the discarded projections. This gives a rapidly decreasing remainder.

Theorem 10.3 (Omitted-set transfer). *Let $n \rightarrow \infty$ through multiples of 1024, and let μ_n be probabilities on S_n such that*

$$\delta_n = \mathbb{E}_{|R|=n/1024} \|(\mu_n)_{-R} - U_{-R}\|_{\text{TV}} \longrightarrow 0.$$

For all sufficiently large n , condition μ_n on the event E of Lemma 10.2, obtaining ν_n . Then $\|\nu_n - \mu_n\|_{\text{TV}} \leq 16\delta_n$, and there is an absolute C such that every irreducible representation ρ of degree $D > 1$ satisfies

$$\|\widehat{\nu}_n(\rho)\|_{\text{op}} \leq CD^{-1/4}.$$

If in addition $\mathbb{E}_{\mu_n} \text{sgn}(g) = o(1)$, then $\|\mu_n^{*8} - U_n\|_{\text{TV}} \rightarrow 0$.

Proof. Suppress n , fix ρ , and use Lemma 10.2 with $q = 1024$ and $\xi = 1/16$. Thus $c \geq 63/64$, and the right side of (10.4) is $AD^{-3/8}$, where $A = 2\sqrt{C_2}$. For $g \in E$ define

$$Q_g = \mathbb{E}_R e_R(g) P_R, \quad F_g = I - Q_g/c = c^{-1} \mathbb{E}_R (1 - e_R(g)) P_R.$$

The equality for F_g uses the scalar identity $\mathbb{E}_R P_R = cI$; it does not require the individual projections to commute. These are positive operators, and $\|F_g\|_{\text{op}} \leq 1/(8c)$. Put $s = \lceil \log D \rceil$, with natural logarithms. The finite identity

$$I = \sum_{j=0}^{s-1} (Q_g/c) F_g^j + F_g^s \quad (10.5)$$

holds because $Q_g/c = I - F_g$.

We now bound the expected coefficient of each term while preserving its projection order. For fixed unit vectors u, w , the j th summand in (10.5) expands as an average over independently chosen $R_0, \dots, R_j$, with scalar factor $c^{-(j+1)}$, indicator

$$e_{R_0}(g) \prod_{i=1}^j (1 - e_{R_i}(g)),$$

and vector $P_{R_0} P_{R_1} \cdots P_{R_j} w$. Fix the sets before integrating over g . The vector $z = P_{R_1} \cdots P_{R_j} w$ is then deterministic with norm at most one. Take absolute values and discard the failure indicators for $i \geq 1$. Equation (10.4) now bounds the expected coefficient by $c^{-(j+1)} AD^{-3/8}$. In particular the argument never exchanges two projections.

The remainder has norm at most $(8c)^{-s}$. Summing gives

$$\mathbb{E}_\nu |\langle u, \rho(g) w \rangle| \leq Asc^{-s} D^{-3/8} + (8c)^{-s} \leq CD^{-1/4}.$$

For the last inequality, $c^{-s} \leq c^{-1} D^{\log(1/c)}$ and $\log(1/c) \leq \log(64/63) < 1/32$; the remaining logarithmic factor is absorbed by $D^{3/32}$. Also $\log(8c) > 1/4$. Taking the supremum over the unit vectors proves the operator bound.

To obtain the convolution conclusion, the sign coefficient of ν_n is $o(1)$, since the original sign coefficient is $o(1)$ and $\|\nu_n - \mu_n\|_{\text{TV}} = o(1)$. Plancherel and $\|B\|_{\text{HS}} \leq \sqrt{D} \|B\|_{\text{op}}$ yield

$$4\|\nu_n^{*8} - U_n\|_{\text{TV}}^2 \leq o(1) + C^{16} \sum_{\rho: \dim \rho > 1} D^{2-16/4} = o(1) + C^{16} \sum_{\rho: \dim \rho > 1} D^{-2} \rightarrow 0$$

by Lemma A.2. Replacing the eight independent factors ν_n by μ_n costs at most $8\|\nu_n - \mu_n\|_{\text{TV}} = o(1)$. $\square$

10.4 A power expansion for 256 omitted blocks

The second transfer uses the unnormalized average of good projections. The preceding lemma supplies its input with $q = 256$ and the same threshold $D^{1/16}$. Here a lower operator bound on

that average controls the remainder, while expanding its complement costs the sum of the absolute polynomial coefficients. This gives the stated, weaker exponent without the scalar normalization used above.

For probabilities p, q on a finite set, write $D(p||q) = \sum_x p(x) \log(p(x)/q(x))$ for relative entropy, using natural logarithms, $0 \log(0/q) = 0$, and value $+\infty$ if p gives positive mass where q vanishes.

Theorem 10.4 (Random-domain transfer). *Let $n \rightarrow \infty$ through multiples of 256, and let μ_n be probabilities on S_n satisfying*

$$\delta_n = \mathbb{E}_{|R|=n/256} \|(\mu_n)_{-R} - U_{-R}\|_{\text{TV}} \rightarrow 0.$$

Condition on the event that at least $7/8$ of the sets R satisfy $\mu_n(gH_R) \leq 2/[S_n : H_R]$, and call the resulting law ν_n . For all sufficiently large n it is defined, satisfies $\|\nu_n - \mu_n\|_{\text{TV}} \leq 16\delta_n$, and obeys

$$\|\hat{\nu}_n(\rho)\|_{\text{op}} \leq CD^{-1/8} \quad (D = \dim \rho > 1)$$

*with an absolute C . If $\mathbb{E}_{\mu_n} \text{sgn}(g) = o(1)$, then $\|\mu_n^{*32} - U_n\|_{\text{TV}} \rightarrow 0$. The averaged relative-entropy hypothesis*

$$\mathbb{E}_{|R|=n/256} D((\mu_n)_{-R} || U_{-R}) = o(1)$$

is sufficient for these conclusions.

Proof. Use Lemma 10.2 with $q = 256$ and $\xi = 1/16$. In particular $c \geq 15/16$. For a retained permutation put

$$M_g = \mathbb{E}_R e_R(g) P_R.$$

Removing at most $1/8$ of the projections from their scalar average gives

$$\frac{1}{2}I \leq \frac{13}{16}I \leq M_g \leq I. \quad (10.6)$$

For fixed unit vectors u, v and every integer $p \geq 1$,

$$\mathbb{E}_\nu |\langle u, \rho(g) M_g^p v \rangle| \leq AD^{-3/8}, \quad A = 2\sqrt{C_2}. \quad (10.7)$$

Indeed, expand M_g^p over $R_1, \dots, R_p$ in that order. After taking absolute values, retain only the indicator $e_{R_1}(g)$. For the fixed sets, apply (10.4) to the deterministic vector $z = P_{R_2} \cdots P_{R_p} v$, of norm at most one. This proves (10.7), even when the omitted sets overlap.

The estimates for positive powers must now recover the original vector. Put $L = \lfloor (\log_2 D)/8 \rfloor$. The identity

$$v = \sum_{j=0}^{L-1} M_g(I - M_g)^j v + (I - M_g)^L v$$

holds also when $L = 0$, with an empty sum. Its remainder has norm at most 2^{-L} by (10.6). Expand each $M_g(I - M_g)^j$ as a polynomial in M_g . The absolute values of its coefficients sum to 2^j , so (10.7) gives

$$\mathbb{E}_\nu |\langle u, \rho(g)v \rangle| \leq 2^{-L} + A \sum_{j=0}^{L-1} 2^j D^{-3/8} \leq 2^{-L} + A2^L D^{-3/8} \leq CD^{-1/8}.$$

Here $2^{-L} \leq 2D^{-1/8}$ and $2^L \leq D^{1/8}$, including $L = 0$. This proves the asserted operator bound.

The sign coefficient is again $o(1)$. For 32 factors the remaining Plancherel sum is at most

$$C^{64} \sum_{\rho: \dim \rho > 1} D^{2-64/8} = C^{64} \sum_{\rho: \dim \rho > 1} D^{-6} \rightarrow 0,$$

because $D^{-6} \leq D^{-2}$ and Lemma A.2 applies. The total-variation cost of replacing the 32 factors is at most $32\|\nu_n - \mu_n\|_{\text{TV}} = o(1)$. Finally Pinsker's inequality and Jensen's inequality give

$$\delta_n \leq \left(\frac{1}{2} \mathbb{E}_R D((\mu_n)_{-R} \| U_{-R}) \right)^{1/2},$$

which proves the entropy version. In this normalization Pinsker's inequality reads $\|p - u\|_{\text{TV}}^2 \leq D(p\|u)/2$. To check the factor, take $A = \{p \geq u\}$, set $a = p(A)$ and $b = u(A)$, and group the entropy sum over A and its complement. The log-sum inequality gives $D(p\|u) \geq D(\text{Ber}(a)\|\text{Ber}(b))$. As a function of a , the binary relative entropy on the right vanishes with zero derivative at $a = b$ and has second derivative $1/[a(1-a)] \geq 4$. It is therefore at least $2(a-b)^2 = 2\|p - u\|_{\text{TV}}^2$, with boundary cases obtained by limits. $\square$

11 Entropy comparisons for every change of measure

The domain transfers use marginal information about one fixed law. Here the hypothesis instead compares entropy under every change of measure supported on one common event. We condition on a large representation coefficient and show that the resulting law puts mass on complementary-restriction events that are rare under uniform measure. The entropy comparison then bounds the probability of that coefficient event. The common event must be chosen before the representation and its two test vectors; an estimate only for the original law would not suffice after this conditioning.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $g : \Omega \rightarrow S_n$ be a measurable endpoint map. For a law σ on this space and $X \subseteq [n]$, denote the law of $g|_X$ by σ_X and its uniform injection reference by U_X . Relative entropy uses natural logarithms:

$$D(\sigma\|\mathbb{P}) = \int \log\left(\frac{d\sigma}{d\mathbb{P}}\right) d\sigma$$

when $\sigma \ll \mathbb{P}$, and is $+\infty$ otherwise.

Theorem 11.1 (Arbitrary-tilt entropy transfer). *Let 128 divide n , let $\mathcal{G} \in \mathcal{F}$, and suppose $a = \mathbb{P}(\mathcal{G}) > 0$. Assume that every probability law σ on $(\Omega, \mathcal{F})$ supported on $\mathcal{G}$ ($\sigma(\mathcal{G}) = 1$) satisfies*

$$D(\sigma\|\mathbb{P}) \geq \mathbb{E}_{X: |X|=n-128} D(\sigma_X\|U_X) - n^{-2}. \quad (11.1)$$

There is an absolute C such that, for every irreducible unitary representation ρ of S_n of degree D and all unit vectors u, z ,

$$\mathbb{P}\{\mathcal{G}, |\langle z, \rho(g)u \rangle| \geq D^{-1/8}\} \leq CD^{-1/256}. \quad (11.2)$$

If η is the endpoint law conditioned on $\mathcal{G}$, then

$$\|\hat{\eta}(\rho)\|_{\text{op}} \leq D^{-1/8} + \frac{C}{a} D^{-1/256}. \quad (11.3)$$

In particular, when $a \geq 1/2$, this is at most $C'D^{-1/256}$ for an absolute C' . The statement therefore applies uniformly to a sequence with $\mathbb{P}(\mathcal{G}) = 1 - o(1)$.

Proof. Put $b = 128$. The claim for $D = 1$ follows by increasing C , so assume $D > 1$. Fix u, z and let A be the event on the left side of (11.2). There is nothing to prove if $\mathbb{P}(A) = 0$. Otherwise set $\sigma = \mathbb{P}(\cdot \mid A)$. Then σ is supported on the same $\mathcal{G}$ as in the hypothesis, and

$$D(\sigma \parallel \mathbb{P}) = \log \frac{1}{\mathbb{P}(A)}. \quad (11.4)$$

The reference law here remains $\mathbb{P}$, not $\mathbb{P}(\cdot \mid \mathcal{G})$. Thus this identity controls the unconditioned probability of A ; the factor a^{-1} enters only when we pass to the endpoint law η at the end. Our goal is to lower-bound this entropy by a positive multiple of $\log D$. We do so using tests visible from complementary restrictions.

Fix an ordered partition $(C_1, \dots, C_b)$ of $[n]$ into equal blocks. Let $X_i = [n] \setminus C_i$ and $K_i = \text{Sym}(C_i)$. On V_ρ , let S_i project onto the sum of the K_i -isotypic spaces with type degree at most $D^{1/b}$. The subgroups K_i commute, as do their isotypic projections. On each tensor type in the product-group restriction, the product of its factor degrees is at most D . At least one factor is therefore at most $D^{1/b}$, giving

$$\prod_{i=1}^b (I - S_i) = 0.$$

Consequently

$$u = \sum_{i=1}^b u_i, \quad u_i = S_i \prod_{j < i} (I - S_j) u, \quad \|u_i\| \leq 1. \quad (11.5)$$

Let $W_i = \text{span}\{\rho(h)u_i : h \in K_i\}$. Lemma 3.1 bounds its dimension by the sum of the squared type degrees, independently of their multiplicities. Lemma A.1 supplies the finite constant

$$C_{20} = \sup_{m \geq 1} \sum_{\tau \in \widehat{S_m}} (\dim \tau)^{-20}.$$

It follows that

$$\dim W_i \leq \sum_{f \leq D^{1/b}} f^2 \leq C_{20} D^{22/b}, \quad (11.6)$$

where the sum counts irreducible types once, not their multiplicities.

Because W_i is K_i -invariant, $\rho(g)W_i$ depends only on $g|_{X_i}$. Let E_i be the subset of injections on X_i for which

$$\|\text{Proj}_{\rho(g)W_i} z\| \geq D^{-1/8}/b.$$

Schur averaging gives uniform mean square projection length $\dim W_i/D$. Markov's inequality and (11.6) imply

$$U_{X_i}(E_i) \leq C_{20} b^2 D^{-1+22/b+1/4} = C_{20} b^2 D^{-37/64} \leq C_{20} b^2 D^{-1/2}. \quad (11.7)$$

For each endpoint on A , the coefficient threshold and (11.5) force at least one of these tests:

$$D^{-1/8} \leq |\langle z, \rho(g)u \rangle| \leq \sum_i \|\text{Proj}_{\rho(g)W_i} z\|.$$

Thus, for this partition,

$$\sum_{i=1}^b \sigma_{X_i}(E_i) \geq 1. \quad (11.8)$$

We have found complementary restrictions on which σ assigns substantial mass to events that are rare under the uniform law. To convert that fact to entropy, put $p_i = \sigma_{X_i}(E_i)$ and $q_i = U_{X_i}(E_i)$. Data processing for the binary test gives

$$D(\sigma_{X_i} \| U_{X_i}) \geq p_i \log \frac{1}{q_i} - \log 2 \geq \frac{1}{2} p_i \log D - \log(2C_{20}b^2).$$

The first inequality follows by expanding binary relative entropy and using that binary Shannon entropy is at most $\log 2$; its term $(1-p_i) \log(1/(1-q_i))$ is nonnegative. Empty tests have $p_i = q_i = 0$ and satisfy the displayed bound as well. The second inequality uses (11.7) and $0 \leq p_i \leq 1$.

Average over i , using (11.8), and then over uniform ordered equal-block partitions. Each X_i is uniform of size $n - n/b$, so

$$\mathbb{E}_{|X|=n-n/b} D(\sigma_X \| U_X) \geq \frac{\log D}{2b} - \log(2C_{20}b^2).$$

Now apply (11.1) to this particular σ . Together with (11.4), it gives

$$\log \frac{1}{\mathbb{P}(A)} \geq \frac{\log D}{256} - \log(2C_{20}b^2) - n^{-2}.$$

This proves (11.2), for example with $C = 2eC_{20}b^2$.

Finally the absolute coefficient is at most one. Split its expectation under $\mathbb{P}(\cdot | \mathcal{G})$ at $D^{-1/8}$, and use (11.2) for the upper part. This proves (11.3) after taking the supremum over the unit vectors. All partitions and tests in the proof were chosen after u, z were fixed; the hypothesis covering every law on the common event is what permits the rare-event conditioning at the first step. $\square$

12 Rare events and signed palettes

An estimate for a typical partial permutation need not remain useful after conditioning on a very rare mapping event. The first result below explains how a single retained event resolves this difficulty. It converts a bound on the accumulated conditional errors, valid at every retained setting, into simultaneous bounds on many coset probabilities. We then choose the cosets by the representation being tested. Both steps are statements about finite probability spaces and permutation groups; no dynamics enter their proofs.

Let $n = 2h$ and partition $[n] = A \sqcup B$, with $|A| = |B| = h$. A *palette on B* is a partition $B = B_1 \sqcup \dots \sqcup B_u$ into nonempty color classes. Its subgroup is

$$H = \prod_{i=1}^u S_{B_i} \leq S_n,$$

acting identically on A . Put $l_i = |B_i|$, $p_i = l_i/h$, and $\mathcal{H}(p) = -\sum_i p_i \log p_i$. We use the analogous definitions for palettes on A . A left coset xH prescribes each image in A separately and prescribes the image set of every color class in B .

12.1 An information bound valid under rare conditioning

Here is the precise reveal experiment. Let (Ω, P) be a finite probability space and $g : \Omega \rightarrow S_n$ an endpoint map. For each fixed ordering $o = (c_1, \dots, c_n)$ of the labels, let $Y_1^o, \dots, Y_n^o$ be observations

on Ω such that Y_j^o determines $g(c_j)$. These observations may contain more information than the endpoint. Write $\mathcal{F}_r^o = \sigma(Y_1^o, \dots, Y_r^o)$. At a feasible history the previously observed endpoints leave exactly $m = n - r$ available sites. Suppose $\Delta_r^o \geq 0$ is $\mathcal{F}_r^o$ -measurable and, for every subset Q of these sites,

$$P\{g(c_{r+1}) \in Q \mid \mathcal{F}_r^o\} \leq \frac{|Q|}{n-r} + n\Delta_r^o. \quad (12.1)$$

All conditional statements concern histories of positive reference probability. Let $\mathbb{E}_{AB}$ denote averaging over independent uniform orders within A and B , with all labels of A first; define $\mathbb{E}_{BA}$ by reversing the two halves.

Lemma 12.1 (Information from a retained reveal experiment). *In the experiment just described, fix an integer $0 \leq v < h$ and an event $\mathcal{G} \subseteq \Omega$ of reference probability $z > 0$. Assume that (12.1) holds for both half-first order conventions and every $0 \leq r < n - v$. Suppose that every $\omega \in \mathcal{G}$ satisfies*

$$\mathbb{E}_{AB} \sum_{r=0}^{n-v-1} \Delta_r^o(\omega) + \mathbb{E}_{BA} \sum_{r=0}^{n-v-1} \Delta_r^o(\omega) \leq a_n. \quad (12.2)$$

Let μ be the law of g under $P(\cdot \mid \mathcal{G})$. For every palette on either half, its subgroup H , and every $x \in S_n$,

$$\mu(xH) \leq \frac{1}{|S_n : H|} \exp\{v\mathcal{H}(p) + n^2 a_n - \log z\}. \quad (12.3)$$

In particular, if $a_n \leq n^{-6}$ and $z = 1 - o(1)$, the last two terms in the exponent are $o(1)$, uniformly over all palettes and cosets.

Proof. The event $\mathcal{G}$ and the retained endpoint law μ are fixed before choosing the palette. Fix now a palette on B and a coset xH , and put $E = \mathcal{G} \cap \{g \in xH\}$. If $P(E) = 0$ the conclusion is immediate. Otherwise write $Q = P(\cdot \mid E)$. The three laws are related by

$$P(E) = z\mu(xH), \quad D(Q\|P) = -\log P(E).$$

Choose the reveal order independently of the setting, using the AB convention under both P and Q . Since E does not involve the order, conditioning the setting on E leaves this order uniformly distributed. For a fixed order, data processing followed by the entropy chain rule gives

$$-\log P(E) \geq \sum_{r=0}^{n-v-1} \mathbb{E}_Q D(\mathcal{L}_Q(Y_{r+1}^o \mid \mathcal{F}_r^o) \parallel \mathcal{L}_P(Y_{r+1}^o \mid \mathcal{F}_r^o)).$$

For labels in A , membership in xH prescribes a singleton target; for a label in B_i it prescribes the target set xB_i . At a history that occurs under Q , let $a_r \geq 1$ be the number of unoccupied sites in this target. The next endpoint lies in that target with Q -probability one. For any probability R supported on an event F of positive reference probability, the identity $D(R\|P') = D(R\|P'(\cdot \mid F)) - \log P'(F)$ shows that its entropy is at least $-\log P'(F)$. Apply this observation to the next observation and then use (12.1). Since $m = n - r \leq n$,

$$\begin{aligned} -\log \left(\frac{a_r}{m} + n\Delta_r^o \right) &= \log(m/a_r) - \log \left(1 + \frac{nm\Delta_r^o}{a_r} \right) \\ &\geq \log(m/a_r) - n^2 \Delta_r^o. \end{aligned}$$

This lower bound is valid even when the displayed upper bound on the reference probability exceeds one. Averaging the chain rule over orders therefore yields

$$-\log P(E) \geq \mathbb{E}_{AB} \mathbb{E}_Q \sum_{r=0}^{n-v-1} \log((n-r)/a_r) - n^2 a_n. \quad (12.4)$$

The error bound uses (12.2) pointwise on E . It does not use an average error estimate under P after the change of law to Q .

It remains to count the information lost by omitting the last v labels. If all n labels were revealed, the numerator counts would multiply to $n!$, while the target counts would multiply to $\prod_i l_i!$. The full logarithmic sum is consequently $\log[S_n : H]$, independently of the compatible observations. The last v labels are in B . For $v = 0$ the missing contribution is zero. For $v > 0$, if their color counts are $t_1, \dots, t_u$, it is

$$\log \frac{v!}{\prod_i t_i!} \leq v \mathcal{H}((t_i/v)_i).$$

The inequality follows because the multinomial probability of these counts under probabilities t_i/v is at most one. The last v labels form a uniform subset of B under the order average, so $\mathbb{E}_{AB}(t_i/v) = p_i$. Concavity of entropy gives expected missing contribution at most $v \mathcal{H}(p)$. Insert this bound in (12.4), exponentiate, and use $\mu(xH) = P(E)/z$. The BA order gives the same result for palettes on A . Every error term is independent of the chosen palette and coset, which proves simultaneous uniformity. $\square$

12.2 Choosing a palette for a representation

We next relate the entropy of a suitable palette to the dimension of a representation. We use the characteristic-zero branching rule [6, Theorem 9.2] and the hook-length formula [4, Theorem 4.53]. The horizontal Pieri rule is the one-row case of the induction rule in [6, Section 16]; its vertical form follows by transposition and sign twist [6, Theorem 6.7]. Write D_γ for the dimension of the irreducible representation indexed by a partition γ .

Lemma 12.2 (A dimension estimate in terms of the longest row or column). *There is an absolute $c_0 > 0$ such that, for all sufficiently large b , if $\gamma \vdash b$, $L = \max(\gamma_1, \gamma'_1)$, and $k = b - L > 0$, then*

$$\log D_\gamma \geq \begin{cases} c_0 k \log(b/k), & k \leq b/2, \\ c_0 b, & k > b/2. \end{cases} \quad (12.5)$$

In particular $\log D_\gamma \geq c_1 k$ for an absolute $c_1 > 0$.

Proof. Transpose the diagram if necessary, so its first row has length L . Apply the first-row hook bound (6.6) with row length L , k lower boxes, and total size b . The dimension of the lower diagram is at least one, so

$$\log D_\gamma \geq \log \binom{b}{L} - \frac{k(1 + \log L)}{L}.$$

If $k \leq b/2$, the correction is $O(k \log b/b)$ and $\binom{b}{k} \geq (b/k)^k$, proving the first case. If $b/10 \leq L < b/2$, the binomial coefficient has logarithm at least $L \log(b/L) \geq (b/10) \log 2$, and the correction is $O(\log b)$. If $L < b/10$, every hook is at most $2L < b/5$, so $D_\gamma \geq (b/e)^b / (b/5)^b = (5/e)^b$. These bounds prove the second case and the stated consequence. $\square$

Lemma 12.3 (Signed palettes with controlled entropy). *There is an absolute $C_0 < \infty$ such that, for every sufficiently large h and every $\beta \vdash h$, there are positive integers $l_1, \dots, l_u$ summing to h and a product character $\theta = \theta_1 \otimes \dots \otimes \theta_u$ of $H = S_{l_1} \times \dots \times S_{l_u}$, each θ_i either trivial or sign, for which*

$$\text{Hom}_H(\theta, V_\beta) \neq 0, \quad h\mathcal{H}((l_i/h)_i) \leq C_0 \log D_\beta. \quad (12.6)$$

The same character occurrence holds for every placement of the classes of these sizes on an h -element set.

Proof. Starting with β , remove a first row or a first column of maximal length, and repeat on the remaining partition until it is empty. The successive lengths are the l_i . Assign a trivial character to a row removal and a sign character to a column removal. If a first row is removed, the remaining partition interlaces the original one: putting back its size is a horizontal-strip addition in the Pieri rule. Transposing gives a vertical-strip addition for a first column. Iterated Pieri, transitivity of induction, and Frobenius reciprocity give the character occurrence. Conjugation supplies every placement.

Let $L = l_1$ and $k = h - L$. Every hook in a strip of length l at the time of its removal is at most $2l$, because l is the longer of the current first row and first column. Its hook product is at most $(2l)^l \leq (2e)^l l!$. For the first strip use the sharper bound $L!e^k$, obtained from $\log(1+x) \leq x$ and the fact that the sum of the numbers of boxes below the strip is k . Remaining hooks are unchanged by each removal. Thus, if $M = h! / \prod_i l_i!$, the hook formula yields

$$\log M \leq \log D_\beta + (1 + \log(2e))k.$$

To compare this multinomial count with entropy, set $f(j) = j \log j - \log j!$, with $f(0) = 0$. Then $f(j) \geq 0$ and $0 \leq f(j) - f(j-1) \leq 1$. Consequently

$$h\mathcal{H}((l_i/h)_i) - \log M = f(h) - \sum_i f(l_i) \leq f(h) - f(L) \leq k.$$

If $k > 0$, Lemma 12.2 absorbs this total $O(k)$ error into $C_0 \log D_\beta$. If $k = 0$, the representation is trivial or sign, and the one-color palette has entropy zero. $\square$

12.3 The transfer from both halves

Fix once and for all

$$0 < \delta < \frac{1}{4}, \quad 2\delta C_0 \leq \frac{1}{10}, \quad v = \lfloor \delta n \rfloor. \quad (12.7)$$

The next theorem receives only coset caps. In particular, it can be applied to the output of Lemma 12.1 without referring again to its reveal experiment.

Theorem 12.4 (Signed-palette transfer). *Let n tend to infinity through even integers. For each n , partition $[n] = A \sqcup B$ into equal halves and let μ be a probability on S_n . Suppose that $\varepsilon_n \rightarrow 0$ and that, for every palette on either half, every placement of its classes, its subgroup H , and every $x \in S_n$,*

$$\mu(xH) \leq [S_n : H]^{-1} \exp\{v\mathcal{H}(p) + \varepsilon_n\}. \quad (12.8)$$

Here δ and v satisfy (12.7). For all sufficiently large n , uniformly over $\lambda \notin \{(n), (1^n)\}$,

$$\|\widehat{\mu}(\lambda)\|_{\text{HS}}^2 \leq D_\lambda^{-1/3}, \quad (12.9)$$

$$\|\widehat{\mu}(\lambda)\|_{\text{op}} \leq D_\lambda^{-1/6}. \quad (12.10)$$

If in addition $|\widehat{\mu}(\text{sgn})| = o(1)$, then $\|\mu^{*4} - U_{S_n}\|_{\text{TV}} = o(1)$, and consequently $\|\mu^{*30} - U_{S_n}\|_{\text{TV}} = o(1)$ as well. Both conclusions also hold with μ replaced in every factor by a probability μ_0 satisfying $\|\mu_0 - \mu\|_{\text{TV}} = o(1)$.

Proof. Put $J = \widehat{\mu}(\lambda)$ and $D = D_\lambda$, using a unitary model ρ_λ . We first bound J on each half-isotypic component, including its multiplicities. Combining the halves will bound each joint component; we then absorb the number of joint types into a small power of D . For a type β of S_B , choose its signed palette from Lemma 12.3. Equations (12.7) and (12.8) give, for large n ,

$$\mu(xH) \leq \frac{2D_\beta^{1/10}}{[S_n : H]}. \quad (12.11)$$

This holds for every conjugate of the chosen subgroup within S_B .

Let $P_{H,\theta}$ be the orthogonal projection onto the θ -character space in the restriction to H . Define the signed mass θ_H to equal $\theta(h)/|H|$ on H and zero elsewhere. Right convolution by θ_H has Fourier matrix $JP_{H,\theta}$. On a coset xH each resulting mass has absolute value at most $\mu(xH)/|H|$. Hence

$$|S_n| \sum_{x \in S_n} |(\mu * \theta_H)(x)|^2 \leq [S_n : H] \sum_{C \in S_n/H} \mu(C)^2 \leq 2D_\beta^{1/10}.$$

Plancherel, retaining the λ term, proves

$$D\|JP_{H,\theta}\|_{\text{HS}}^2 \leq 2D_\beta^{1/10}. \quad (12.12)$$

Let $\Pi_{B,\beta}$ denote the entire β -isotypic projection. Average $P_{H,\theta}$ over conjugation by S_B . On the β carrier its rank is at least one, so Schur's lemma makes this average at least $D_\beta^{-1}I$ on that carrier. The group algebra acts as the identity on its multiplicity space. The average is therefore at least $D_\beta^{-1}\Pi_{B,\beta}$ in positive semidefinite order on the full representation. Tracing against J^*J in (12.12) gives

$$\|J\Pi_{B,\beta}\|_{\text{HS}}^2 \leq \frac{2D_\beta^{11/10}}{D}. \quad (12.13)$$

The identical estimate holds for a type α of S_A .

Every constituent of the product of the two halves has a small carrier on at least one side. Indeed, the commuting projections $\Pi_{A,\alpha}\Pi_{B,\beta}$ decompose the identity orthogonally, and every nonzero such component has $D_\alpha D_\beta \leq D$. Use (12.13) for the smaller degree to obtain

$$\|J\Pi_{A,\alpha}\Pi_{B,\beta}\|_{\text{HS}}^2 \leq 2D^{-9/20}. \quad (12.14)$$

We verify that the number of type pairs costs only a vanishing power of D . Let $L = \max(\lambda_1, \lambda'_1)$ and $k = n - L > 0$. Branching forces every half-type diagram to be contained in λ . If the longest line is a column, transpose λ and every contained half-type diagram for this count. Containment and dimensions are preserved, and each resulting tail beyond the first row has at most k boxes. The number of choices for either half-type is at most $B_k = \sum_{j=0}^k p(j)$, where $p(j)$ is the number of partitions of j . The partition bound (6.1) gives

$$B_k \leq (k+1)e^{3\sqrt{k}} = e^{O(\sqrt{k})}.$$

Lemma 12.2 implies $\log B_k / \log D \rightarrow 0$ uniformly: if $k \leq \sqrt{n}$, its denominator is at least $(c_0/2)k \log n$, and if $k > \sqrt{n}$, it is at least $c_1 k$. Thus $B_k^2 = D^{o(1)}$ uniformly over the nontrivial, nonsign diagrams. Orthogonality of the domain projections in (12.14) now yields

$$\|J\|_{\text{op}}^2 \leq \|J\|_{\text{HS}}^2 \leq 2B_k^2 D^{-9/20} \leq D^{-1/3}$$

for sufficiently large n . This proves both (12.9) and (12.10).

For four factors, Hilbert–Schmidt submultiplicativity, Plancherel and Cauchy–Schwarz give

$$4\|\mu^{*4} - U_{S_n}\|_{\text{TV}}^2 \leq |\widehat{\mu}(\text{sgn})|^8 + \sum_{\lambda \notin \{(n), (1^n)\}} D_\lambda^{1-4/3} = o(1),$$

since the nonlinear sum is $Z_{1/3}(n) \rightarrow 0$ by Theorem 6.3. Replacing the four factors by μ_0 costs at most $4\|\mu_0 - \mu\|_{\text{TV}} = o(1)$. Convolution contracts total variation and preserves uniform measure, so each four-factor conclusion implies its thirty-factor counterpart.

There is also a direct thirty-factor completion using only the operator bound (12.10). It gives

$$4\|\mu^{*30} - U_{S_n}\|_{\text{TV}}^2 \leq |\widehat{\mu}(\text{sgn})|^{60} + \sum_{\lambda \notin \{(n), (1^n)\}} D_\lambda^{-8}.$$

The sum tends to zero. To see this directly, group the diagrams by $k = n - \max(\lambda_1, \lambda'_1)$. There are at most $2p(k)$ diagrams in each group. Lemma 12.2 gives the summable majorant $2e^{3\sqrt{k}-8c_1k}$, while each fixed $k > 0$ has $D_\lambda \rightarrow \infty$ as $n \rightarrow \infty$. Dominated convergence applies. Replacing the thirty factors one at a time changes total variation by at most $30\|\mu_0 - \mu\|_{\text{TV}}$. $\square$

The order of choices is now visible. First choose the absolute C_0 in Lemma 12.3, then δ in (12.7). A reveal construction supplies the reference conditional bound (12.1), together with a common event of mass $1 - o(1)$ on which (12.2) holds with $a_n \leq n^{-6}$ down to v unrevealed labels. Lemma 12.1 supplies all the caps required by Theorem 12.4 with $\varepsilon_n = n^{-4} - \log z$. The sign expectation remains a separate scalar input.

13 Entropy clipping and large representation dimensions

Small total variation error is one way to obtain a useful coset cap. A weaker entropy estimate can also suffice when only representations above a prescribed dimension threshold need to be controlled. The cutoff below preserves the original, unnormalized retained measure. Its loss of mass is separated from its Fourier estimate.

For probabilities p, u on a finite set, with u positive, write

$$D(p\|u) = \sum_x p(x) \log \frac{p(x)}{u(x)};$$

zero summands have value zero and logarithms are natural. For $G = S_n$ and $H \leq G$, write $w_H(gH)$ for the mass of the left coset gH under a probability w on G , and $U_{G/H}$ for the uniform law on left cosets. These are the cosets that record the images of a list whose pointwise stabilizer is H .

Theorem 13.1 (Entropy clipping and isotypic transfer). *Partition $\{1, \dots, n\}$ into $q \geq 1$ nonempty blocks, and let H_i be the subgroup supported on block i . Suppose a probability w on S_n satisfies*

$$\sum_{i=1}^q D(w_{H_i} \| U_{G/H_i}) \leq K. \tag{13.1}$$

For every $t > 0$ there is a subprobability $v \leq w$, of mass m , such that

$$1 - m \leq \frac{K + q/e}{t}, \quad (13.2)$$

$$v(gH_i) \leq e^t \frac{|H_i|}{|G|} \quad (g \in G, 1 \leq i \leq q), \quad (13.3)$$

$$\sum_{h \in H_i} (\check{v} * v)(h) \leq e^t \frac{|H_i|}{|G|}, \quad (13.4)$$

$$\|\hat{v}(\lambda)\|_{\text{HS}}^2 \leq q C_u e^t D_\lambda^{-1+(u+2)/q} \quad (\lambda \vdash n, u > 0). \quad (13.5)$$

Here $\check{v}(g) = v(g^{-1})$, C_u is the constant in Theorem 6.3, the Fourier transform is the mass transform $\hat{v}(\lambda) = \sum_g v(g) \rho_\lambda(g)$, and the Hilbert–Schmidt norm is unnormalized.

Proof. For a density p relative to a probability u , the negative part of the logarithm has expectation at most $1/e$ under pu , since $p \max(-\log p, 0) \leq 1/e$. Consequently

$$\mathbb{E}_{pu}(\log p)_+ \leq D(pu \| u) + 1/e.$$

For each i let p_i be the density of w_{H_i} relative to U_{G/H_i} . Restrict w to those g for which $p_i(gH_i) \leq e^t$ for every i , and call the result v . Markov's inequality and a union bound prove (13.2); the restriction proves (13.3). Zero-density cosets have zero w -mass and create no logarithmic exception.

The convolution $\check{v} * v$ is the law, with its subprobability mass, of $g_1^{-1}g_2$. Membership in H_i means that g_1, g_2 belong to the same left coset. Thus

$$\sum_{h \in H_i} (\check{v} * v)(h) = \sum_{gH_i} v(gH_i)^2 \leq e^t \frac{|H_i|}{|G|} \sum_{gH_i} v(gH_i) \leq e^t \frac{|H_i|}{|G|}.$$

We spell out the isotypic step to show exactly what the collision cap controls. Put $E_\lambda = \hat{v}(\lambda) * \hat{v}(\lambda)$. For an irreducible τ of H_i of dimension d_τ and character χ_τ , let $P_{\lambda; \tau}^{H_i}$ be the full τ -isotypic projection in $\rho_\lambda|_{H_i}$, including all multiplicity copies. Fourier inversion and the central projection formula give

$$\sum_{h \in H_i} (\check{v} * v)(h) \chi_\tau(h) = \frac{|H_i|}{|G| d_\tau} \sum_{\lambda \vdash n} D_\lambda \text{tr}(E_\lambda P_{\lambda; \tau}^{H_i}). \quad (13.6)$$

Indeed $\sum_{h \in H_i} \chi_\tau(h) \rho_\lambda(h^{-1}) = (|H_i|/d_\tau) P_{\lambda; \tau}^{H_i}$. Every trace on the right is nonnegative, since E_λ and $P_{\lambda; \tau}^{H_i}$ are positive. Since $|\chi_\tau(h)| \leq d_\tau$, (13.4) yields

$$\text{tr}(E_\lambda P_{\lambda; \tau}^{H_i}) \leq e^t \frac{d_\tau^2}{D_\lambda}. \quad (13.7)$$

This is the collision form of Lemma 3.4.

The block subgroups form a direct product. Every irreducible constituent of its restriction has factor dimensions whose product is at most D_λ . At least one factor dimension is therefore at most $D_\lambda^{1/q}$. The commuting isotypic projections satisfy

$$I \leq \sum_{i=1}^q \sum_{\tau: d_\tau \leq D_\lambda^{1/q}} P_{\lambda; \tau}^{H_i}.$$

There are at most $C_u D_\lambda^{u/q}$ such types in each subgroup: each contributes at least $D_\lambda^{-u/q}$ to its inverse- u degree sum. Taking traces against E_λ and using (13.7) proves (13.5). $\square$

Corollary 13.2 (Selecting complementary lists by averaging). *Let q divide n , put $r = n/q$, and suppose the image law w_I of a uniformly chosen ordered list I of $n - r$ distinct inputs satisfies*

$$\mathbb{E}_I D(w_I \| U_I) \leq K_0,$$

where U_I is uniform on ordered distinct output lists of that length. There is a partition into q blocks of size r for which (13.1) holds with $K = qK_0$.

Proof. Choose the equal partition uniformly. Each block complement is uniform among sets of size $n - r$. Entropy of an image list is unchanged by reordering its coordinates. Therefore the expected sum of the q complement entropies is at most qK_0 ; one deterministic partition has sum no greater than that expectation. Agreement on the complement is exactly membership in the same left coset of its block subgroup. $\square$

Corollary 13.3 (The logarithmic cutoff with 256 blocks). *Let $n = 2^d$ and suppose, for each sufficiently large integer d , that a probability w has an equal partition into $q = 256$ blocks with total complement entropy $O(d^4)$. There is $v \leq w$ of mass $1 - O(d^{-2})$ such that*

$$\|\hat{v}(\lambda)\|_{\text{HS}}^2 \leq 256C_{32}e^{d^6}D_\lambda^{-1+34/256}. \quad (13.8)$$

In particular, whenever $\log D_\lambda > d^8$, this is at most $D_\lambda^{-1/2}$ for all sufficiently large d , uniformly in λ .

Proof. Apply Theorem 13.1 with $t = d^6$ and $u = 32$. The loss bound is $O(d^4)/d^6 = O(d^{-2})$. The logarithm of the prefactor is $d^6 + \log(256C_{32})$, while

$$\left(\frac{1}{2} - \frac{34}{256}\right) \log D_\lambda > \left(\frac{1}{2} - \frac{34}{256}\right) d^8.$$

The latter eventually exceeds the former, proving the final bound. $\square$

The hypotheses here concern only the entropy of a group law. Controlling dimensions below the cutoff requires an additional input on that law; the clipping theorem itself makes no assertion about those representations.

14 Weak entropy and an independent sparse estimate

A complementary marginal may have relative entropy from uniform of order $(\log n)^4$ and still be far from uniform in total variation. Such information is nevertheless sufficient for representations whose dimensions are large enough to absorb a growing coset cap. This section proves that assertion and combines it with an independent Fourier estimate for representations with a long first row. The second law must also annihilate every representation whose diagram has more than $n/2$ rows. These two Fourier properties are explicit hypotheses; the construction of the second law plays no role in the transfer.

Throughout this section, 16 divides n and $[n] = B_1 \sqcup \cdots \sqcup B_{16}$ is a fixed partition into equal blocks. Let $H_j = S_{B_j}$, acting identically off B_j . For a probability μ on S_n , let μ_j be its pushforward to the cosets S_n/H_j , and let U_j be uniform on this quotient. The coset xH_j specifies exactly the images under x of all labels outside B_j .

14.1 Clipping a weak entropy bound

Lemma 14.1 (One retained law for sixteen entropy bounds). *Suppose*

$$E_n := \sum_{j=1}^{16} D(\mu_j \| U_j) \leq C(1 + \log n)^4 \quad (14.1)$$

for a constant C independent of n . Set $b_n = n^{1/50}$ and $\eta_n = (E_n + 16/e)/b_n$. For all sufficiently large n , there is an event $\mathcal{G} \subseteq S_n$ with $z := \mu(\mathcal{G}) \geq 1 - \eta_n > 0$ such that $\nu = \mu(\cdot | \mathcal{G})$ satisfies

$$\|\nu - \mu\|_{\text{TV}} = 1 - z \leq \eta_n = o(1), \quad \nu(xH_j) \leq \frac{M_n}{[S_n : H_j]}, \quad M_n = 2 \exp(n^{1/50}). \quad (14.2)$$

The cap holds for every $x \in S_n$ and every j under this same law ν .

Proof. Apply the clipping construction of Theorem 13.1 with $q = 16$, $w = \mu$, $K = E_n$, and $t = b_n$. It restricts μ to the event $\mathcal{G}$ on which all sixteen original quotient densities are at most e^{b_n} . The retained subprobability has mass $z \geq 1 - \eta_n$ and satisfies every coset cap with constant e^{b_n} . Here $\eta_n = o(1)$ by (14.1), so eventually $z \geq 1/2$. Normalizing this one retained measure increases all caps by the same factor $z^{-1} \leq 2$, giving (14.2). Since the construction is an event restriction, its normalization is exactly $\mu(\cdot | \mathcal{G})$, at total-variation distance $1 - z$ from μ . $\square$

14.2 The representation range controlled by these caps

Write D_λ for the dimension of the irreducible representation V_λ of S_n , and put $k = n - \lambda_1$. Here k measures the number of boxes below the first row, without transposing the diagram. The restriction on the number of rows in the next lemma prevents a shape with a long first column from having small dimension while k is large.

Lemma 14.2 (Dimension with at most $n/2$ rows). *There is an absolute $c > 0$ such that, for all sufficiently large n , every $\lambda \vdash n$ with $\lambda'_1 \leq n/2$ and $k > 0$ satisfies*

$$\log D_\lambda \geq ck. \quad (14.3)$$

Proof. If both the first row and the first column are shorter than $n/8$, every hook has length less than $n/4$. The hook formula then gives $D_\lambda \geq (4/e)^n$, which suffices. Otherwise transpose if necessary so the first row has length $m = \max(\lambda_1, \lambda'_1) \geq n/8$. The number $n - m$ of remaining boxes is at least $k/2$: it is k without transposition, and after transposition the assumed bound on the number of rows gives $n - m \geq n/2 \geq k/2$.

Retain this first row and a Young subdiagram of $s = \min(n - m, \lfloor m/2 \rfloor)$ boxes below it. For large n , $s \geq k/17$. To construct standard tableaux of the retained diagram, put $1, \dots, s$ in the first s boxes of its top row. Choose any s of the remaining m labels for the lower diagram, placed in a fixed standard order; fill the remaining top row in increasing order. Every column of the lower diagram is under one of the first s top boxes, so these are valid tableaux. There are $\binom{m}{s} \geq (m/s)^s \geq 2^s$ choices, and each extends to the full diagram by successively adding the omitted boxes in a Young order. Thus $D_\lambda \geq 2^s$, proving the lemma. $\square$

Proposition 14.3 (Sixteen groups in the large-degree range). *Let ν be any probability on S_n satisfying the sixteen coset caps in (14.2). Uniformly over all partitions with*

$$k = n - \lambda_1 > n^{1/20}, \quad \lambda'_1 \leq n/2,$$

one has, for sufficiently large n ,

$$\|\widehat{\nu}(\lambda)\|_{\text{op}} \leq D_\lambda^{-1/3}. \quad (14.4)$$

Proof. Put $D = D_\lambda$ and restrict V_λ to $H_1 \times \cdots \times H_{16}$. Its irreducible types are tensor products of sixteen carriers, with arbitrary multiplicities. For an occurring tensor type, the product of its carrier dimensions is at most D , so at least one carrier has dimension at most $D^{1/16}$. Assign the whole isotypic component of that type to one such index j . This decomposes every unit vector v into an orthogonal sum $v = \sum_{j=1}^{16} v_j$, where v_j is supported only on H_j -types of dimension at most $D^{1/16}$.

Branching shows that each H_j -type diagram is a subdiagram of λ ; its tail below its first row has at most k boxes. Let $p(i)$ count the partitions of i , with $p(0) = 1$. There are at most

$$B_k = \sum_{i=0}^k p(i), \quad \log B_k = O(\sqrt{k} \log(k+1)),$$

such types, by the elementary multiplicity-and-list count in (6.1). Its sharper exponential bound also suffices.

Let W_j be the span of the H_j -orbit of v_j . For one type of degree a , collect all equivalent copies as $V_a \otimes \mathcal{M}_a$. The orbit of the component of v_j lies in the image of

$$\text{End}(V_a) \longrightarrow V_a \otimes \mathcal{M}_a, \quad T \longmapsto (T \otimes I)v_j.$$

Its dimension is at most a^2 , regardless of the multiplicity. Consequently

$$\dim W_j \leq B_k D^{1/8}. \quad (14.5)$$

This is the point at which grouping equivalent copies is essential: counting each copy separately would lose the degree saving.

We now use the caps to average these small orbit spaces over the full group. Since W_j is H_j -invariant, the space $\rho_\lambda(x)W_j$ depends only on xH_j . Its projection, averaged uniformly over the group, is $(\dim W_j/D)I$ by Schur's lemma and the trace. For a unit vector w , Cauchy–Schwarz and the coset cap therefore give

$$\begin{aligned} |\langle w, \widehat{\nu}(\lambda)v_j \rangle| &\leq \|v_j\| \left(\mathbb{E}_{x \sim \nu} \|P_{\rho_\lambda(x)W_j} w\|^2 \right)^{1/2} \\ &\leq \|v_j\| \sqrt{M_n B_k D^{-7/8}}. \end{aligned}$$

Because $\sum_j \|v_j\| \leq 4$, summing yields the quantitative bound

$$\|\widehat{\nu}(\lambda)\|_{\text{op}} \leq 4\sqrt{M_n B_k} D^{-7/16}. \quad (14.6)$$

It remains to absorb the growing cap and the type count. Lemma 14.2 gives $\log D \geq ck$, whereas $\log M_n = O(n^{1/50})$ and $\log B_k = O(\sqrt{k} \log(k+1))$. For $k > n^{1/20}$, both of the latter quantities are $o(k)$ uniformly. Thus, for large n , $\log(16M_n B_k) \leq (5/24) \log D$. Squaring (14.6) and using this inequality gives $16M_n B_k D^{-7/8} \leq D^{-2/3}$, as required. $\square$

14.3 Combining the large and small degrees

The preceding proposition leaves two classes untreated: diagrams with more than $n/2$ rows, and diagrams with at most $n^{1/20}$ boxes below the first row. The following theorem states exactly what a second law must provide on those classes.

Theorem 14.4 (Entropy and sparse-degree hybrid). *Let $n \rightarrow \infty$ through multiples of 16. For each n , let μ and K be probabilities on S_n . Suppose a fixed equal sixteen-block partition satisfies (14.1) for μ . Suppose also that K satisfies*

$$\|\widehat{K}(\lambda)\|_{\text{op}} \leq n^{-k/10} \quad \left(1 \leq k = n - \lambda_1 \leq n^{1/20}\right), \quad (14.7)$$

$$\widehat{K}(\lambda) = 0 \quad (\lambda'_1 > n/2). \quad (14.8)$$

Then the retained probability ν from Lemma 14.1 satisfies

$$\|\nu^{*12} * K^{*40} - U_{S_n}\|_{\text{TV}} = o(1), \quad \|\mu^{*12} * K^{*40} - U_{S_n}\|_{\text{TV}} = o(1). \quad (14.9)$$

All factors in these convolutions are independent. There is no assumed identity between K and a convolution root or power of μ .

Proof. Put $\sigma = \nu^{*12} * K^{*40}$. The Fourier matrices multiply in this order. Each factor is a contraction, since it is an average of unitary matrices. For $\lambda'_1 \leq n/2$ and $k > n^{1/20}$, Proposition 14.3 gives $\|\widehat{\sigma}(\lambda)\|_{\text{op}} \leq D_{\lambda}^{-4}$. For more than $n/2$ rows, (14.8) gives zero. For $1 \leq k \leq n^{1/20}$, (14.7) gives $\|\widehat{\sigma}(\lambda)\|_{\text{op}} \leq n^{-4k}$. These statements use submultiplicativity and do not require normality of any Fourier matrix.

Plancherel bounds four times the squared total-variation distance by $\sum_{\lambda \neq (n)} D_{\lambda}^2 \|\widehat{\sigma}(\lambda)\|_{\text{op}}^2$. For each fixed k , there are at most $p(k)$ diagrams with $n - \lambda_1 = k$, since the boxes below the first row form a partition of k . By Lemma 14.2, the contribution from the large-degree range is at most

$$\sum_{k > n^{1/20}} p(k) e^{-6ck} = o(1).$$

For the small-degree range, branching shows that V_{λ} occurs in the permutation representation on ordered distinct k -tuples: its restriction to S_{n-k} contains the trivial representation because its first row has length $n - k$. Thus $D_{\lambda} \leq (n)_k \leq n^k$, and that contribution is at most

$$\sum_{1 \leq k \leq n^{1/20}} p(k) n^{2k} n^{-8k} \leq \sum_{k \geq 1} e^{3\sqrt{k}} n^{-6k} = o(1).$$

This also explains why no separate sign hypothesis is needed here: the sign diagram has more than $n/2$ rows and is annihilated by K . The first assertion follows. Convolution contracts total variation, so replacing twelve factors changes the distance by at most $12\|\mu - \nu\|_{\text{TV}} = o(1)$, proving the second. $\square$

15 Signed tabloids and conditional products

Information about the images of almost all positions can control a representation through a larger space with a particularly simple basis. In a signed tabloid representation, each permutation permutes the basis vectors and changes some signs. A cap on tuple probabilities therefore becomes a cap on matrix entries. We construct such a space containing each irreducible, with dimension

bounded by a fixed power of the irreducible dimension. This gives a normalized Hilbert–Schmidt bound. For two half-permutations which are independent after conditioning on an environment, a second averaging step converts that bound into an operator-norm estimate on the full symmetric group.

For a finite set E , write S_E for its permutation group and U_E for uniform measure on that group. An ordering of any subset $I \subseteq E$ is fixed when we write $g|_I$ or $g^{-1}|_I$ as a tuple. The uniform distribution on the possible tuples is independent of this choice of ordering. A subprobability is a nonnegative measure of total mass at most one. For a unitary representation ρ_λ of dimension D_λ , our Fourier convention is

$$\hat{\nu}(\lambda) = \sum_g \nu(g) \rho_\lambda(g), \quad \|B\|_{\text{HS}}^2 = \text{Tr}(B^* B).$$

Thus the Hilbert–Schmidt norm uses the ordinary, unnormalized trace. Composition yx applies x first, and convolution has this same order.

Theorem 15.1 (Conditional-product transfer). *Put $C_0 = 2,000,000$, and fix an integer $L \geq 2C_0$. Let $V = A \sqcup C$, where $|A| = |C| = h$ and L divides h . Partition each half into L buffers of size h/L , and put $H = S_A \times S_C \subseteq S_V$. Let ξ be a subprobability on S_V such that, for $J = A, C$ and every ordered injection $\mathbf{v} : J \rightarrow V$,*

$$\xi\{x : x^{-1}|_J = \mathbf{v}\} \leq 2U_V\{x : x^{-1}|_J = \mathbf{v}\}. \quad (15.1)$$

Let Z have any probability distribution. Measurably in z , let $t_z \in S_V$ and let ν_A^z, ν_C^z be subprobabilities on S_A, S_C , respectively. Suppose that for almost every z , each $J = A, C$, each of its buffers B , and every ordered injection $\mathbf{w} : J \setminus B \rightarrow J$,

$$\nu_J^z\{y : y|_{J \setminus B} = \mathbf{w}\} \leq 2U_J\{y : y|_{J \setminus B} = \mathbf{w}\}. \quad (15.2)$$

Regard the product $\nu_A^z \otimes \nu_C^z$ as a measure on H and set

$$\omega = \mathbb{E}_Z[\delta_{t_Z} * (\nu_A^Z \otimes \nu_C^Z)], \quad \theta = \omega * \xi.$$

Then, for all sufficiently large h , every irreducible representation of S_V with $D_\lambda > 1$ satisfies

$$\|\hat{\theta}(\lambda)\|_{\text{op}} \leq K_0 D_\lambda^{-1/40}, \quad K_0 = \sqrt{22}. \quad (15.3)$$

The threshold on h and the constant K_0 are independent of the environment, the transporters and all the measures.

The product in the definition of ω is part of the hypothesis: after fixing z , the two factors are separate subprobabilities. The theorem allows their laws to depend arbitrarily on z , while the convolution with ξ uses the same measure for every environment. The first measure controls inverse images of whole halves; the conditional factors control images of buffer complements. The proof below uses these two orientations at different steps.

15.1 A signed tabloid containing each irreducible

For positive part sizes $s_1, \dots, s_r$ summing to m , choose on each factor of $S_{s_1} \times \dots \times S_{s_r}$ either its trivial or its sign representation. Inducing their tensor product to S_m gives a signed tabloid representation. It has an orthonormal basis indexed by ordered set partitions $(T_1, \dots, T_r)$ of

$\{1, \dots, m\}$ with $|T_i| = s_i$, and each group element acts by a signed permutation matrix. Its dimension is

$$M = \frac{m!}{\prod_i s_i!}.$$

Tensoring the whole representation with sign preserves this basis property and interchanges the trivial and sign choices on all factors. We use the usual tableau description of irreducibles, the hook-length formula and the Pieri rule; see [13, 4].

Lemma 15.2 (Polynomial-size signed tabloid embedding). *For every $m \geq 1$ and every partition $\lambda \vdash m$, the irreducible representation V_λ occurs in a signed tabloid representation of dimension M with*

$$M \leq D_\lambda^{C_0}, \quad C_0 = 2,000,000.$$

Proof. The subgroup constructed in Lemma 7.1 is a product of symmetric groups on disjoint assigned row or column sets. Those sets partition the m labels, so after omitting empty sets it is conjugate to $J = S_{s_1} \times \dots \times S_{s_r}$. The occurring character χ is trivial or sign on each factor. The full-group and trivial-subgroup choices in that lemma have the same form.

Frobenius reciprocity now places V_λ in $\text{Ind}_J^{S_m} \chi$. Its coset basis is indexed by the ordered set partitions of sizes $s_1, \dots, s_r$, and each group element permutes this basis with signs because χ takes values in $\{1, -1\}$. It is therefore a signed tabloid representation, with

$$M = [S_m : J] \leq D_\lambda^{1024} \leq D_\lambda^{C_0}.$$

□

Appendix B gives an independent construction from diagonal arms and legs, together with a degree-sum proof using those same part sizes. The tuple estimate below needs only the signed basis and the polynomial dimension bound just proved.

15.2 Tuple caps give a normalized Hilbert–Schmidt bound

Lemma 15.3 (Buffer-complement estimate). *Let E have m elements, partitioned into L buffers, and let ν be a subprobability on S_E . Suppose that for every buffer B , the marginal of ν on the ordered image tuple of $E \setminus B$ is pointwise at most twice its uniform marginal. If $L \geq 2C_0$, then for every irreducible γ of S_E ,*

$$\frac{\|\widehat{\nu}(\gamma)\|_{\text{HS}}^2}{D_\gamma} \leq \min\{1, 2D_\gamma^{-1/2}\}. \quad (15.4)$$

Proof. Fix two tabloids of part sizes $s_1, \dots, s_r$, and let E_0 be the event that a permutation takes the first to the second. For buffer B_z , let s_i^z count the elements of the first tabloid's part i lying in B_z , and set

$$M = \frac{m!}{\prod_i s_i!}, \quad M_z = \frac{|B_z|!}{\prod_i s_i^z!}.$$

Relax E_0 by requiring the correct target part only for positions outside B_z . This relaxed event is determined by the corresponding image tuple. Conditional on any successful outside assignment, the unassigned target parts have sizes s_i^z , so a uniform completion succeeds with probability $1/M_z$. Since $U_E(E_0) = 1/M$, the relaxed event has uniform probability M_z/M . Its probability under ν , and hence $\nu(E_0)$, is at most $2M_z/M$.

The product $\prod_z M_z$ counts the ways to assign the part names inside the buffers with these fixed counts; all such assignments are among the M unrestricted assignments with totals s_i . Thus $\prod_z M_z \leq M$, and some buffer has $M_z \leq M^{1/L}$. We have proved the uniform entry bound

$$\nu(E_0) \leq 2M^{-1+1/L}.$$

In the signed tabloid representation, each absolute entry of the averaged matrix is bounded by this probability. Each column has absolute sum at most $\nu(S_E) \leq 1$, since every group element acts by a signed permutation matrix. Its squared Hilbert–Schmidt norm is consequently at most $2M^{1/L}$.

Choose the signed tabloid containing γ from Lemma 15.2. Orthogonal decomposition into irreducibles shows that its squared Hilbert–Schmidt norm bounds the contribution of any one copy of γ . Hence

$$\frac{\|\widehat{\nu}(\gamma)\|_{\text{HS}}^2}{D_\gamma} \leq 2D_\gamma^{C_0/L-1} \leq 2D_\gamma^{-1/2}.$$

The bound by one follows separately because the average of unitaries against a subprobability is a contraction. $\square$

We also need to retain the multiplicity in restriction to two halves. The following elementary consequence of the Littlewood–Richardson rule is stronger than a count of the occurring types alone.

Lemma 15.4 (Two-factor multiplicity bound). *For $\lambda \vdash 2h$ and $\alpha, \beta \vdash h$, let $c_{\alpha\beta}^\lambda$ be the multiplicity of $V_\alpha \otimes V_\beta$ in the restriction of V_λ to $S_h \times S_h$. Then*

$$c_{\alpha\beta}^\lambda \leq \min(D_\alpha, D_\beta).$$

Proof. The Littlewood–Richardson rule counts certain tableaux of skew shape λ/α and content β whose reading words satisfy the lattice condition. The word determines the tableau. Every prefix of such a word has weakly decreasing counts of the successive letters. All words of content β with this prefix condition are in bijection with standard tableaux of shape β : put the successive integers in the next available box of the row specified by the letter. The prefix condition makes each partial shape a partition. There are D_β such words, so the multiplicity is at most D_β . Interchanging the two factors gives the bound by D_α . $\square$

15.3 Positive conjugation and the product transfer

We now have the two ingredients needed for Theorem 15.1: a small normalized Hilbert–Schmidt norm for each conditional factor, and a bound on the multiplicity of each pair of factors. The remaining step is to use the first law’s inverse-image caps to average positive operators.

Proof of Theorem 15.1. Fix an irreducible $\rho = \rho_\lambda$ of S_V , and put $D = D_\lambda > 1$. For almost every z , let

$$B_z = \sum_{y \in H} (\nu_A^z \otimes \nu_C^z)(y) \rho(y).$$

Then

$$\widehat{\theta}(\lambda) = \mathbb{E}_Z \sum_x \xi(x) \rho(t_Z) B_Z \rho(x).$$

Jensen's inequality remains valid for a subprobability by adding the zero vector with its missing mass. Since $\rho(t_Z)$ is unitary, for every vector u we obtain

$$\|\widehat{\theta}(\lambda)u\|^2 \leq \mathbb{E}_Z \sum_x \xi(x) \|B_Z \rho(x)u\|^2. \quad (15.5)$$

It is enough to bound the positive operator in the right side.

Fix z for the moment. The restriction to H decomposes as

$$V_\lambda|_H = \bigoplus_{\alpha, \beta \vdash h} V_\alpha \otimes V_\beta \otimes \mathbb{C}^{c_{\alpha\beta}^\lambda}.$$

On one occurring block write $a = D_\alpha$, $b = D_\beta$ and $c = c_{\alpha\beta}^\lambda$. Because the conditional measure is a product, the block of B_z is exactly

$$B_A(\alpha) \otimes B_C(\beta) \otimes I_c, \quad B_J(\gamma) = \widehat{\nu_J^z}(\gamma).$$

Both factor matrices are contractions. The contribution of this block to $B_z^* B_z$ is therefore bounded in positive-semidefinite order by the operator Q_A whose block is

$$B_A(\alpha)^* B_A(\alpha) \otimes I_b \otimes I_c$$

and which is zero on the other blocks. In particular Q_A commutes with $\rho(S_C)$.

If x and x' have the same ordered inverse images on A , then $x' = kx$ for some $k \in S_C$. Thus $\rho(x)^* Q_A \rho(x)$ depends only on $x^{-1}|_A$. It is positive, so the pointwise cap (15.1) gives an operator inequality:

$$\begin{aligned} \sum_x \xi(x) \rho(x)^* Q_A \rho(x) &\preceq 2 \mathbb{E}_{x \sim U_V} \rho(x)^* Q_A \rho(x) \\ &= 2 \frac{\text{Tr } Q_A}{D} I = 2 \frac{cab}{D} \frac{\|B_A(\alpha)\|_{\text{HS}}^2}{a} I. \end{aligned}$$

Here $P \preceq Q$ means that $Q - P$ is positive semidefinite; the uniform average is scalar by irreducibility, and its trace determines the scalar. Dropping the A factor instead gives the same estimate with $\|B_C(\beta)\|_{\text{HS}}^2/b$, using the cap on $x^{-1}|_C$. Consequently the averaged contribution of this block is bounded by

$$2 \frac{cab}{D} \min \left\{ 1, 2a^{-1/2}, 2b^{-1/2} \right\} I, \quad (15.6)$$

where Lemma 15.3 supplies the two normalized norms.

The weights cab/D sum to one. For blocks with $\max(a, b) \geq D^{1/10}$, summing (15.6) therefore costs at most $4D^{-1/20}I$. On each remaining block, Lemma 15.4 gives $c \leq D^{1/10}$, so $cab/D \leq D^{-7/10}$. By Theorem 6.3, for all sufficiently large h the number of partitions of h with degree less than $D^{1/10}$ is at most

$$D^{1/10} \sum_{\alpha \vdash h} D_\alpha^{-1} \leq 3D^{1/10}.$$

There are at most $9D^{1/5}$ such pairs. Their total contribution to (15.6) is at most $18D^{-1/2}I$. Thus, uniformly in z ,

$$\sum_x \xi(x) \rho(x)^* B_z^* B_z \rho(x) \preceq (4D^{-1/20} + 18D^{-1/2})I \preceq 22D^{-1/20}I.$$

Insert this into (15.5) and take square roots. This proves (15.3). $\square$

15.4 Mass, sign and a fixed convolution power

Corollary 15.5 (Completion of the conditional-product bound). *Identify V with $\{1, \dots, 2h\}$, and consider a sequence of the data of Theorem 15.1 with $h \rightarrow \infty$ and fixed L . Suppose that μ is a probability on S_{2h} satisfying $\theta \leq \mu$ pointwise,*

$$\theta(S_{2h}) = 1 - o(1), \quad |\hat{\mu}(\text{sgn})| = o(1).$$

Then

$$\|\mu^{*100} - U_{\{1, \dots, 2h\}}\|_{\text{TV}} \rightarrow 0.$$

In particular the sign hypothesis holds if μ has exactly zero expected sign.

Proof. Write $q = \theta(S_{2h})$ and $j = 100$. Convolution preserves pointwise domination of nonnegative measures, so $\theta^{*j} \leq \mu^{*j}$, and the missing mass is $1 - q^j = o(1)$. Also

$$|\hat{\theta}(\text{sgn})| \leq |\hat{\mu}(\text{sgn})| + (1 - q) = o(1).$$

The trivial coefficient of $\theta^{*j} - U$ is $q^j - 1 = o(1)$, and its sign coefficient is $\hat{\theta}(\text{sgn})^j = o(1)$. For all other irreducibles, operator-norm submultiplicativity and $\|B\|_{\text{HS}} \leq \sqrt{D}\|B\|_{\text{op}}$ give

$$\begin{aligned} \sum_{D_\lambda > 1} D_\lambda \|\hat{\theta}(\lambda)^j\|_{\text{HS}}^2 &\leq K_0^{2j} \sum_{D_\lambda > 1} D_\lambda^{2-2j/40} \\ &= K_0^{200} \sum_{D_\lambda > 1} D_\lambda^{-3} = o(1), \end{aligned}$$

by Theorem 6.3. No normality assumption on the Fourier matrices is used.

Finite-group Plancherel, including the trivial and sign terms, now yields

$$(2h)! \sum_g \left| \theta^{*j}(g) - \frac{1}{(2h)!} \right|^2 = o(1).$$

Cauchy–Schwarz gives vanishing absolute sum of this difference. Adding the nonnegative missing measure of mass $1 - q^j$ proves the asserted total-variation convergence for μ^{*j} . $\square$

A Independent degree-summation arguments

The quantitative toolkit in Section 6 proves every inverse-power limit needed by the transfers. The arguments below retain distinct ways to obtain those limits: some use only the composition bound on partitions, and some avoid the hook formula. We use the same notation $L(\lambda)$, $k(\lambda)$, D_λ , C_u and $Z_u(n)$ as in that section. The common tableau and hook estimates are Lemmas 6.1 and 6.2.

A.1 A hook proof at the twentieth power

The following proof needs only the composition bound on the number of partitions. Its explicit exponent is useful when an argument counts small subgroup types without the sharper partition estimate.

Lemma A.1 (Inverse-twentieth powers and type counts). *One has $Z_{20}(n) \rightarrow 0$. There is an absolute constant K_0 such that, for all $n \geq 1$ and $R \geq 1$,*

$$\#\{\lambda \vdash n : D_\lambda \leq R\} \leq K_0 R^{20}. \tag{A.1}$$

Proof. Orient the first row to have length $L = L(\lambda)$ and put $k = n - L$. If $L < n/8$, then $D_\lambda \geq (4/e)^n$, so the contribution is at most $[2(e/4)^{20}]^n = o(1)$. For $L \geq n/8$ and $k \geq 1$, (6.6) gives

$$D_\lambda \geq \binom{n}{k} \exp\left(-\frac{k(1 + \log L)}{L}\right).$$

There are at most $2 \cdot 2^k$ shapes at each k . For $1 \leq k \leq \sqrt{n}$, the exponent is uniformly $o(1)$ and $\binom{n}{k} \geq (n/k)^k \geq n^{k/2}$. The contribution is at most

$$2e^{o(1)} \sum_{k=1}^{\lfloor \sqrt{n} \rfloor} (2n^{-10})^k = o(1).$$

For $\sqrt{n} < k \leq 7n/8$, use $\binom{n}{k} \geq (8/7)^k$ and $k(1 + \log L)/L \leq 7(1 + \log n)$. This contribution is at most

$$2e^{140} n^{140} \sum_{k > \sqrt{n}} [2(7/8)^{20}]^k = o(1),$$

since $2(7/8)^{20} < 1$. The full inverse-twentieth sum is uniformly bounded after adjoining the row and column and the finitely many small sizes. Each type of dimension at most R contributes at least R^{-20} to that sum, proving (A.1). $\square$

A.2 Direct reciprocal and inverse-square proofs

These proofs share a summation step, not a dimension estimate. At deficit $k \geq 1$ there are at most $2p(k)$ shapes. A lower bound that diverges at each fixed k and has a summable inverse-power majorant therefore gives a vanishing near-row and near-column sum. In the remaining ranges, $D_\lambda \geq e^{cn}$, or even $e^{cn^{0.6}}$, beats $p(n) \leq e^{3\sqrt{n}}$. Adding the row and column and finitely many small sizes then gives the corresponding constant C_u . Each argument below specifies its dimension bounds and the partition count it uses.

Lemma A.2 (Direct inverse-square summability). *One has $C_2 < \infty$ and $Z_2(n) \rightarrow 0$. This conclusion admits each of the tableau, hook-ratio, and second-row proofs below.*

Initial-row interleavings and diagonals. Orient a longest line as the first row $L = n - k$. For $1 \leq k \leq n/3$, the bound $D_\lambda \geq \binom{n-k}{k} \geq 2^k$ and the composition count $2p(k) \leq 2 \cdot 2^k$ give the majorant $2 \cdot 2^{-k}$, with fixed- k divergence. If $k > n/3$ and $L \geq n/8$, retain $\lfloor L/2 \rfloor$ lower boxes to obtain

$$D_\lambda \geq \binom{L}{\lfloor L/2 \rfloor} \geq \frac{2^L}{L+1} \geq \frac{2^{n/8}}{n+1}.$$

If $L < n/8$, diagonal filling gives $D_\lambda \geq 2^{3n/4}$. The shared summation step completes this hook-free proof.

The alternative diagonal cutoff $L < n/4$ gives $D_\lambda \geq 2^{n/2}$. For $L \geq n/4$, retain $b = \min(k, \lfloor L/2 \rfloor)$: $k \leq L/2$ gives the same 2^k majorant, while $k > L/2$ gives the exponential central-binomial bound. $\square$

Initial-row interleavings with factorial or hook tails. At $k \leq n/4$, use $\binom{n-k}{k} \geq 3^k$, with inverse-square majorant $2(2/9)^k$; at $k \leq n/3$, use 2^k and $2 \cdot 2^{-k}$. Either binomial diverges for fixed positive

k . The following rows give alternative completions of these near-row bounds. In each middle range retain the first row and b lower boxes:

middle range	b	lower bound for D_λ
$k > n/4, L \geq n/10$	$\lfloor L/4 \rfloor$	4^b
$k > n/4, L \geq n/8$	$\lfloor L/4 \rfloor$	4^b
$k > n/4, L \geq n/8$	$\lfloor n/32 \rfloor$	4^b
$k > n/4, L \geq n/8$	$\lfloor n/16 \rfloor$	2^b
$k > n/3, L \geq n/10$	$\lfloor L/3 \rfloor$	3^b
$k > n/4, L \geq n^{0.6}$	$\lfloor L/4 \rfloor$	4^b

Every row follows from $D_\lambda \geq \binom{L}{b} \geq (L/b)^b$. There are enough lower boxes: $k > n/4$ implies $k > L/3$, and $k > n/3$ implies $k > L/2$; the stated cutoffs also ensure $b \leq L$. The fixed-cutoff rows are exponential in n . Below $L = n/10$ or $n/8$, the factorial form of diagonal filling gives, respectively, $(5/e)^n$ or $(4/e)^n$. These are hook-free completions; using the hook formula gives the same two bounds independently. For the moving cutoff, the middle bound is $e^{cn^{0.6}}$ and hooks below the cutoff give $n!/(2n^{0.6})^n$. Thus every row completes the inverse-square sum. $\square$

First-row hook ratios. For $k \leq n/4$, (6.7) gives $D_\lambda \geq e^{-1} \binom{n}{k}$, so the inverse-square sum is at most

$$2e^2 \sum_{1 \leq k \leq n/4} 2^k (k/n)^{2k} = o(1).$$

The summands inside the sum are at most 8^{-k} and vanish at fixed k . In the remainder $L < 3n/4$, use $(5/e)^n$ for $L \leq n/10$; otherwise retain $b = \lfloor L/4 \rfloor$ lower boxes. The hook ratio gives $D_\lambda \geq e^{-1} \binom{L+b}{b} \geq e^{-1} 4^b$, where $b \geq n/50$ for large n .

Two alternative hook cutoffs retain different quantitative losses. For $L \geq n/2$, opposite rearrangement gives

$$D_\lambda \geq \binom{n}{k} e^{-k(1+\log L)/L} \geq 2^k e^{-k(1+\log L)/L} \geq (\sqrt{2})^k$$

for large n . Here use $2p(k) = e^{O(\sqrt{k})}$ for domination; the binomial gives fixed- k divergence. For $n/10 \leq L < n/2$, $\binom{n}{L}$ is exponential in n and the logarithmic correction is $O(\log n)$; below $n/10$ use $(5/e)^n$. Alternatively, split at $L = 3n/4$ and $n/8$: the first range has short-tail loss $k/(L-k+1) \leq 1$, the middle uses the same $b = \lfloor L/4 \rfloor$ hook subdiagram, and the last uses $(4/e)^n$. Each choice proves the limit by the shared summation argument. $\square$

Second-row width and two-row rectangles. Write $c = \lambda_2$ after orientation. At $1 \leq k \leq n/4$,

$$D_\lambda \geq \binom{n-c}{k} \geq \binom{n-k}{k}.$$

Its ratio base is at least $\sqrt{n}/2$ for $k \leq \sqrt{n}$ and at least three for $\sqrt{n} < k \leq n/4$, so the composition count gives vanishing geometric sums. In the remainder, hooks handle $L < n/10$. For $L \geq n/10$ and $c \leq L/2$,

$$D_\lambda \geq \binom{k+L-c}{k} \geq 2^{\min(k, L-c)} \geq 2^{n/20}.$$

The middle inequality follows by retaining $\min(k, L-c)$ factors in the binomial product. If $c > L/2$, the diagram contains a two-row rectangle of width $\lfloor n/20 \rfloor$, whose Catalan number is exponential in n . The subexponential total count finishes. The alternative near-row cutoff $k \leq n/3$ gives majorant $2 \cdot 2^{-k}$; in its remainder $n/10 \leq L < 2n/3$, the same minimum is at least $n/20$, and the rectangle may have width c . $\square$

Direct reciprocal bounds by interleaving and ballot words. Interleaving gives $D_\lambda \geq \binom{n-k}{k} \geq 3^k$ on $1 \leq k \leq n/4$. Use the sharper deficit count $2p(k) \leq 2e^{3\sqrt{k}}$ to sum $2p(k)3^{-k}$. For $k > n/4$, hooks give 2^n if $L < n/(4e)$; otherwise $b = \min(k, \lfloor L/2 \rfloor)$ is a positive linear fraction of n and $\binom{L}{b} \geq 2^b$. Equivalently, split at $L = n/8$, using $b = \lfloor n/16 \rfloor$ above and $(4/e)^n$ below. Both give $C_1 < \infty$ and $Z_1(n) \rightarrow 0$ without a near-row hook ratio.

Ballot words permit the coarser composition count near a long row. For $L \geq n/2$, the binomial difference in (6.3) diverges at fixed k , while its Catalan bound is at least $4^k/((k+1)(2k+1))$. The reciprocal contribution is therefore dominated by a constant times $(k+1)(2k+1)2^{-k}$. For $n/10 \leq L < n/2$, retain L lower boxes and use their Catalan bound; below $n/10$, hooks give $(5/e)^n$.

A quarter-tail ballot completion instead splits at $L = n/8$ and $3n/4$. The first range has hook bound $(4/e)^n$; in the middle, $b = \lfloor L/4 \rfloor$ lower boxes give

$$D_\lambda \geq \binom{L+b}{b} \frac{L-b+1}{L+1},$$

exponential in n . For $L > 3n/4$, the ballot fraction is at least $1/2$, so $D_\lambda \geq \frac{1}{2} \binom{n}{k} \geq \frac{1}{2} 4^k$. The deficit count and fixed- k divergence finish this version as well. $\square$

A reciprocal hook proof. The hook estimates in Lemma 7.1 give $D_\lambda \geq e^{n/64}$ for $n \geq 64$ and $L \leq 3n/4$, which beats the subexponential total count. For $L > 3n/4$ and $k \geq 1$,

$$D_\lambda \geq e^{-1}(n/k)^k \geq e^{-1}4^k.$$

The composition count gives the reciprocal majorant $2e2^{-k}$, and the first bound diverges at fixed k . Thus $Z_1(n) \rightarrow 0$ and $C_1 < \infty$ follow independently from the hook comparisons used to control the subgroup index. $\square$

A.3 Coarse partition counts and square-root dimension bounds

Lemma A.3 (Elementary sixteenth and thirteenth powers). *The conclusions $C_u < \infty$ and $Z_u(n) \rightarrow 0$ for $u = 13, 16$ follow from tableau counting and $p(n) \leq 2^n$ alone.*

Proof. If $1 \leq k \leq n/3$, the initial-row construction gives $D_\lambda \geq \binom{n-k}{k} \geq 2^k$, with at most $2 \cdot 2^k$ shapes. The contribution is bounded by $2 \cdot 2^{-(u-1)k}$, and each fixed- k contribution tends to zero. For $k > n/3$ and $L \geq n/8$, retain $\lfloor L/2 \rfloor$ lower boxes to get $D_\lambda \geq 2^{n/8}/(n+1)$. The remaining contribution in this range is at most

$$(n+1)^u 2^{(1-u/8)n} = o(1).$$

For $L < n/8$, diagonal filling gives $D_\lambda \geq 2^{3n/4}$, and the contribution is at most $2^{(1-3u/4)n} = o(1)$. At $u = 16$ these two bounds are exactly $(n+1)^{16}2^{-n}$ and 2^{-11n} . At $u = 13$ they vanish because $13/8 > 1$ and $39/4 > 1$. The finitely many small-size sums give the uniform constants. $\square$

Lemma A.4 (A tail-adapted diagonal bound). *For all sufficiently large n and every shape of deficit $k = k(\lambda) \geq 1$,*

$$D_\lambda \geq 2^{k/2}, \quad D_\lambda \geq L(\lambda) \geq \sqrt{n}.$$

These inequalities and $p(k) \leq 2^k$ prove $C_4 < \infty$ and $Z_4(n) \rightarrow 0$.

Proof. Let a be the largest value of $i + j - 1$ over cells (i, j) . Diagonal filling gives $D_\lambda \geq 2^{n-a}$. Every cell satisfies $ij \leq n$, so its smaller index is at most $\sqrt{n}$ and $a \leq L + \sqrt{n}$. If $k < n/2$, orient the first row as L . A lower cell satisfies $(i-1)j \leq k$, whence $i + j - 1 \leq k + 1 \leq L$; thus $a = L$. If $k \geq n/2$, then $n - a \geq k - \sqrt{n} \geq k/2$ for large n . This proves the first bound in both cases. A non-row, non-column shape contains $(L, 1)$ after orientation; that hook has L tableaux, proving the second bound. For each k , at most $2p(k) \leq 2 \cdot 2^k$ shapes occur, so their inverse fourth powers are bounded by $2 \cdot 2^{-k}$. At fixed positive k they tend to zero by $D_\lambda \geq \sqrt{n}$. Dominated convergence and the finite-small-size argument finish. $\square$

Lemma A.5 (Inverse-tenth powers from a square-root first line). *One has $C_{10} < \infty$ and $Z_{10}(n) \rightarrow 0$, using only the first-row interleaving construction and the bound $p(n) \leq e^{3\sqrt{n}}$.*

Proof. Orient the longest line as the first row $L \geq \sqrt{n}$. For $1 \leq k < L/2$, use $D_\lambda \geq \binom{L}{k} \geq 2^k$; the contribution at k is at most $2 \cdot 2^{-9k}$ and tends to zero for fixed k . For $k \geq L/2$, retain $\lfloor L/2 \rfloor$ lower boxes. The bound $D_\lambda \geq 2^L/(L+1)$ gives total contribution at most

$$e^{3\sqrt{n}}(n+1)^{10}2^{-10\sqrt{n}} = o(1).$$

This proves the limit and the uniform bound. $\square$

Central-binomial powers eight and twenty. Divide at $k \leq n/3$. Initial-row interleaving gives $D_\lambda \geq \binom{n-k}{k} \geq 2^k$, with fixed- k divergence and near-row majorants $2 \cdot 2^{-7k}$ and $2 \cdot 2^{-19k}$. For $k > n/3$, one has $k > L/2$ and $L \geq \sqrt{n}$. Retaining $\lfloor L/2 \rfloor$ lower boxes gives $D_\lambda \geq 2^L/(L+1)$, so the remaining sums are at most

$$e^{3\sqrt{n}}(n+1)^u 2^{-u\sqrt{n}} = o(1) \quad (u = 8, 20),$$

because $8 \log 2 > 3$. This proves both limits directly from central-binomial interleavings, without ballot words.

Weaker central-binomial estimates. For powers 16 and 32, the same near-row cutoff and $\binom{L}{\lfloor L/2 \rfloor} \geq 2^{\lfloor L/2 \rfloor}$ give

$$e^{3\sqrt{n}} 2^{-u\lfloor \sqrt{n}/2 \rfloor} = o(1) \quad (u = 16, 32). \quad (\text{A.2})$$

Unlike Lemma A.3, these proofs use only a square-root lower bound on L and no diagonal filling. For power 12, divide at $k \leq \lfloor L/2 \rfloor$: the near-row bound $\binom{L}{k} \geq 2^k$ gives majorant $2 \cdot 2^{-11k}$ and fixed- k divergence. In the remainder, $\binom{L}{\lfloor L/2 \rfloor} \geq 2^{L/2}$ for large L gives $e^{3\sqrt{n}-6(\log 2)\sqrt{n}} = o(1)$.

The ballot inverse-eighth route. For $1 \leq k \leq L$, the ballot fraction is at least $1/(k+1)$. Thus $D_\lambda \geq 2^k/(k+1)$ gives summable majorant $2^k(k+1)^8 2^{-8k}$, using $2p(k) \leq 2^k$; the full binomial difference diverges at fixed k . For $k > L$, retain L lower boxes to get $D_\lambda \geq \binom{2L}{L}/(L+1) \geq 2^L/(L+1)$. The remaining sum is at most $e^{3\sqrt{n}}(n+1)^8 2^{-8\sqrt{n}} = o(1)$. For every route, adjoining the two linear types and finitely many small sizes gives the complete sum's uniform boundedness.

Lemma A.6 (A square-root lower bound in the deficit). *For every $n \geq 1$ and $\lambda \vdash n$,*

$$\log D_\lambda \geq \frac{\log 2}{2} \sqrt{k(\lambda)}. \quad (\text{A.3})$$

In particular $C_{32} < \infty$ follows from this bound alone and $p(k) \leq e^{3\sqrt{k}}$.

Proof. Orient a longest line as the first row. Put $b = \lambda_2$ and $h = \lambda'_1$. For $k \geq 1$, $b(h-1) \geq k$. If $h-1 \geq \sqrt{k}$, the diagram contains a hook with row and column length h , because $L \geq h$. That hook has $\binom{2h-2}{h-1} \geq 2^{h-1}$ tableaux. Otherwise $b > \sqrt{k}$ and the diagram contains a two-row rectangle of width b . Its Catalan number is at least 2^{b-1} : encode the 2^{b-1} compositions of b as distinct Dyck paths obtained by concatenating pyramids of those lengths. In this second case $b \geq 2$, so $b-1 \geq b/2 > \sqrt{k}/2$. Subdiagram extension proves (A.3); the case $k = 0$ is immediate. Finally sum over deficits using at most $2p(k)$ shapes:

$$\sum_{\lambda \vdash n} D_\lambda^{-32} \leq 2 + \sum_{k \geq 1} 2e^{3\sqrt{k}} e^{-16(\log 2)\sqrt{k}} < \infty.$$

□

A.4 Further direct fixed-power constructions

Two final counts retain the larger powers that arise when only especially rough estimates are used. They are independent specializations of the tableau constructions, not deductions from an already proved smaller inverse power.

For power 30 and $1 \leq k \leq n/4$, choose all k lower labels from $\{2k+1, \dots, n\}$. Fill them in one fixed standard order and put the other labels increasingly in the first row. Since the first $2k$ labels then lie in the top row, every lower predecessor condition holds. This gives

$$D_\lambda \geq \binom{n-2k}{k} \geq (n/(2k))^k \geq 2^k. \quad (\text{A.4})$$

The $2 \cdot 2^k$ shape count supplies the summable majorant $2 \cdot 2^{-29k}$ and fixed- k convergence. For $k > n/4$, put $b = \lfloor \sqrt{n}/4 \rfloor$ and retain the first row $L \geq \sqrt{n}$ together with b lower boxes. Applying the same over-reserving construction to this subdiagram gives $D_\lambda \geq \binom{L-b}{b} \geq 3^b$. Since $30 \log 3/4 > 3$, its inverse thirtieth power dominates $p(n) \leq e^{3\sqrt{n}}$. Hence $Z_{30}(n) \rightarrow 0$ and $C_{30} < \infty$.

For power 32, a proof using the rough count 2^n separates three ranges. If $L \leq n/8$, hooks give $(4/e)^n$, which is enough because $32 \log(4/e) > \log 2$. If $L > n/8$ and $k > n/3$, then $k > L/2$ and retaining $\lfloor L/2 \rfloor$ lower boxes gives $D_\lambda \geq 2^{\lfloor L/2 \rfloor}$; its inverse thirty-second power also dominates 2^n . For $1 \leq k \leq n/3$, use

$$D_\lambda \geq \binom{n-k}{k}, \quad 2^{k+1} \left(\frac{k}{n-k} \right)^{32k} \leq 2^{1-31k},$$

with fixed- k convergence. This proves the same complete inverse-thirty-second assertion without using the square-root partition bound.

For power 6, the hook ratio gives $D_\lambda \geq e^{-1} \binom{n}{k}$ for $1 \leq k \leq n/3$. The near-row and near-column contribution is bounded by

$$2e^6 \sum_{1 \leq k \leq n/3} [2(k/n)^6]^k = o(1). \quad (\text{A.5})$$

For $k \leq \sqrt{n}$ compare with the geometric series of base $2n^{-3}$; for $k > \sqrt{n}$ use $2(k/n)^6 \leq 2/3^6 < 1$. In the remainder $L \leq 2n/3$. If $L \geq n/(4e)$, retain $b = \lfloor L/2 \rfloor$ lower boxes and apply the hook ratio to the resulting diagram, obtaining $e^{-1} \binom{L+b}{b} \geq e^{-1} 3^b$. If $L < n/(4e)$, hooks give $(n/(2eL))^n > 2^n$. These exponential bounds finish the sum with $p(n) = e^{O(\sqrt{n})}$.

A direct fourth-power proof. For power 4, one may also use the near-row majorant $2 \cdot 2^{-3k}$ on $k \leq n/3$, the middle bound $2^L/(L+1)$ on $n/8 \leq L < 2n/3$, and the hook bound $(4/e)^n$ on $L < n/8$. This is a different use of the fourth power from the tail-adapted diagonal argument in Lemma A.4. It uses the subexponential total partition count in its last two ranges. $\square$

The sharper quantitative estimate

$$\sum_{\lambda \vdash n} D_{\lambda}^{-u} = 2 + O(n^{-u}) \quad (u > 0 \text{ fixed})$$

is due to Liebeck and Shalev [7, Theorem 2.6]. The arguments above supply the boundedness and vanishing assertions used here without that rate.

The corresponding alternating-group statement is

$$\sum_{\chi \in \text{Irr}(A_n)} \chi(1)^{-u} = 1 + o(1) \quad (u > 0 \text{ fixed}). \quad (\text{A.6})$$

Here A_n is the subgroup of even permutations. For completeness, the index-two restriction rule makes the two vanishing assertions equivalent. If V is irreducible for S_n , Frobenius reciprocity and $\text{Ind}_{A_n}^{S_n} \text{Res}_{A_n}^{S_n} V \cong V \oplus (V \otimes \text{sgn})$ show that the endomorphism algebra of its restriction has dimension one or two. Thus the restriction is irreducible or is a sum of two distinct irreducibles. In the latter case an odd permutation exchanges the two summands, since otherwise either would be an S_n -invariant proper subspace. Both therefore have dimension $\dim V/2$. Every irreducible of A_n occurs in some restriction, by induction and Frobenius reciprocity; it occurs in restrictions of at most two S_n -types, since its induced representation has dimension twice its own and each such type has at least its dimension. For $n \geq 5$, write $Z_u^A(n)$ for the sum over the nontrivial A_n -types. The only S_n -types restricting to the trivial type are the trivial and sign representations, so the preceding counts give

$$Z_u(n) \leq 2Z_u^A(n), \quad Z_u^A(n) \leq 2^{u+1} Z_u(n).$$

This proves the equivalence and (A.6), while retaining the sharper cited symmetric-group rate separately.

B An independent signed-tabloid construction

The proof of Lemma 15.2 in the main text uses the row-column assignment from Lemma 7.1. Here we construct the signed tabloid directly from diagonal arms and legs and prove its dimension bound from the hook formula. The same part sizes then give an independent reciprocal-degree argument. Throughout, $C_0 = 2,000,000$ as in Theorem 15.1.

An arm-leg proof of Lemma 15.2. Transposing λ tensors its representation with sign and does not change its dimension. We may therefore orient the diagram so that $\lambda_1 \geq \lambda'_1$, where the prime denotes transpose, and undo this orientation by a sign twist at the end. Let k be its diagonal length, and define

$$a_i = \lambda_i - i + 1, \quad b_i = \lambda'_i - i \quad (1 \leq i \leq k).$$

The a_i count horizontal arms including their diagonal boxes; the b_i count vertical legs below those boxes. They sum to m . Use the sizes $a_1, b_1, \dots, a_k, b_k$, omitting zero parts, with trivial factors on the arms and sign factors on the legs.

To check occurrence, add arm 1, leg 1, arm 2, leg 2, and so on. Immediately before arm i , rows above i have their final lengths, while row $r \geq i$ has length $\min(i-1, \lambda_r)$. Adding arm i completes row i and is a horizontal strip. Adding leg i then extends column i down to height λ'_i and is a vertical strip. Each intermediate shape is a partition. Repeated application of the horizontal and vertical Pieri rules therefore shows that the resulting induced representation contains V_λ .

It remains to compare its dimension with D_λ . The hook-length formula gives

$$\frac{M}{D_\lambda} = \frac{\prod_{1 \leq i, j \leq k} (a_i + b_j)}{\left(\prod_{i=1}^k a_i \right) \prod_{1 \leq i < j \leq k} (a_i - a_j)(b_i - b_j)}. \quad (\text{B.1})$$

Here is a direct verification of the identity. The hooks in the diagonal k by k square have lengths $a_i + b_j$. For any partition with k rows of lengths r_i , allowing zero rows, the hook lengths in row i are the integers from 1 to $r_i + k - i$ with $r_i - r_j + j - i$ omitted for every $j > i$. Indeed its hook at column t has length $r_i - t + 1 + \#\{j > i : r_j \geq t\}$; the gaps as t increases are precisely those omitted values. Apply this observation to the diagram to the right of the square, whose row lengths are $\lambda_i - k$, and to the transpose of the diagram below it, whose row lengths are $\lambda'_i - k$. Their hook products are, respectively,

$$\frac{\prod_i (a_i - 1)!}{\prod_{i < j} (a_i - a_j)}, \quad \frac{\prod_i b_i!}{\prod_{i < j} (b_i - b_j)}.$$

Division by $\prod_i a_i! b_i!$ proves (B.1).

When $k = 1$, orientation gives $a_1 \geq b_1 + 1$, so the ratio in (B.1) is at most two. A one-row diagram has $M = D_\lambda = 1$. Every other such hook has $D_\lambda \geq 2$, since the entry 2 can be placed either to the right of or below 1 and the resulting partial tableaux can be completed. The claimed bound follows in this case.

Assume now that $k \geq 2$. All logarithms in this proof are natural. Arithmetic-geometric mean bounds the numerator in (B.1) by $(m/k)^{k^2}$. The two sequences a_i and b_i are strictly decreasing integers, so their two Vandermonde products are at least $\prod_{j=1}^k ((j-1)!)^2$. For $1 \leq j \leq k$,

$$\log((j-1)!) \geq (j-1) \log k - k.$$

To see this, sum $\log(r/k)$ for $1 \leq r < j$, and bound that sum below by the sum for $1 \leq r \leq k$, which is at least $-k$ by $k! \geq (k/e)^k$. Consequently

$$\log(M/D_\lambda) \leq k^2(\log(m/k^2) + 3). \quad (\text{B.2})$$

We need two lower bounds on D_λ to pay for this ratio. Every standard tableau of a contained diagram extends to λ by adding the remaining boxes in a fixed order compatible with rows and columns. In particular the contained k by k square gives

$$D_\lambda \geq \frac{(k^2)!}{(2k)^{k^2}} \geq \left(\frac{k}{2e}\right)^{k^2}.$$

It follows that

$$\log D_\lambda \geq c_0 k^2 \log k, \quad c_0 = 10^{-5}. \quad (\text{B.3})$$

For $k \geq 64$ the displayed factorial estimate proves this directly. For $2 \leq k < 64$, the contained two-square gives $D_\lambda \geq 2$, and $10^{-5} 63^2 \log 63 < \log 2$ suffices.

Orientation also gives $a_1 \geq m/(2k)$, since every arm and leg has size at most a_1 . The diagram contains $(a_1, k, \dots, k)$ with k rows. Fill its first k top-row boxes with the smallest entries. Then interleave the rest of the top row with any fixed standard order of the $k(k-1)$ lower boxes. Each interleaving is a standard tableau, so

$$D_\lambda \geq \binom{a_1 - k + k(k-1)}{k(k-1)}.$$

If $m \geq 16k^3$, put $r = m/k^3$, $u = a_1 - k$ and $v = k(k-1)$. Then $u \geq m/(4k)$ and $k^2/2 \leq v \leq k^2$. Using $\binom{u+v}{v} \geq ((u+v)/v)^v$ gives

$$\log D_\lambda \geq \frac{k^2}{2} \log(r/4) \geq \frac{k^2}{4} \log r. \quad (\text{B.4})$$

If $m < 16k^3$, then $\log(m/k^2) + 3 \leq 13 \log k$, and (B.2)–(B.3) give $\log M \leq (1 + 13/c_0) \log D_\lambda$. If $m \geq 16k^3$, write $\log(m/k^2) + 3 = \log(m/k^3) + \log k + 3$ and use both degree bounds. This gives

$$\log M \leq \left(5 + \frac{1 + 3/\log 2}{c_0}\right) \log D_\lambda.$$

Both coefficients are less than 2,000,000, completing the proof. $\square$

This arm–leg construction also gives an independent way to sum reciprocal representation degrees. The same tabloid sizes therefore supply both the embedding and the degree estimates used for counting small restriction types and summing the final Fourier bounds.

Lemma B.1 (Degree sums from the tabloid embedding). *For every fixed $s > 0$,*

$$\sum_{\substack{\lambda \vdash m \\ D_\lambda > 1}} D_\lambda^{-s} \rightarrow 0 \quad (m \rightarrow \infty).$$

In particular $\sum_{\lambda \vdash m} D_\lambda^{-1} \leq 3$ for all sufficiently large m .

Proof. Let $p(u)$ count partitions of u , with $p(0) = 1$. For $u \geq 1$, evaluating its generating product at $x = e^{-1/\sqrt{u}}$ gives

$$p(u)x^u \leq \prod_{j \geq 1} (1 - x^j)^{-1}, \quad \log \prod_{j \geq 1} (1 - x^j)^{-1} = \sum_{i \geq 1} \frac{1}{i(e^{i/\sqrt{u}} - 1)} \leq \sqrt{u} \sum_{i \geq 1} i^{-2}.$$

Thus $p(u) \leq e^{3\sqrt{u}}$. Orient λ so that $\lambda_1 \geq \lambda'_1$, and use the preceding arm–leg construction with $a_1 = \lambda_1$ and $j = m - a_1$. For $1 \leq j \leq m/2$, at most $2p(j)$ original diagrams have this value: delete the oriented first row, and remember whether a transpose was used. The tabloid dimension satisfies $M \geq \binom{m}{j}$, and that construction gives

$$D_\lambda^{-s} \leq M^{-s/C_0} \leq \binom{m}{j}^{-s/C_0}.$$

For each fixed $j \geq 1$ this tends to zero. Since $\binom{m}{j} \geq (m/j)^j \geq 2^j$ on this range, the summable sequence $2p(j)2^{-js/C_0}$ dominates the contributions. They therefore sum to $o(1)$.

If $j > m/2$, every arm or leg used in the tabloid has size at most $a_1 < m/2$. Add some of these parts until their total w first reaches $m/4$. Then $m/4 \leq w \leq 3m/4$ and

$$M \geq \binom{m}{w} \geq (m/w)^w \geq (4/3)^{m/4}.$$

The total contribution of this range is at most $p(m)(4/3)^{-ms/(4C_0)} = o(1)$. The omitted $j = 0$ diagrams are the row and column, whose representations are trivial and sign. They are the only one-dimensional representations for $m \geq 2$, and their contribution to the inverse-first-degree sum is two. $\square$

C Further fixed-coset transfer arguments

The main transfer assigns each joint restriction type to one small factor, or covers the representation by small isotypic projections. The alternatives here retain different information. The first two arguments average many low-type projections before recovering a test vector; the third counts whole joint types. We also record parameter choices for these and the main estimates. Finally, a separate character construction removes whole longest lines instead of assigning individual boxes to rows or columns. Its dimension ratio, reciprocal-degree proof, and transfer are kept together.

For the projection arguments and substitutions below, use the fixed-coset data of Section 4. Namely, let $G = S_n$, partition $[n] = M_1 \sqcup \cdots \sqcup M_b$ into nonempty blocks, and let $H_i = \text{Sym}(M_i)$ fix the complement. Let $f \geq 0$ have mass at most one and satisfy

$$f(gH_i) \leq \frac{B}{[G : H_i]} \quad (g \in G, 1 \leq i \leq b).$$

Write $D = \dim \rho$ for the degree of the ambient irreducible and $C_u = \sup_{m \geq 1} \sum_{\lambda \vdash m} D_\lambda^{-u}$. The peeling construction concerns a single symmetric group first; its final transfer states its own equal-block data.

C.1 Averages of many low-type projections

Proposition C.1 (Coefficient-space average). *For $b = 32$ and cap $B = 4$,*

$$\|\hat{f}(\rho)\|_{\text{op}} \leq \frac{8}{3} \sqrt{C_1} D^{-5/16}.$$

Proof. Let P_i now select factor degrees at most $D^{1/8}$. At most eight factors of any product type can exceed that threshold, so $\bar{P} = 32^{-1} \sum_i P_i \succeq 3I/4$. Lemma 4.1, the coset cap, and full-group Schur orthogonality give

$$|\langle w, \hat{f}(\rho) P_i v \rangle|^2 \leq 4C_1 D^{-5/8} \|w\|^2 \|P_i v\|^2.$$

Here the selected coefficient space has dimension at most $C_1 D^{3/8}$. Thus $\|\hat{f} P_i\|_{\text{op}} \leq 2\sqrt{C_1} D^{-5/16}$. Average over i and multiply by $\bar{P}^{-1}$, whose norm is at most $4/3$. $\square$

Proposition C.2 (Orbit-space average). *For $b = 256$ and cap $B = 4$,*

$$\|\hat{f}(\rho)\|_{\text{op}} \leq 4\sqrt{C_{32}} D^{-1/4}.$$

Proof. Let P_i select degrees at most $D^{2/b}$. At least half the factors of every product type meet this condition, so $\bar{P} = b^{-1} \sum_i P_i \succeq I/2$. A vector in the range of P_i has orbit span of dimension at most $C_{32} D^{68/b}$ by Lemma 3.1. The coset cap and Schur averaging, exactly as in Proposition 3.2, give

$$\|\hat{f}P_i\|_{\text{op}} \leq 2\sqrt{C_{32}} D^{-1/2+34/b} \leq 2\sqrt{C_{32}} D^{-1/4},$$

since $68/256 \leq 1/2$. Average and use $\|\bar{P}^{-1}\|_{\text{op}} \leq 2$. This proof retains the geometric averaging mechanism and only uses the reciprocal power 32. $\square$

Proposition C.3 (Four-factor coefficient decomposition). *If there are four blocks and the subprobability cap is $B = 2$, then*

$$\|\hat{f}(\rho)\|_{\text{op}} \leq 2C_{1/4}^2 D^{-1/8}.$$

Proof. Decompose a vector orthogonally into full product types $v = \sum_{\tau} v_{\tau}$. If d_i are the four factor degrees, their product is at most D . The sum of $(\prod_i d_i)^{-1/4}$ over all possible product types is at most $C_{1/4}^4$. Thus the number T of occurring types is at most $C_{1/4}^4 D^{1/4}$.

For one type choose i with $d_i \leq D^{1/4}$. The coefficient function on gH_i belongs to a single type's coefficient space, even when that type occurs repeatedly. Lemma 4.1 bounds its supremum by d_i times its uniform root mean square. Sum over cosets with cap two, then apply Cauchy–Schwarz over the uniform cosets. Full-group Schur orthogonality yields

$$\sum_g f(g) |\langle w, \rho(g)v_{\tau} \rangle| \leq 2d_i D^{-1/2} \|w\| \|v_{\tau}\| \leq 2D^{-1/4} \|w\| \|v_{\tau}\|.$$

Finally $\sum_{\tau} \|v_{\tau}\| \leq \sqrt{T} \|v\|$, giving the claimed exponent and constant. The count here is of entire product types, unlike the blockwise assignment in Proposition 4.2. $\square$

C.2 Exact substitutions

For applications using a prescribed block count or reciprocal-degree power, the following substitutions give explicit constants and exponents. They are convenient choices, not optimal convolution counts. In Proposition 3.2, $u = 16$ gives exponent $-(1 - 18/b)/2$; $u = 2$ gives $-(1 - 4/b)/2$; $u = 1$, $b = 8$ gives $-5/16$. Each substitution changes only the inequality $\sum_{a \leq D^{1/b}} a^2 \leq C_u D^{(u+2)/b}$.

For Hilbert–Schmidt bounds, the coefficient proof with $u = 2$, $b = 16$, $B = 3$ gives $D \|\hat{f}\|_{\text{HS}}^2 \leq 9bC_2 D^{4/b}$. The pointwise character proof with $u = 2$, $b = 8$, $B = 1$ gives $\|\hat{f}\|_{\text{HS}}^2 \leq bC_2 D^{-1+6/b}$. The central-isotypic proof with $u = 16$, $b = 64$, $B = 4$ gives $4bC_{16} D^{-1+18/b}$. With respectively four, sixteen, and thirty-two convolutions, the nonsign exponents are -2 , -3 , and -22 . The coefficient proof with $u = 10$ gives $B^2 b C_{10} D^{-1+12/b}$; its sharper branching alternative is Proposition 4.6. These substitutions preserve the distinct proofs of the estimates even when a stronger estimate is also available.

C.3 A separate construction by peeling longest lines

The next construction chooses whole lines successively, instead of assigning individual boxes. Its index estimate is weaker, but its stepwise dimension inequality gives another proof of inverse-degree summability. The occurrence argument uses the horizontal- and vertical-strip cases of the Pieri rule [13].

Lemma C.4 (A subcharacter obtained by peeling). *Put $C_{\text{peel}} = 512^2$. Every irreducible representation of S_m of degree D has a one-dimensional character in its restriction to some subgroup J with $[S_m : J] \leq D^{C_{\text{peel}}}$. At a step removing a longest first row or first column of length r , leaving size s and degree D' , one has*

$$\frac{D}{D'} \geq \binom{r+s}{r}^{1/C_{\text{peel}}}. \quad (\text{C.1})$$

The degree of the empty remainder is one.

Proof. Remove the longer of the first row and first column of the current diagram. In a row removal, the realigned lower diagram μ is contained in the original diagram λ , and λ/μ is a horizontal strip: every column loses exactly its bottom box. The Pieri rule and Frobenius reciprocity show that restriction to $S_s \times S_r$ contains V_μ tensored with the trivial representation of S_r . For a column removal use the sign representation and the transposed argument. Iteration supplies a product subgroup and a product of trivial and sign characters occurring in the original representation.

To prove the index bound, it remains to compare dimensions at each step. Transpose so that a row is removed; then the diagram has height at most r . If c_j counts its lower boxes in column j , the hook formula gives

$$\frac{D}{D'} = \frac{\binom{r+s}{r}}{R}, \quad R = \prod_{j=1}^r \left(1 + \frac{c_j}{r-j+1}\right), \quad \sum_j c_j = s.$$

The case $s = 0$ is immediate. For $r \geq 512$ and $s \geq 16r$, the bound $c_j \leq r$ gives

$$R \leq \binom{2r}{r} \leq 4^r, \quad \binom{r+s}{r} \geq (1 + s/r)^r \geq 17^r.$$

Thus $R \leq \binom{r+s}{r}^{1/2}$. For $1 \leq s \leq 16r$, the opposite ordering of c_j and $(r-j+1)^{-1}$ instead gives

$$\log R \leq \frac{s}{r} \sum_{j=1}^r \frac{1}{j} \leq \frac{s(1 + \log r)}{r}, \quad \log \binom{r+s}{r} \geq s \log(1 + r/s).$$

Since $(1 + \log r)/r$ decreases for $r \geq 512$ and is at most $\frac{1}{2} \log(17/16)$ there, this also gives $D/D' \geq \binom{r+s}{r}^{1/2}$.

If $r < 512$ and $s > 0$, then $r \geq 2$ and $r + s \leq r^2$. Each standard ordering of the lower diagram yields two distinct standard tableaux: fill the first row before that ordering, or fill its first $r-1$ boxes, the first lower box, the last top-row box, and then the remaining lower boxes in that ordering. The first lower box is in column one, so the second filling is also valid. These two constructions are injective and have disjoint images. Hence $D/D' \geq 2$, while $\binom{r+s}{r} \leq 2^{r^2} \leq 2^{C_{\text{peel}}}$. This proves (C.1) in every case. Multiplying it through the successive steps telescopes the dimension ratios. The product of the binomial coefficients is exactly the index of the product subgroup constructed above, proving the assertion. $\square$

Lemma C.5 (Inverse-first-degree summability from peeling). *The preceding stepwise estimate independently implies*

$$\sum_{\lambda \vdash m: D_\lambda > 1} D_\lambda^{-1} \rightarrow 0.$$

Consequently, for all sufficiently large m and every $Y \geq 1$, S_m has at most $3Y$ irreducible types of degree at most Y .

Proof. We use $p(h) \leq e^{3\sqrt{h}}$, which follows by evaluating the partition generating function at $e^{-1/\sqrt{h}}$: the logarithm of the product is at most $\sqrt{h} \sum_{j \geq 1} j^{-2} \leq 2\sqrt{h}$, and the coefficient bound adds $\sqrt{h}$.

The first removed length r is at least $\sqrt{m}$ because both width and height are at most r . If $r < m/2$, (C.1) gives

$$\log D_\lambda \geq \frac{r \log(m/r)}{C_{\text{peel}}}.$$

The concavity of $r \log(m/r)$ bounds it below on $[\sqrt{m}, m/2]$ by the smaller of $\frac{1}{2}\sqrt{m} \log m$ and $\frac{1}{2}m \log 2$. After division by $\sqrt{m}$, this lower bound tends to infinity. It therefore dominates the logarithm $3\sqrt{m}$ of the number of all diagrams, proving a vanishing inverse sum in this range.

If $m/2 \leq r < m$, put $h = m - r \geq 1$. At most $2p(h)$ diagrams have this value, counting both orientations, and (C.1) gives $D_\lambda \geq (m/h)^{h/C_{\text{peel}}}$. For fixed h their inverse-degree contribution tends to zero. It is bounded by $2e^{3\sqrt{h}}2^{-h/C_{\text{peel}}}$, a summable sequence. Dominated convergence handles this range. The remaining diagrams are the row and column, of degree one. Once the nonlinear reciprocal sum is at most one, at most Y nonlinear types have degree at most Y ; adding the two linear types gives $Y + 2 \leq 3Y$. $\square$

Corollary C.6 (The peeling character transfer). *Let $b = 2^{22}$ and partition $n = bm$ points into b sets of size m . Let H_i permute the i th set. If $f \geq 0$ has mass at most one and $f(gH_i) \leq 2/[S_n : H_i]$ for every g, i , then for all sufficiently large m and every irreducible of degree D ,*

$$\|\hat{f}(\rho)\|_{\text{HS}}^2 \leq 6bD^{-1+(C_{\text{peel}}+2)/b}. \quad (\text{C.2})$$

Proof. Lemma 7.2, using Lemma C.4, bounds a full H_i -isotypic projection of degree d_τ by $2d_\tau^{C_{\text{peel}}+1}/D$ in squared Hilbert–Schmidt norm. As in Theorem 7.3, the projections with $d_\tau \leq Y = D^{1/b}$ cover the identity. By Lemma C.5 there are at most $3Y$ such types for each factor. Summing gives $D^{-1} \cdot 2b \cdot 3Y \cdot Y^{C_{\text{peel}}+1}$, which is (C.2). $\square$

Here $(C_{\text{peel}} + 2)/b = (262144 + 2)/4194304 < 1/8$. Thus four convolutions again tend to uniform whenever the retained mass tends to one and the original sign mean tends to zero, by Proposition 2.5. The peeling and box-assignment proofs supply the subcharacter hypothesis separately; neither uses the other’s subgroup construction.

References

- [1] A. Czumaj, Random permutations using switching networks, in *Proceedings of the Forty-Seventh Annual ACM Symposium on Theory of Computing (STOC 2015)*, ACM, 2015, 703–712. doi:10.1145/2746539.2746629.
- [2] A. Czumaj and B. Vöcking, Thorp shuffling, butterflies, and non-Markovian couplings, in *Automata, Languages, and Programming (ICALP 2014)*, Part I, Lecture Notes in Computer Science, vol. 8572, Springer, 2014, 344–355. doi:10.1007/978-3-662-43948-7_29.
- [3] P. Diaconis and M. Shahshahani, Generating a random permutation with random transpositions, *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* **57** (1981), 159–179. doi:10.1007/BF00535487.

- [4] P. Etingof, O. Golberg, S. Hensel, T. Liu, A. Schwendner, D. Vaintrob, and E. Yudovina, *Introduction to Representation Theory*, Student Mathematical Library, vol. 59, American Mathematical Society, 2011. Numbered references use the author’s lecture notes of January 10, 2011: <https://math.mit.edu/~etingof/replect.pdf>.
- [5] J. S. Frame, G. de B. Robinson, and R. M. Thrall, The hook graphs of the symmetric group, *Canadian Journal of Mathematics* **6** (1954), 316–324. doi:10.4153/CJM-1954-030-1.
- [6] G. D. James, *The Representation Theory of the Symmetric Groups*, Lecture Notes in Mathematics, vol. 682, Springer-Verlag, 1978.
- [7] M. W. Liebeck and A. Shalev, Fuchsian groups, coverings of Riemann surfaces, subgroup growth, random quotients and random walks, *Journal of Algebra* **276** (2004), no. 2, 552–601. Proposition 2.5 and Theorem 2.6 appear on pages 15–16 of the [author’s text](#).
- [8] B. Morris, Improved mixing time bounds for the Thorp shuffle and L -reversal chain, *Annals of Probability* **37** (2009), no. 2, 453–477. doi:10.1214/08-AOP409.
- [9] B. Morris, Improved mixing time bounds for the Thorp shuffle, *Combinatorics, Probability and Computing* **22** (2013), no. 1, 118–132. doi:10.1017/S0963548312000478.
- [10] OpenAI, *Conditional information under deterministic coordinate sweeps*, OpenAI Math Release preprint [OAI:Conditional-information-under-deterministic-coordinate-sweeps-September-26-2026](#), 2026.
- [11] OpenAI, *Conditional permutations in a revealed switching environment*, OpenAI Math Release preprint [OAI:Conditional-permutations-in-a-revealed-switching-environment-September-26-2026](#), 2026.
- [12] OpenAI, *Random coordinate frames and partial permutation laws*, OpenAI Math Release preprint [OAI:Random-coordinate-frames-and-partial-permutation-laws-September-26-2026](#), 2026.
- [13] B. E. Sagan, *The Symmetric Group: Representations, Combinatorial Algorithms, and Symmetric Functions*, second edition, Graduate Texts in Mathematics, vol. 203, Springer, 2001. doi:10.1007/978-1-4757-6804-6.
- [14] E. O. Thorp, Nonrandom shuffling with applications to the game of faro, *Journal of the American Statistical Association* **68** (1973), no. 344, 842–847. doi:10.1080/01621459.1973.10481434.